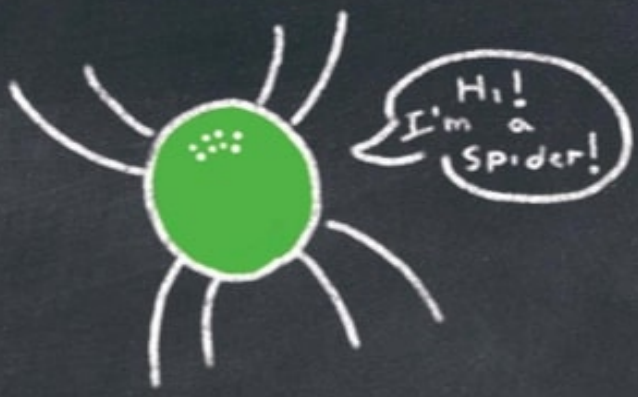
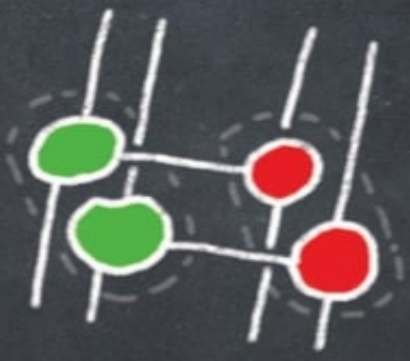
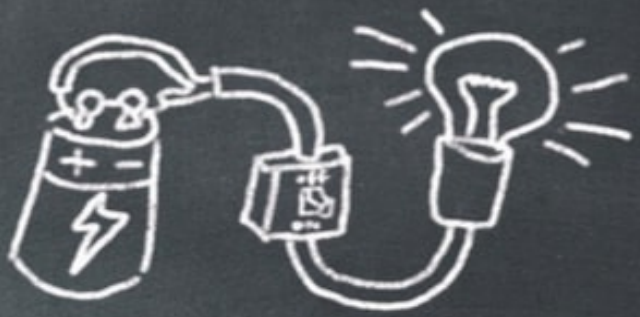
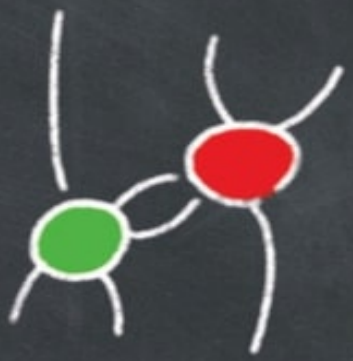


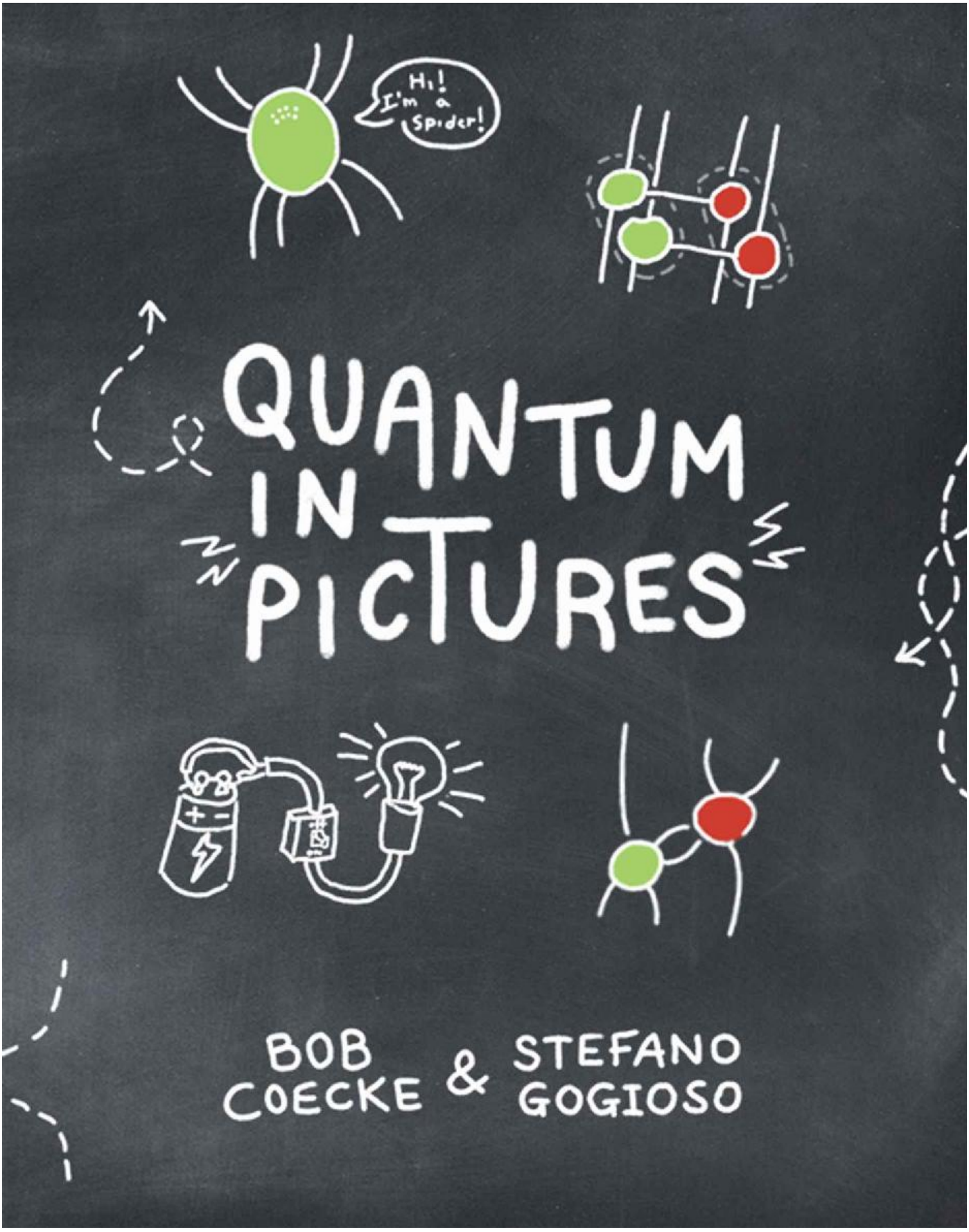


QUANTUM IN PICTURES





BOB COECKE & STEFANO GOGIOSO


Hi! I'm a spider!
QUANTUM IN PICTURES
BOB COECKE & STEFANO GOGIOSO

QUANTUM IN PICTURES

BOB COECKE & STEFANO GOGIOSO

ABOUT THIS BOOK AND ITS BIG BROTHER

Using pictures alone, this book will equip you with the tools you need to understand the quantum world—including recent developments in quantum computing—and prove things about it, both known and new. You'll learn that our pictures aren't mere illustrations, but rather a new kind of rigorous mathematics, tailor-made to talk about quantum things. This new maths was developed and refined over several years by many researchers, including the authors, and it is now being adopted by the quantum computing industry.

Because of the novelty in presentation, and the novel results that are included, this book will be of interest to both the young (and not-so-young) amateur and the quantum specialist. To keep things approachable, we came up with simple, friendly names for many of the concepts involved. At the end of the book, we will tell you how professionals describe things, so you can relate what you learn here to what you read in other books that talk about quantum.

This book has a big brother at university. Sometimes called the "dodo book", and officially called "Picturing Quantum Processes", it was published in 2017 by Bob Coecke and Aleks Kissinger. If you can't get enough of the stuff you read in this book, you may want to move on to the "dodo book" next. At the end of this book, we will give you a brief history on how this new way of talking about quantum things came about.

All the chapters of "Quantum In Pictures" are also available as videos on Quantinuum's YouTube channel, featuring some special guests.

ABOUT THE AUTHORS

Prof. Bob Coecke

Bob Coecke spent 20 years at the University of Oxford, where he built a large group of researchers dedicated to developing a new pictorial way to talk about quantum things. When this work started to find real-world applications in emerging quantum technology, he moved to Cambridge Quantum Computing (now Quantinuum). There, he is now Chief Scientist, leading the development of quantum natural language processing and quantum artificial intelligence.

Dr. Stefano Gogioso

Stefano Gogioso is a mathematician and computer scientist. He did his PhD in Oxford under the supervision of Bob Coecke, developing new pictorial ways to talk about quantum particles and the way they move. He is now a faculty member at the University of Oxford, where he teaches advanced courses on quantum computing. He also co-founded the UK startup Hashberg, where he directs the development of high-level quantum programming tools.



THANKS

We would very much like to thank young J. J. Skinner for his insightful contributions to the book, as well as Belgian teacher Dirk Vandycke and Bob's academic grandchildren Lia Yeh and John van de Wetering for their detailed reading of the final manuscript. We would also like to thank Quanlong Wang, his daughter Katie, Dan Marsden and Thomas Ehmer for their feedback, and to Ferdi Tomassini and Pete Sigrist for making the book look as good as it does, and having it ready on time. We are grateful to Ilyas Khan for his continued encouragement and the invaluable help he provided in the book's publication.



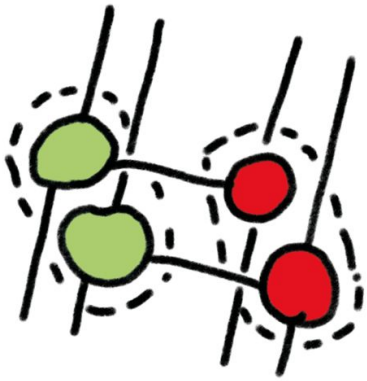


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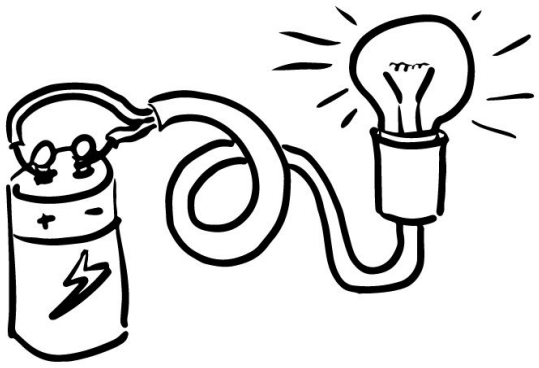
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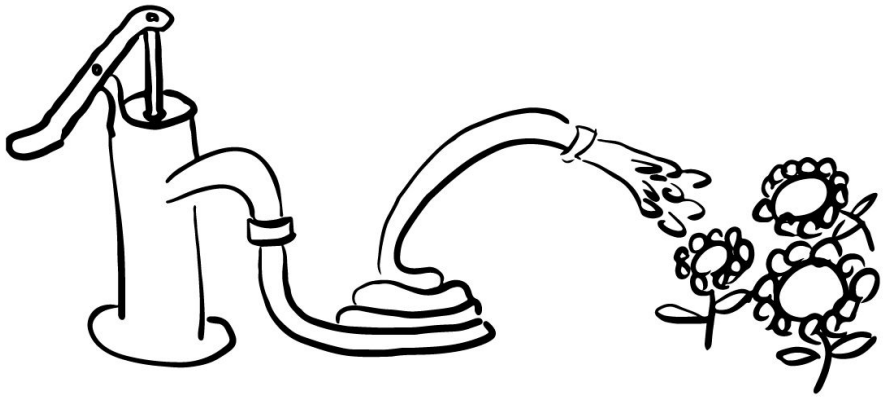
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1.
WIRES AND BOXES

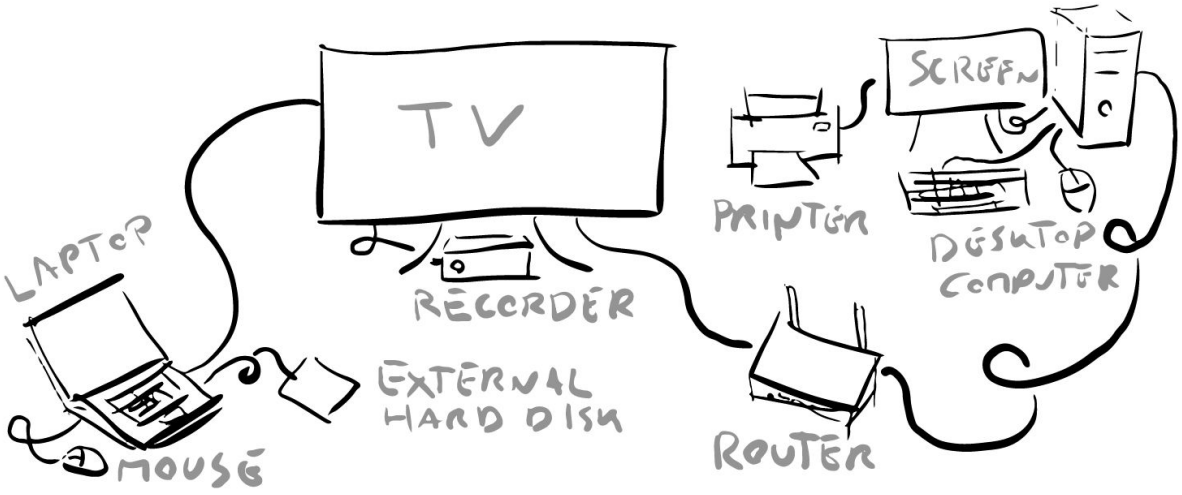
When talking quantum physics, we use wires and boxes. What is a wire? A wire is a place where things travel from somewhere to somewhere else. For example, electrical wires are the place where electricity travels, say from a battery to a light bulb:



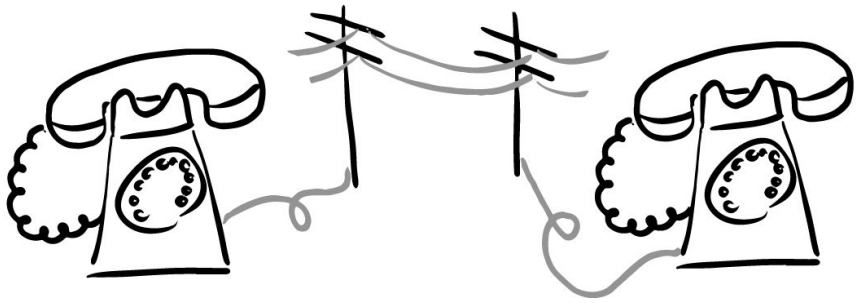
As you may already know, electricity actually consists of particles, called electrons, which travel in those wires. Similarly, water particles travel from a pump to your flowers through a hose, which we can also think of as a wire:



Computer wires are the place were 0's and 1's travel:

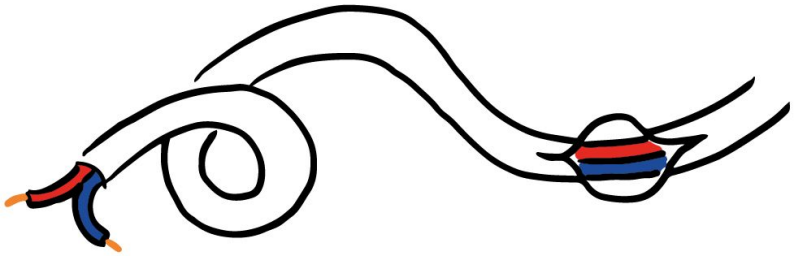


In the old days, we had telephone wires (and telephone poles), which worked just like electrical wires:



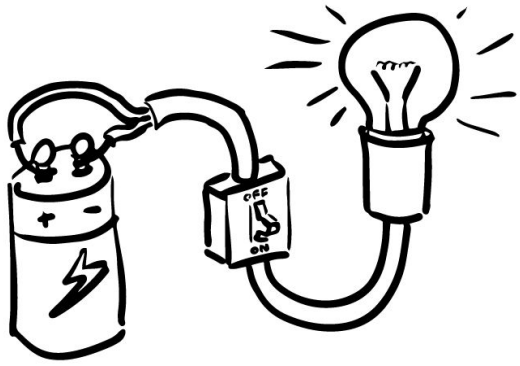
These days we can think of the air as connecting different mobile phones: the particles are travelling through it as invisible waves. So we can even think of the air around us as a wire.

Sometimes wires are composed of a number of smaller wires bunched together, each carrying a different part of the stuff that travels in the main wire. For example, electrical wires are composed of two smaller wires:



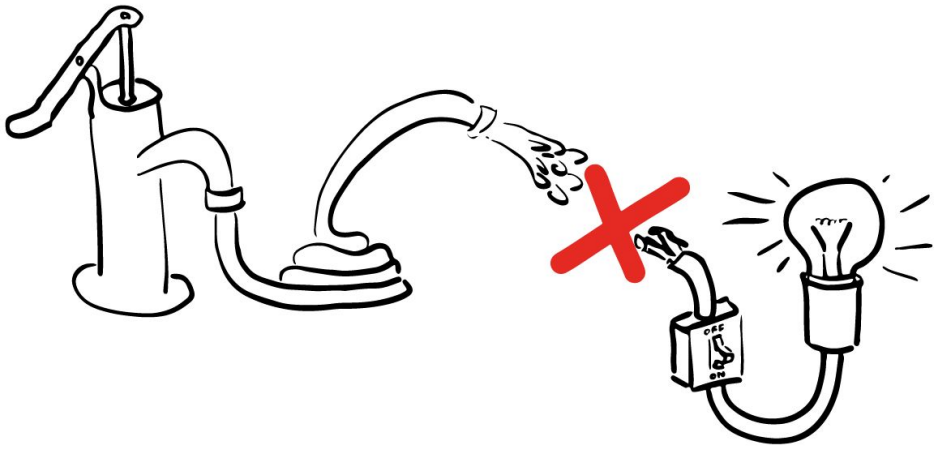
Similarly, computer wires are actually formed of many tiny wires, each carrying a single 0 or 1 at a time.

So, particles travel through wires. But what do we actually do with them? That's where boxes come into play. What is a box? A box is a place where particles in wires go to, get modified, and leave into other wires. For example, an electrical switch is a box with electrical wires going in one side and coming out the other:

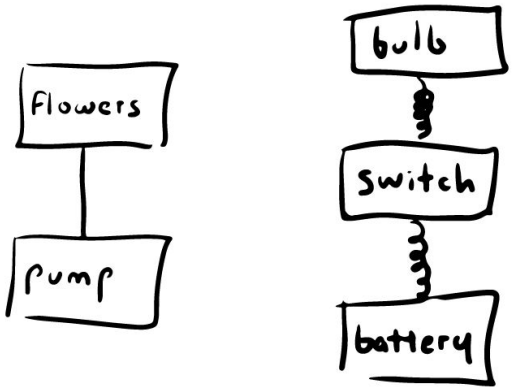


Electrical particles come into the switch, but they only come out if the switch is on. When the switch is off, no particles come out. Batteries are boxes with electricity coming out of them, while light bulbs are boxes with electricity disappearing into them.

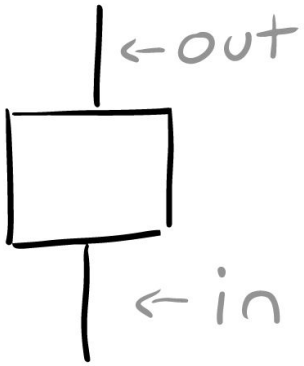
One thing to be careful about when connecting boxes with wires is to only connect wires which fit together. For example, you can't fit an EU plug into a UK socket, and you definitely don't want to connect your water pipe to an electricity wire!



There are different kinds of wires which shouldn't be mixed up, and each kind of wire has its own kinds of boxes. We can depict wires carrying water and electricity differently to distinguish them. Also, rather than having different fancy pictures for each kind of box, we will simply represent them by rectangles (or similar simple shapes):



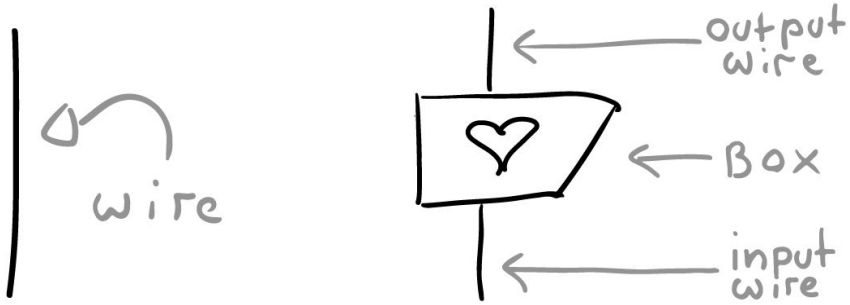
We read these pictures from the bottom to the top, particles go in at the bottom of boxes, and come out from the top:



But this book is not about flowers or about electricity: it's a book about quantum particles. These are tiny, weirdly behaving particles like electrons, protons and neutrons, which we may find in a physics laboratory or inside a quantum computer. Quantum theory is the theory of quantum wires and quantum boxes: the wires tell us where quantum particles travel, while the boxes tell us what happens to them. It's that simple!

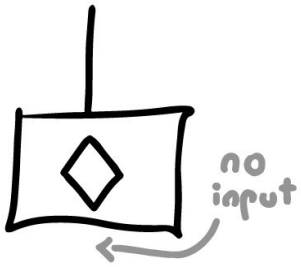
1.1
The mathematics of wires and boxes

The ingredients of our theory are wires and boxes:

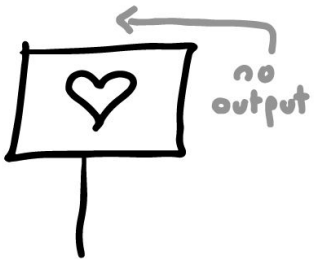


The reason why our boxes are a bit skewed will become clear later. (Some naysayers might object that these are “rectangles”, or “trapezoids”. But we learned to love them as “boxes”, regardless of what other people think!)

There are certain special boxes which either lack input wires or output wires. We call boxes without input wires states:



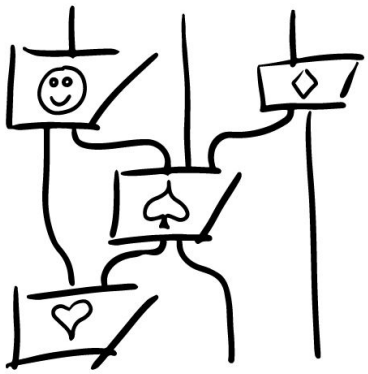
and those without output wires tests:



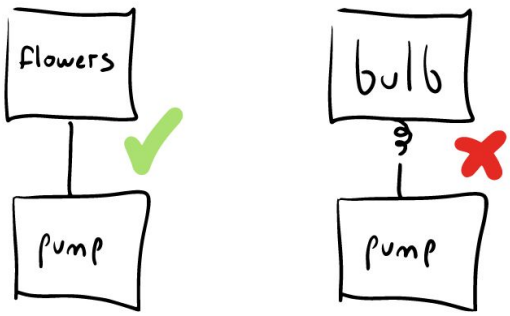
Make sure you remember these names: states and tests are really important in quantum theory!

Diagrams

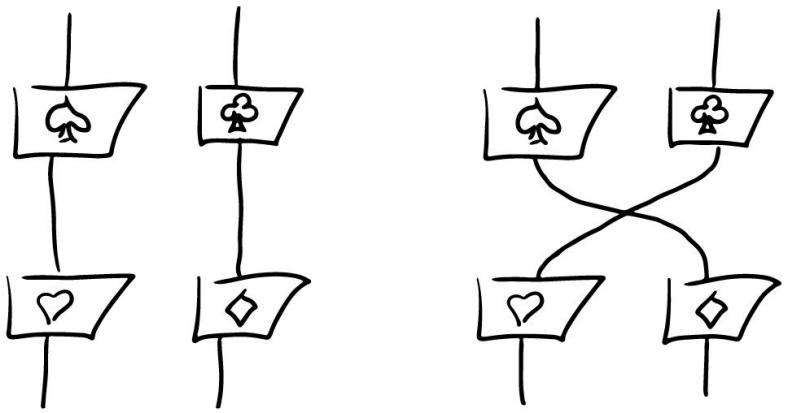
We can now wire several boxes together and form diagrams:



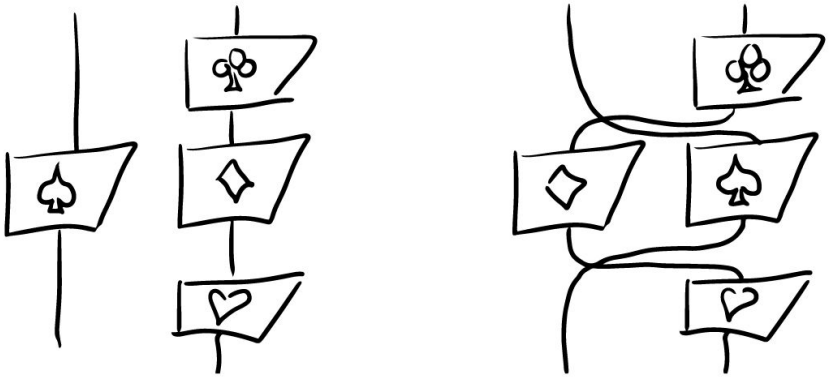
Of course, when connecting boxes together we should be very careful to only connect matching wires:



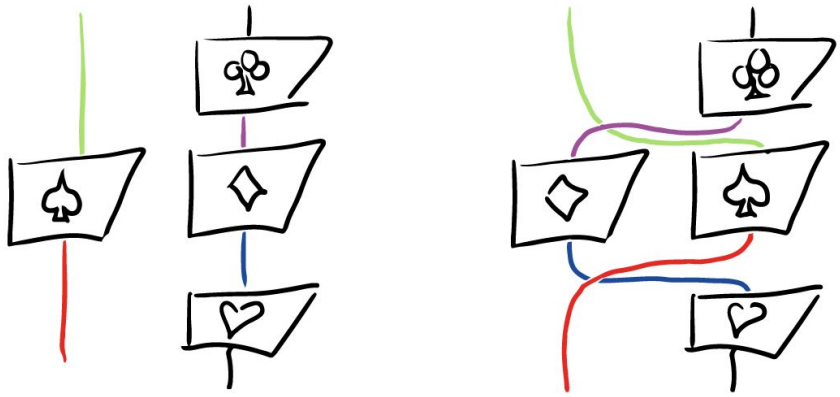
We can connect the same boxes together in many different manners, provided that wires fit of course:



Different ways of connecting wires usually will not result in the same diagram. On the other hand, it is possible that, even though the following diagrams don't look the same at first:



The boxes might be connected in exactly the same way. To make it easier to follow the wires, we have colour-coded them here:



Equal diagrams

So, two different-looking diagrams may actually be equal, something which we denote by means of an equals sign:



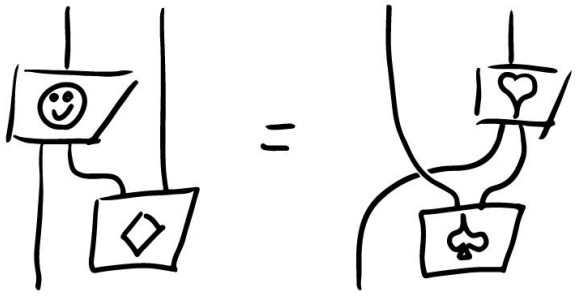
As in traditional mathematics, we call this an equation. This is somewhat similar to when we write:

$$\mathbf{1 + 3 = 3 + 1}$$

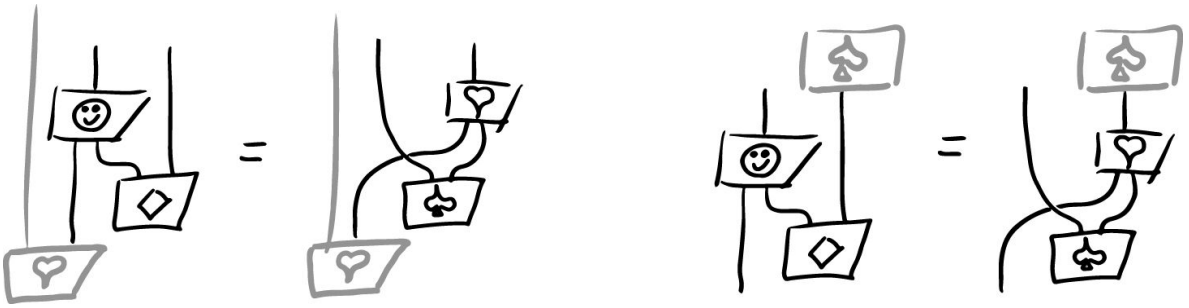
But there is more: while $1 \neq 2$ and $3 \neq 2$, we know that $1 + 3 = 2 + 2$. This is because $3 = 1 + 1 + 1$ and $2 = 1 + 1$, so:

$$\mathbf{1 + 3 = 1 + 1 + 1 + 1 = 2 + 2}$$

Similarly, diagrams with different boxes may turn out to be equal:

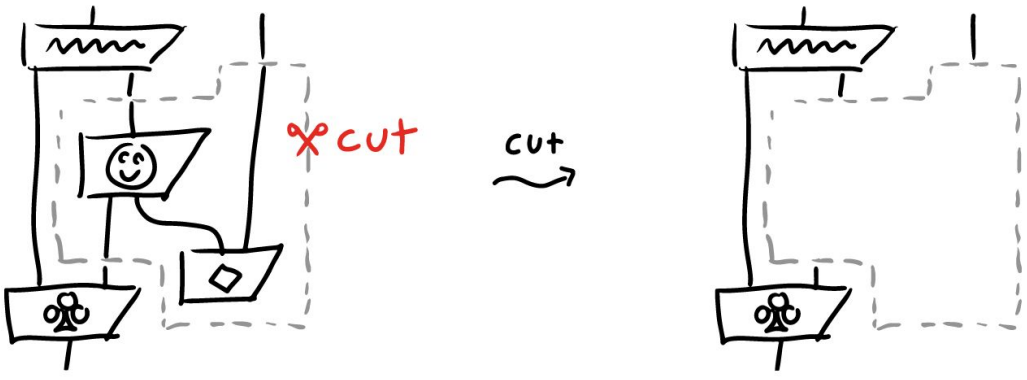


Given such an equation between diagrams, we can create many other equations. For example, we can connect new boxes at the bottom or at the top (or anywhere else we feel like) and still get equal diagrams:



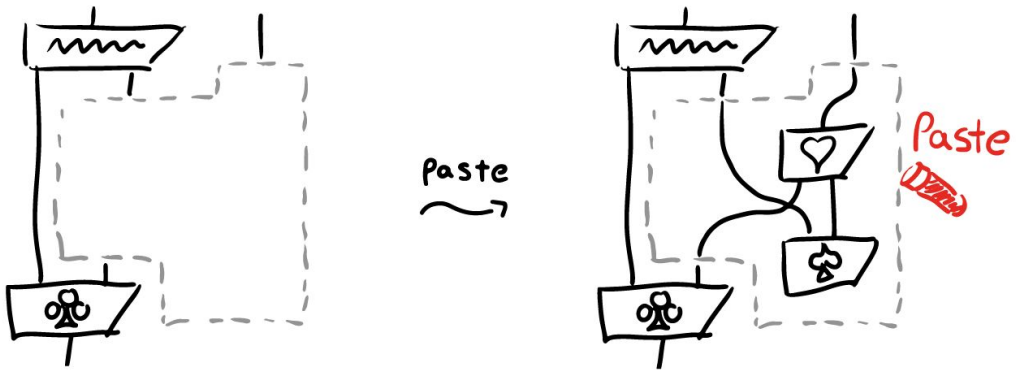
This is the same as ordinary calculations with numbers, where if we have that $x = y$, then we also have that $x + 1 = y + 1$.

We can also do some old-fashioned cut & paste! Given a diagram where one side of the equation appears, we can cut it away:

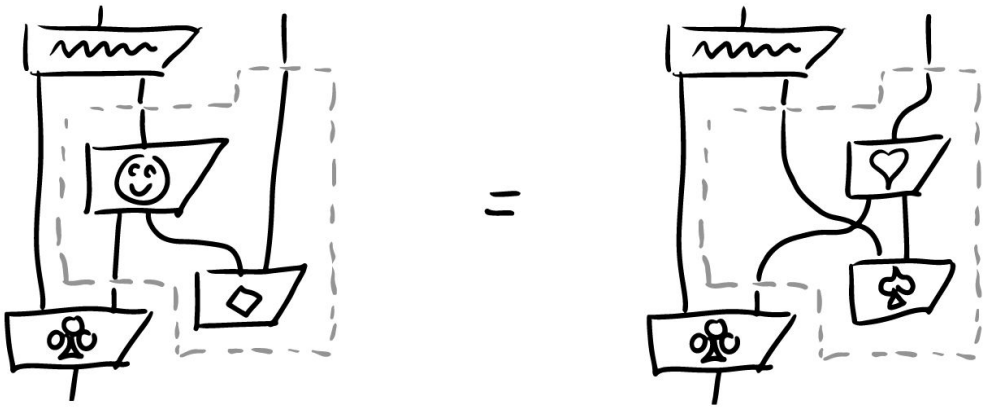


and paste the other side of the equation, making sure that we connect the wires in the

correct places:



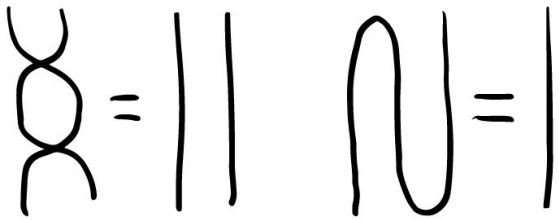
This gives the following equation between diagrams:



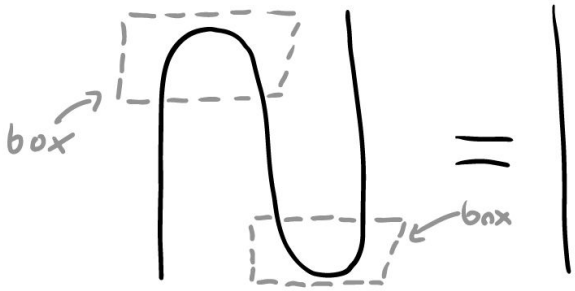
What we are really doing here is starting from x + 1, then cutting x out to obtain □ + 1, and then pasting y into the dotted box. We can do this since x=y and we conclude that x + 1 = y + 1.

Cup-boxes and cap-boxes

Here are some equations which obviously hold and which don't even involve any boxes at all:



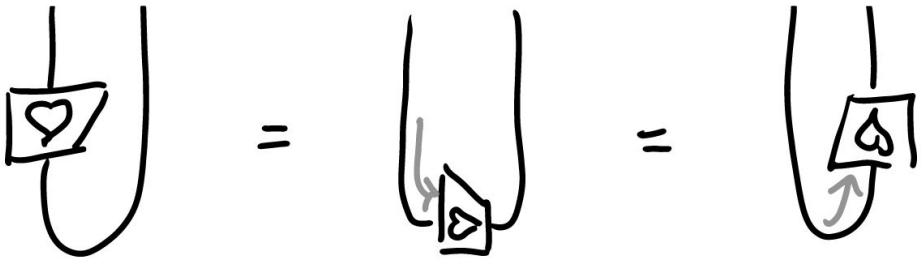
The second of these says that we can always “yank” a wire straight. But we can also think of it as involving two boxes:



Namely a state for the cup-shaped wire bend at the bottom, and a test for the cap-shaped wire bend at the top:



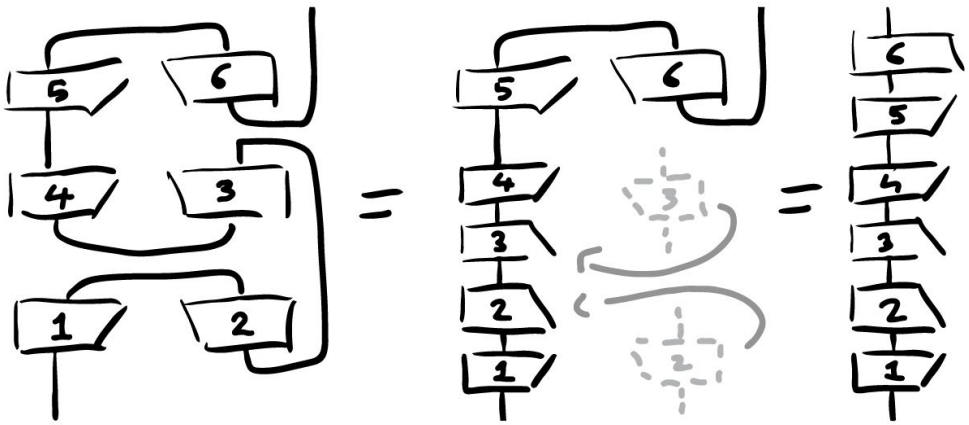
These cups and caps play a very important role in quantum theory. The fact that some boxes are actually wires in disguise is super useful, as it makes certain calculations very easy. For example, we can now obtain equal diagrams by “sliding” boxes along such wire bends:



The same can also be done for the cap test:



By combining sliding and yanking, we can come up with some crazy examples of equal diagrams:



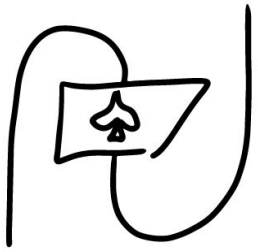
It should now also be clear why we didn't draw boxes as rectangles: this way, we can easily see the difference between the original box and the one that has slid along the wire bend.

Rotating boxes

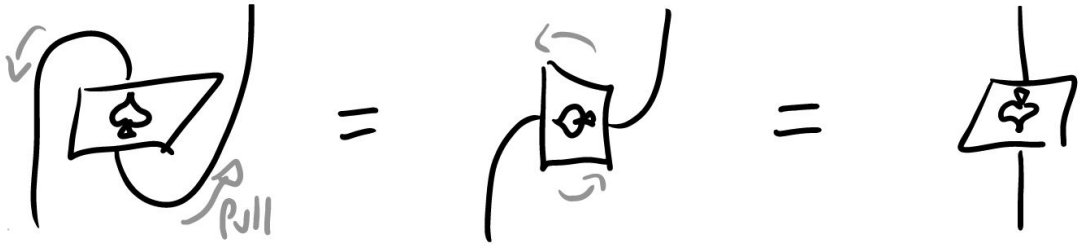
We can also use cups and caps to bend input wires into output wires and vice versa:



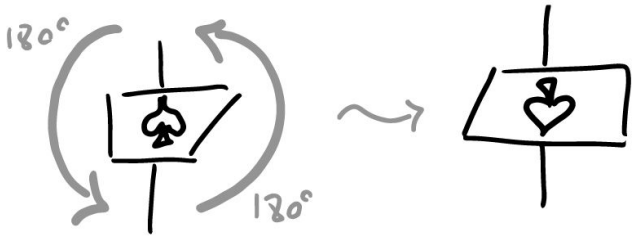
If we bend both the input wire and the output wire of a box, we obtain a new box with inputs and outputs exchanged:



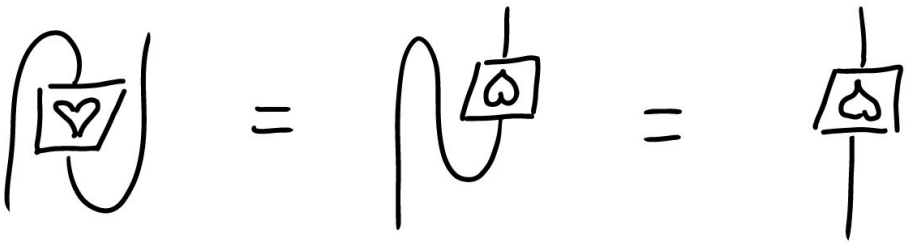
Now, when we yank the wires the following happens:



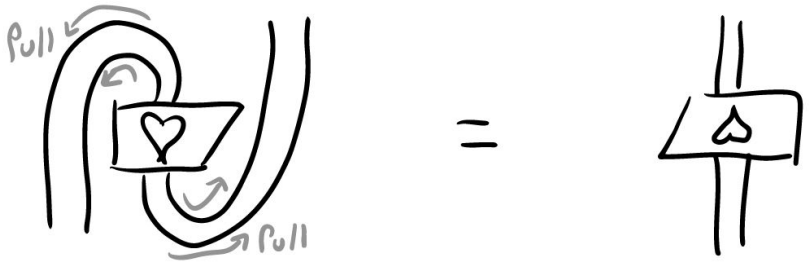
So this bending of both the input wire and the output wire is exactly the same as rotating the entire picture by 180°:



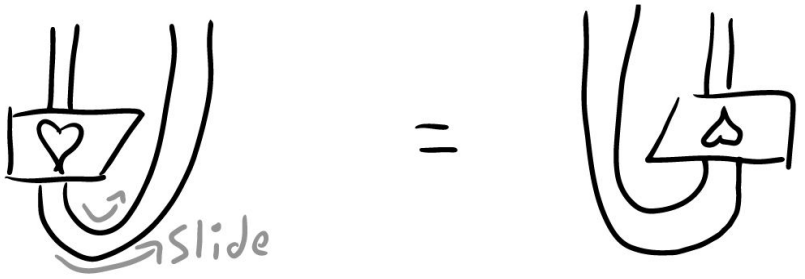
In fact, we could also have already concluded this by sliding the box along the bend and then yanking:



What if we want to rotate a box with two inputs and/or outputs? We simply use two nested cups/caps:

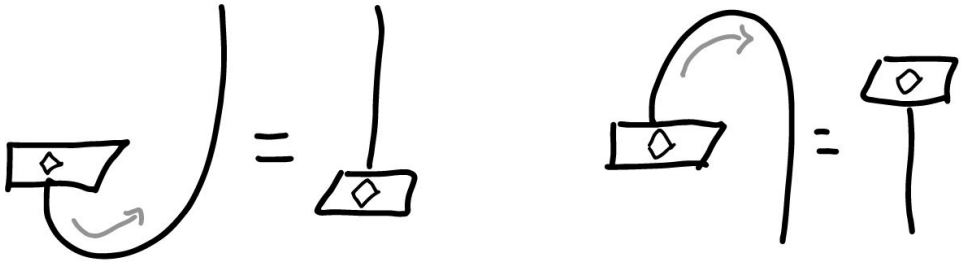


Sliding also still works the same with more than one wire:



Sliding states and tests

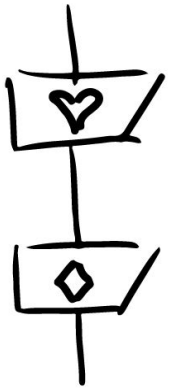
In the special case of tests, which lack a wire at one end, sliding looks as follows:



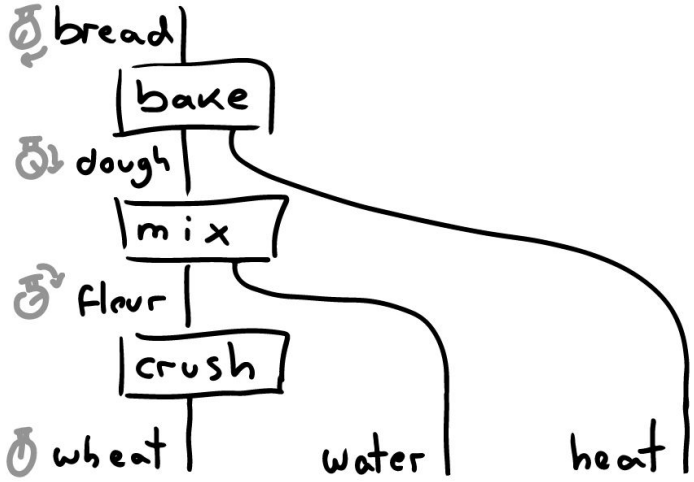
So these boxes “eat the wire”, as there is no place for the wire to come out at the other end.

1.2
Introducing space and time

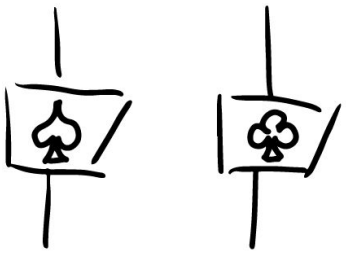
Our diagrams may not, at first, look anything like what you would expect physics to look like. In particular, space and time don't seem to make an appearance. But this couldn't be further from the truth: as we shall now see, whenever you draw a diagram you can always think of it as taking place in space and time. To begin with, let's figure out where time is hiding in our diagrams. Whenever you draw two boxes stacked above each other and connected by a wire:



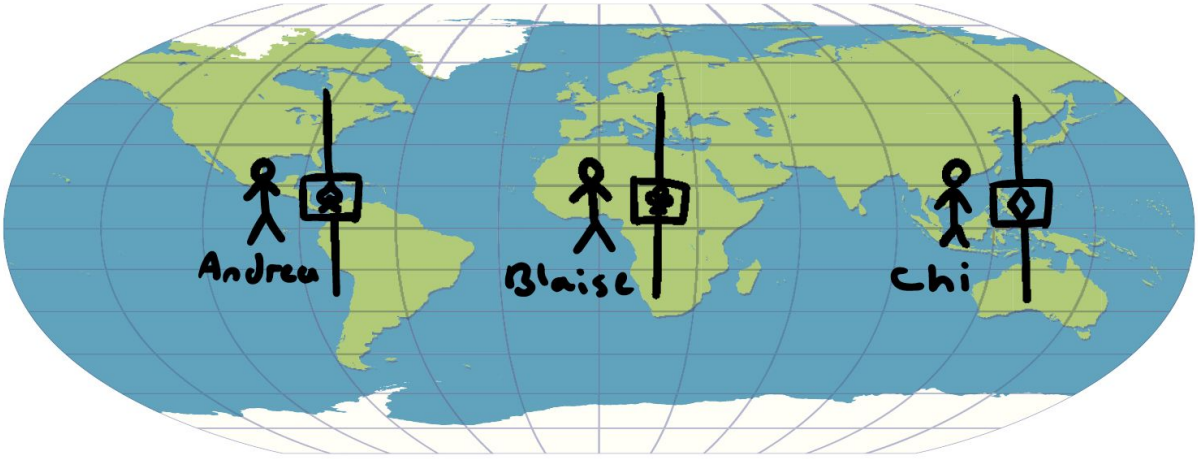
you can always think of box ♢ being applied first and then box ♡ being applied to the output of box ♢. For example, here is a simplified recipe for turning wheat, water and heat into bread:



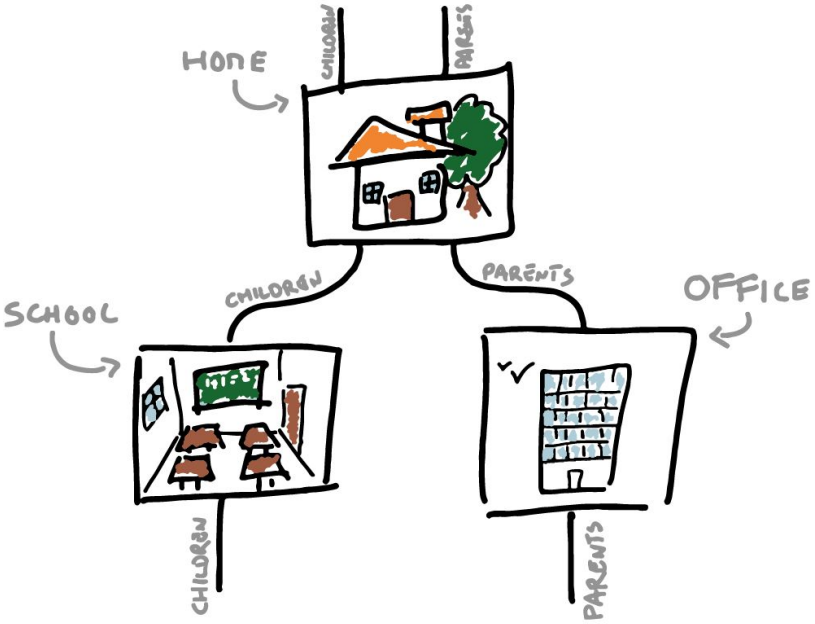
First you need to crush the wheat, then you need to mix the resulting flour with water, and finally you need to bake the resulting dough using the heat from an oven. The order in which you perform these steps is of course important: if you first bake the flour then mix the result with water you'll get some soggy mud instead of fragrant bread. Now let's figure out where space is hiding in our diagrams. Whenever you draw two boxes side-by-side:



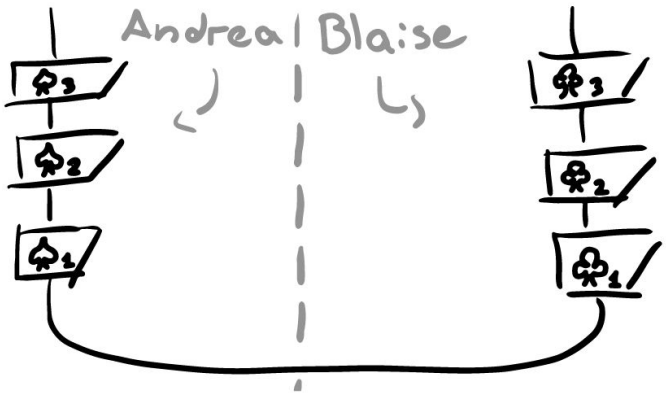
you can always think that the two boxes [♠] and [♣] are being applied to their inputs at the same time but in different places. For example, Andrea is doing [♠] in Mexico, while Blaise is doing [♣] in Burkina Faso and Chi is doing [♢] in Vietnam:



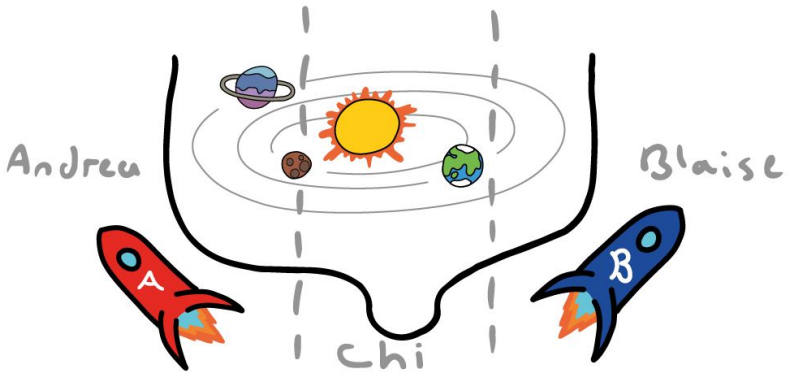
And now, both time and space together in a single diagram:



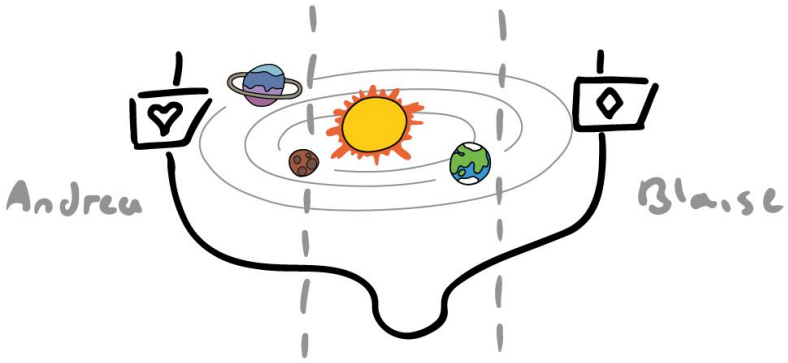
During the day, the children study at school and the parents work in the office. Later in the day, everyone comes home and has dinner together. Just as in the previous example, it can be useful to think of diagrams as involving people doing things in different places, such as bedroom, kitchen or living room. For example, we could think of the following diagram as a girl called Andrea and a boy named Blaise applying some boxes one after the other to two particles:



Just as we did in the diagram above, we can use dashed lines to group together the boxes applied by the same friend. We haven't explained yet how we should think of the cup in terms of time and space. Imagine that, long ago, another friend called Chi sent two particles on rocket ships, one to Andrea and one to Blaise, possibly even at opposite ends of the solar system:



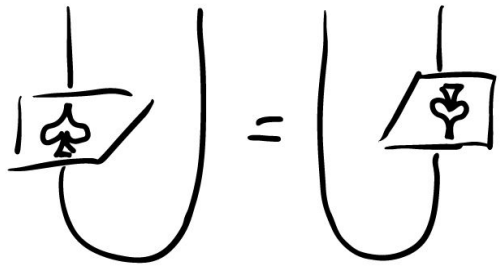
Once they receive the particles, Andrea and Blaise can do whatever they want with them. For example, they could keep the particles as they are, or they could apply boxes to them:



Even though they are now separated by vast stretches of space, the two particles remain tightly connected, as the wires in the diagram clearly show. In fact, they are so tightly connected that they can be used to do some crazy quantum stuff!

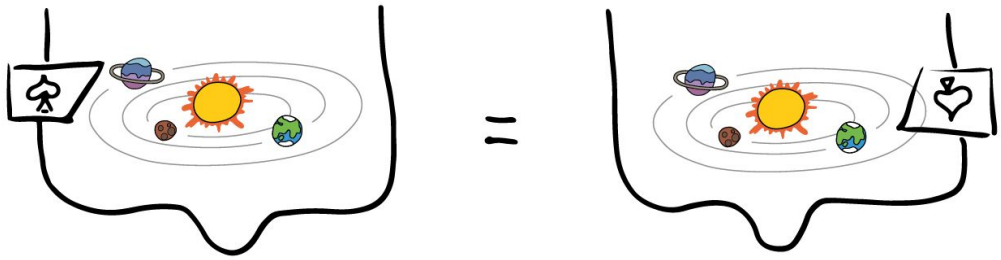
1.3
Quantum teleportation, Part I

The tight connection maintained by two particles in a cup state can be used to teleport particles from Andrea to Blaise. Let's go back to boxes sliding along a cup:



Here we have two particles in a cup state and two diagrams, one where a box is applied to the left particle and one where it is applied to the right particle. The equation tells us that applying a box ♤ to the left particle is the same as applying the box ♡ to the other particle.

For example, assume that Andrea has one of the particles and Blaise the other one. We will, from now on, say that they share a cup state. If Andrea applies the box ♤ to her particles, this produces the same result as Blaise applying the box ♡ to his particle, at the other end of the solar system:



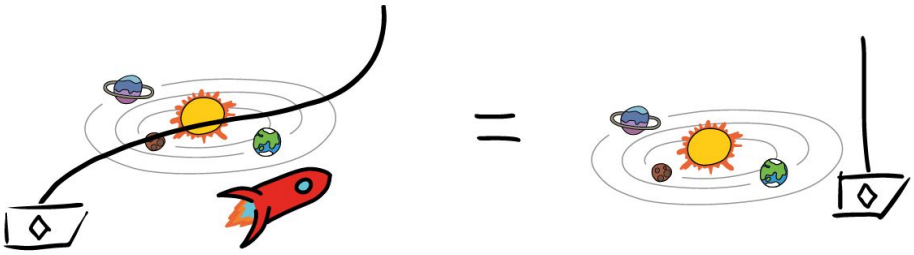
So doing something at one end of a solar system to one particle, or doing something else at the other end of the solar system to the other, has exactly the same effect on these

particles!

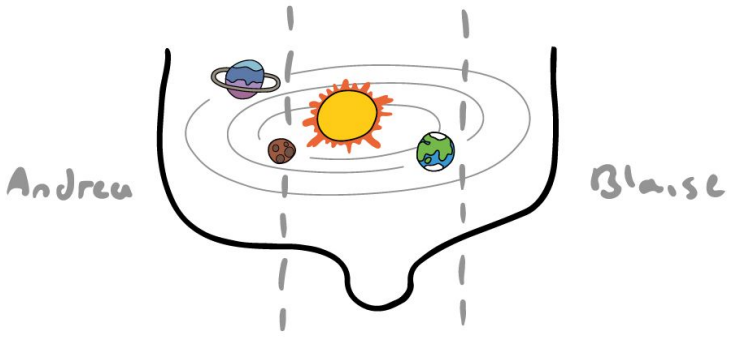
Do you realise how crazy this is? Assume that Andrea and Blaise each have a chocolate cake, the two cakes together being in a cup state. Let the box ♤ be “adding cream”. We now have two possibilities. In one case, Andrea adds cream and Blaise does nothing. In the other case, Andrea does nothing and Blaise does the 180° rotation of “adding cream”, whatever that may be. Either way, Andrea ends up eating the same cake with cream!

Let’s go even crazier!

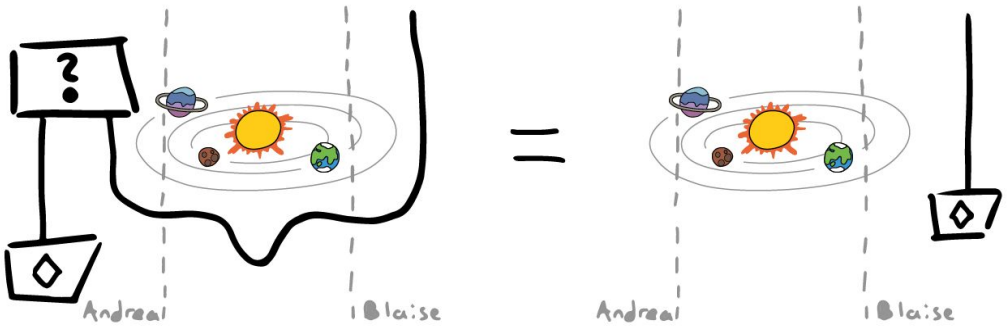
Suppose that Andrea has some quantum particle, in state ◇, which she wants to send to Blaise. One way to do so would be to put the particle in a rocket ship and send it across the solar system:



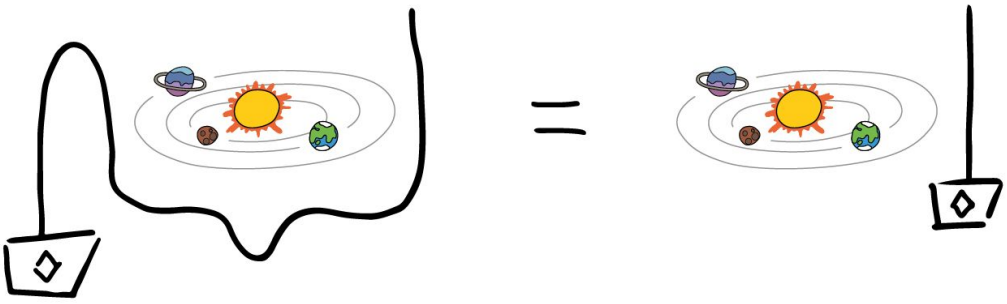
This solution would certainly work, but it would take a lot of fuel and an expensive rocket. Moreover, it is very difficult to move quantum particles around without badly affecting them, so she would rather let the particle sit where it is. Instead, assume that Andrea shares a cup state with Blaise:



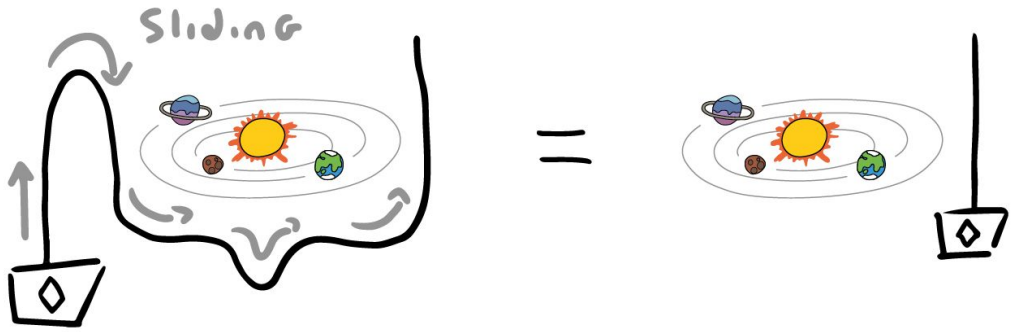
So now Andrea has two particles, the one in state ◇, and one from those in a cup state. Is there something Andrea can do to her two particles:



so that Blaise obtains the state ◇? Of course there is! All she has to do is to apply a cap test to her two particles:



We can now simply slide the box ◇ along the wire:



and as a result Andrea's particle in state ◇ has been teleported all the way across the solar system to Blaise.

Putting things in order: Andrea begins with a quantum particle in state ◇ and shares a cup state with Blaise; then, Andrea applies a cap test to her two particles; as a result, Blaise's particle ends up in state ◇.

And it all happens instantly! But—and this is a very big but!— quantum theory also teaches us that Andrea has no full control over which tests are applied to her particles: the cap test that she needs for teleportation only happens about 1 time out of 4. This uncertainty about tests, which we call the quantum lottery, is at the very heart of quantum theory.

1.4
The quantum lottery

Some quantum boxes can be applied whenever one wants to, while others will only happen when one is a bit lucky. To understand what we mean by this, assume that you flip a coin and look at it. Sometimes it happens that you see heads, sometimes it happens that you see tails, but you have no control over which you'll see:

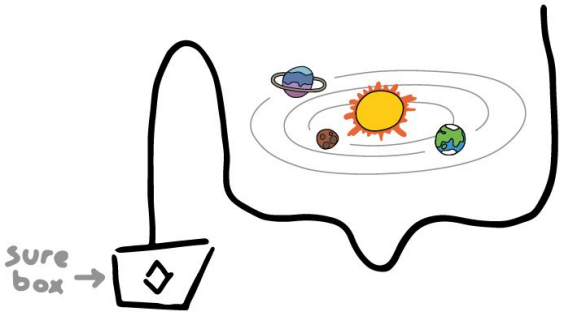


Something similar happens if you roll a die and look at which face came up. You know that you will see one of the six faces, but you have no control over which you'll see:



The boxes above only happen with a bit of luck, you have no control over them. (Unless you cheat, that is, but you can't cheat when it comes to quantum particles). We call boxes such as those, which only happen with some luck, maybe-boxes.

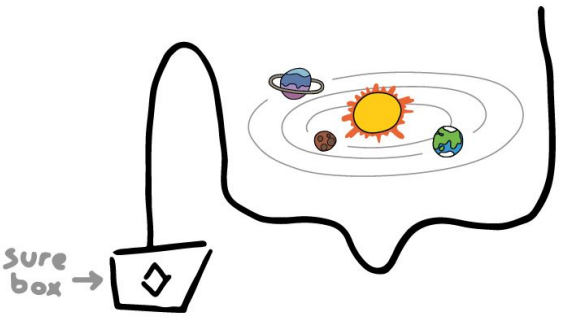
In addition to maybe-boxes, we also have sure-boxes: these are boxes which we can apply with absolute certainty, no luck involved. For example, Alice could choose exactly which particle state ◇ she wants to teleport to Blaise, so that's a sure-box:



On the other hand, Alice cannot make sure that the cap test happens on her quantum particles, so the cap test is a maybe-box.

If we plug many sure-boxes together we get a sure-diagram: if every box in the diagram can be made to happen with certainty, so can the diagram as a whole. If we add maybe-boxes to the mix, however, we often end up with a maybe-diagram: because the maybe-boxes in the diagram require some luck, the whole diagram might not always be made to happen.

Because the cap test is a maybe-box, the teleportation diagram above turns out to be a maybe-diagram:



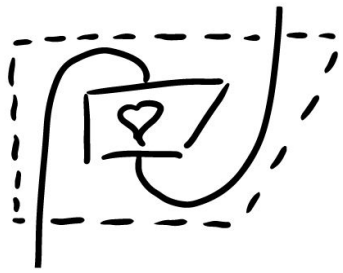
This is the reason why teleportation will turn out not to be instantaneous. And it better not be! Otherwise it would violate Einstein's theory of relativity.

We will talk more about maybe-boxes and how to spot them in the following chapters. For now, keep in mind that almost all tests in quantum theory are maybe-boxes, so that applying tests always involves a little luck, a bit like winning at rock-paper-scissors.

1.5
More maths with pictures

Turning boxes into states and vice versa

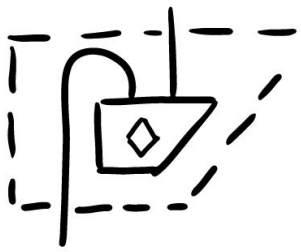
We already saw that we get a new box by bending both the input and the output wire of a box:



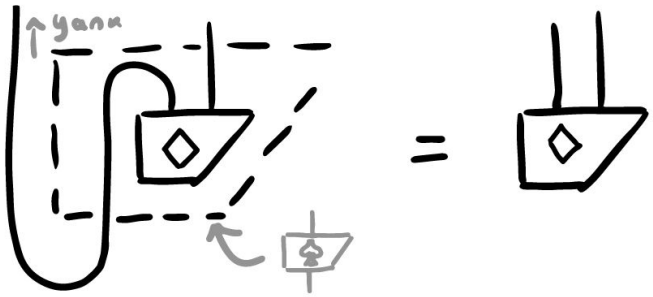
When you see a dotted box like this one here, it indicates that we're looking inside the box as a whole. Another important consequence of being able to bend wires is that every box [♤] can be turned into a two-wire state:



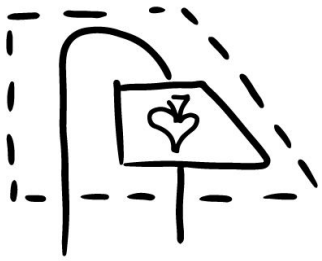
The same statement is true the other way around, i.e. every two-wire state [♢] can be turned into a normal box with input and output wires:



Because of this, we can now think of every two-wire state ♢ as a "normal box" ♤ which has had its input bent up into an output:



In the same way we can think of any two-wire test as actually being a box ♡ which has had its output wire bent down:

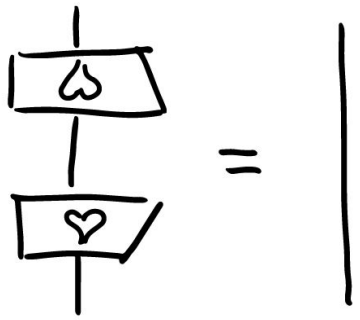


Flip-undoable boxes

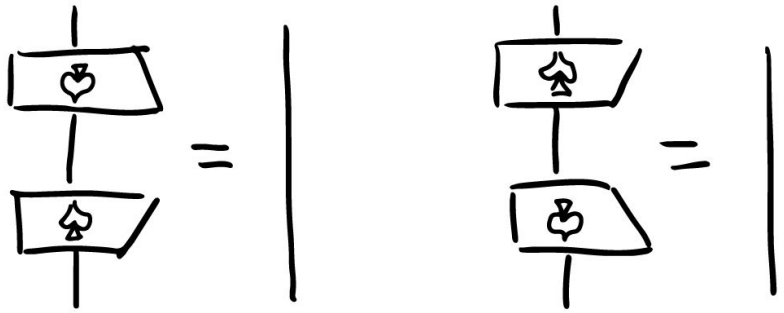
We have seen that boxes can be rotated around 180° by bending outputs into inputs and inputs into outputs. Another thing that one can do with a box is flipping it vertically:



Flip-undoable boxes are boxes ♡ which can be undone by connecting their vertically flipped counterparts ♤ after them:



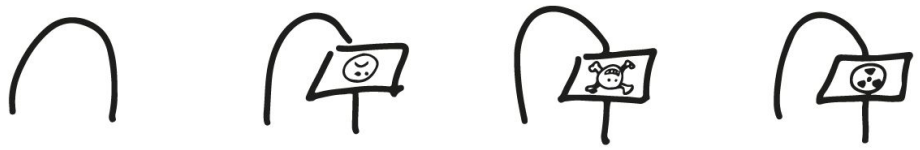
Flip-undoable boxes are special, but there are even more special boxes, which can be also undone before they actually happen:



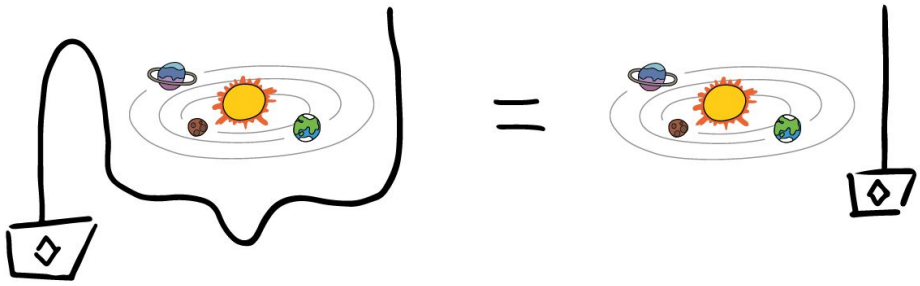
We call such a box [♤] doubly-flip-undoable, as it can be undone by either connecting its vertically flipped counterpart [♡] after it or by connecting [♡] before it. In particular, doubly-flip-undoable boxes are flip-undoable, but the converse is not necessarily true. Flip-undoable and doubly-flip-undoable boxes are extremely common in quantum theory, and we shall now see that they are crucial for quantum teleportation.

1.6
Quantum teleportation, Part II

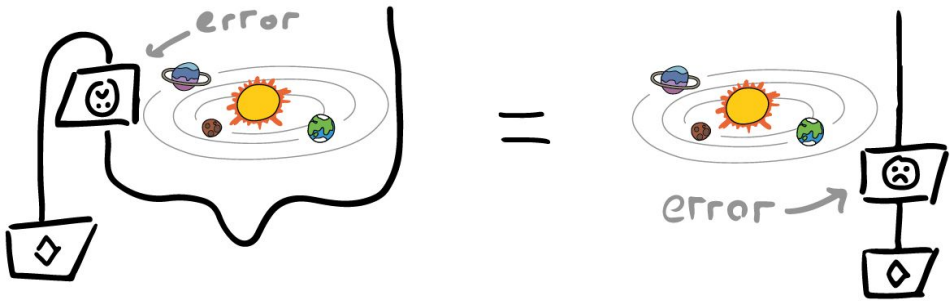
As we've said, the cap test is a maybe-box. As we shall see later, whenever we try to apply it any one out of four tests can happen:



The first test is our desired cap test, while the other three involve some sort of error, represented by the three boxes , and . If the cap test happens, we get the desired teleportation scenario:

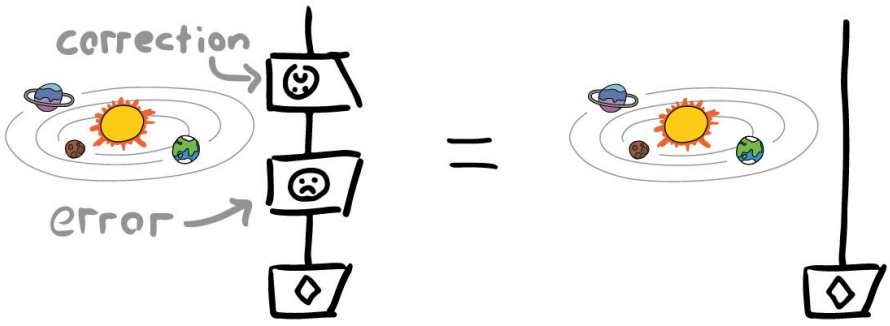


On the other hand, if one of the other three tests happens, we get an error which we can slide to Blaise's end:



Luckily, as we'll show later, the error that was introduced turns out to always be a flip-

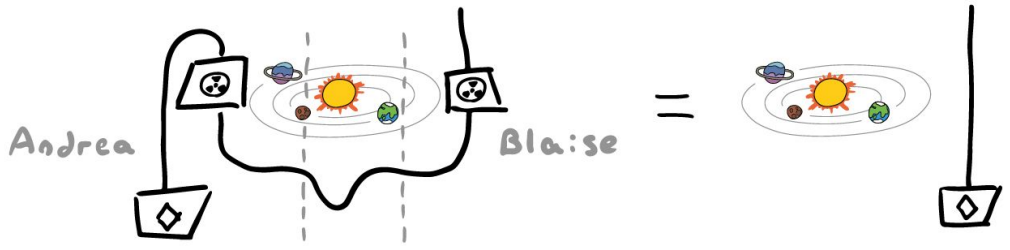
undoable box, which Blaise can correct:



So here is what happens in the other three cases, showing that Andrea's state always ends up correctly with Blaise:

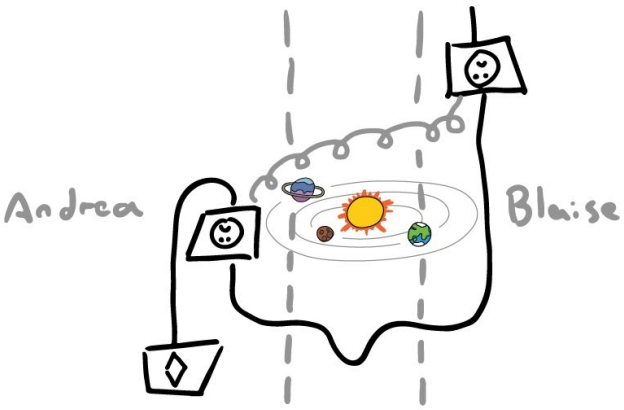






It all simply boils down to a bit of sliding and a bit of yanking.

But wait a minute: if Blaise is to know which flip-undoable box he has to use, Andrea needs to somehow tell him which of the four tests happened. Andrea could do so by making a long-distance phone-call, which we can draw as an old-fashioned telephone wire:



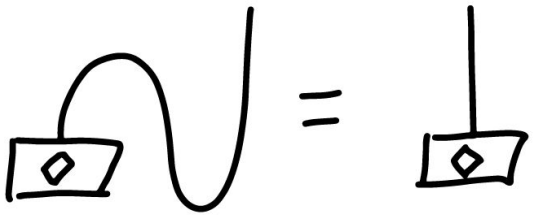
It is important that this telephone wire connects Andrea to Blaise: it takes some amount of time for the messages to travel up the wire, however fast the message might be.

This phone call is the reason why teleportation is not instantaneous after all: her voice takes time to travel through the phone wire, especially if an entire solar system has to be crossed.

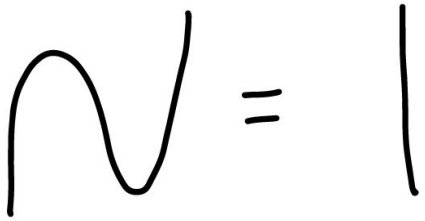
Since we can't really use quantum teleportation to instantaneously send Andrea's state to Blaise, why would we even care about it? As we'll see in the following Chapters, Andrea only has to tell Blaise which of the four tests have happened—that is, a single number 1 to 4—instead of having to send the quantum particle itself. That's why teleportation is so interesting: by making a simple phone call we can send a quantum state!

1.7
Cup fusion

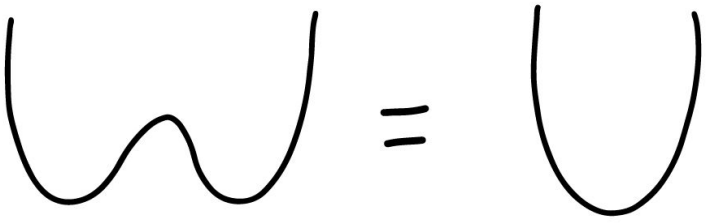
What really got us to teleportation was the equation:



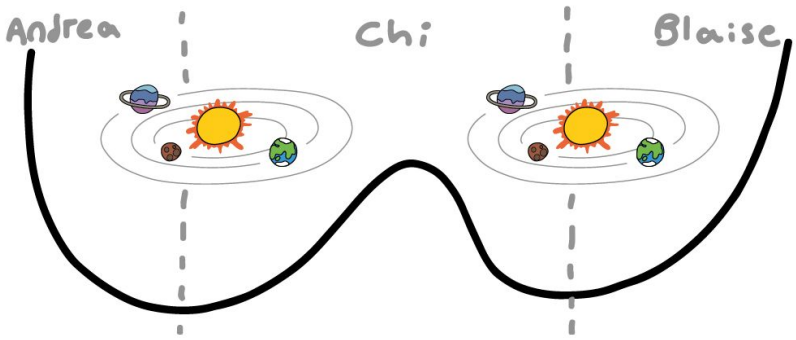
which gets even simpler when we drop the state at both sides:



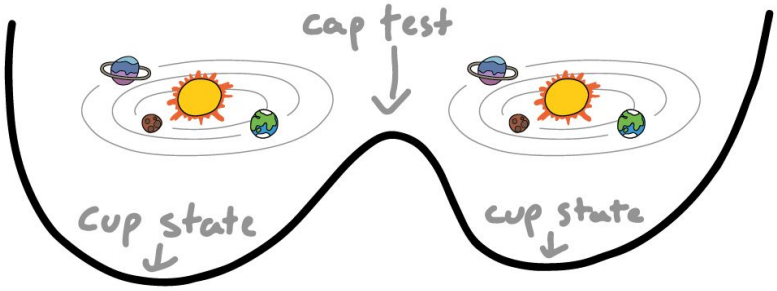
We can always plug ◇ in again whenever we feel like it. Now, a similarly simple equation is:



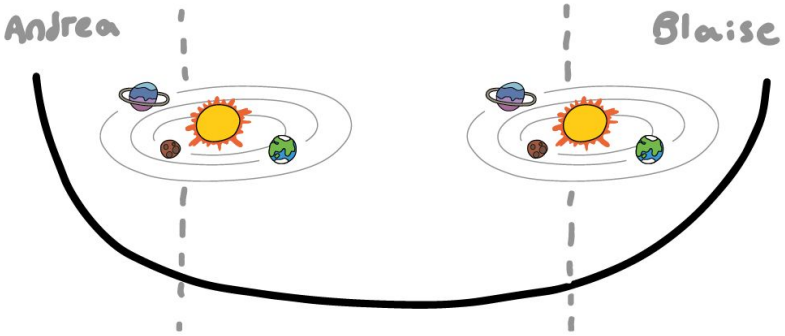
So what does this one do for us? Again, something really cool! As before, let's throw some solar systems into the picture:



We see that Andrea shares a cup state with Chi, that Blaise shares a cup state with Chi, and that Chi then applies a cap test to his two particles:



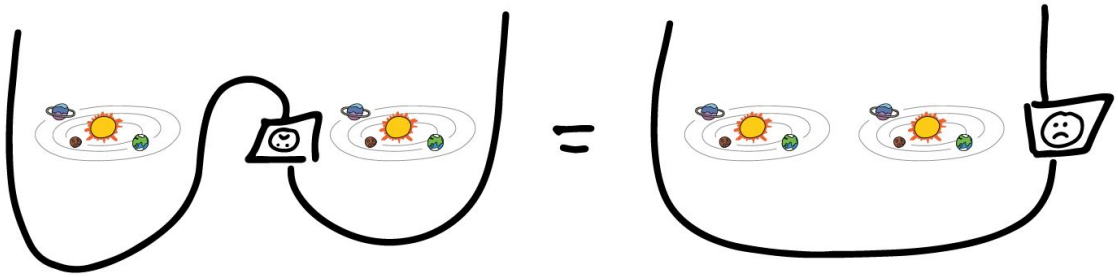
The equation tells us that the result of doing so is as follows:



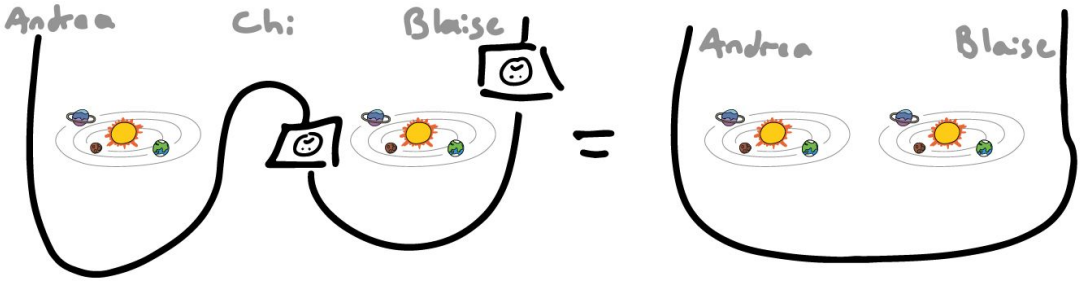
That is, Andrea and Blaise now share a cup state. This is kind of crazy: neither Andrea nor Blaise did anything! Chi didn't send them anything either: he merely performed a test. So two friends can get a shared cup state—which they can use, for example, to perform quantum teleportation—simply by having a third friend perform a cap test.

Remembering that a cap test is a maybe-box, Andrea and Blaise might end up with

another state instead:

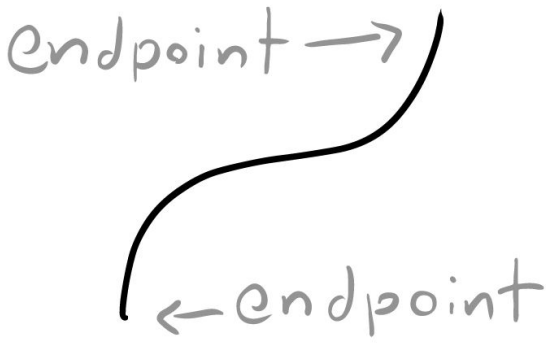


The error is the same from quantum teleportation, and it can be corrected just as easily:

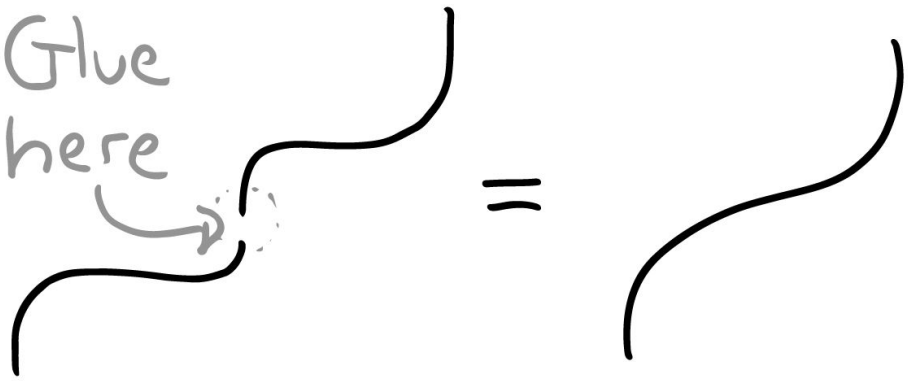


2.
A WORLD OF SPIDERS

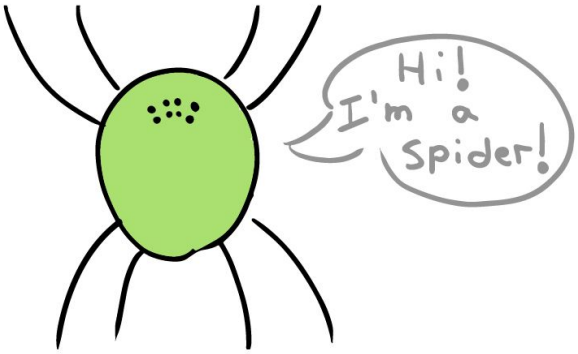
What boxes are available for us to play with, in quantum theory? We have already encountered some: cup states and cap tests, and these arise from wires. One upshot of wires is that they are super simple to understand: they have two ends that are connected. That's pretty much all there is to know about a wire:



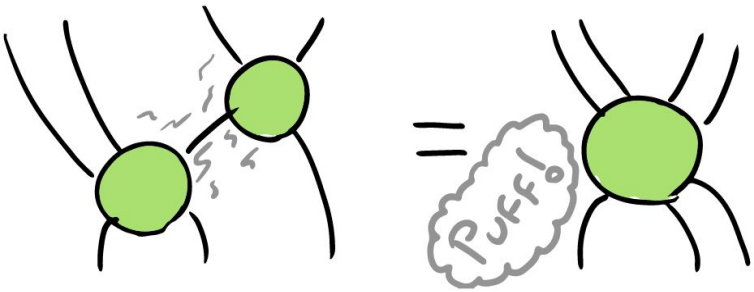
If we connect two wires together we get a single, longer wire:



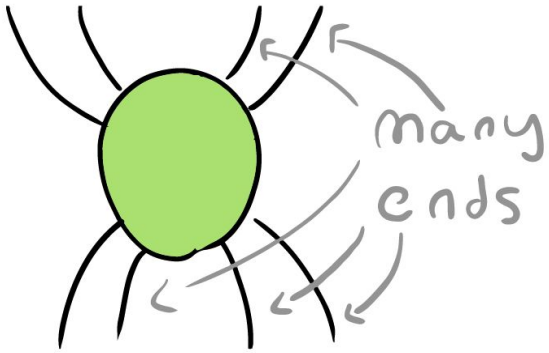
It would be amazing if something almost as simple could be used to build all the quantum boxes. As it turns out, such a thing does exist. It's called a spider:



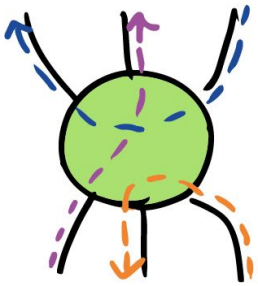
These spiders are almost as simple as wires, in the sense that if we connect two spiders we get again a spider:



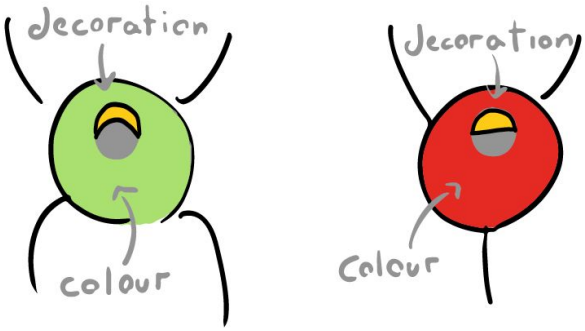
The main difference from wires is that spiders can connect any number of ends:



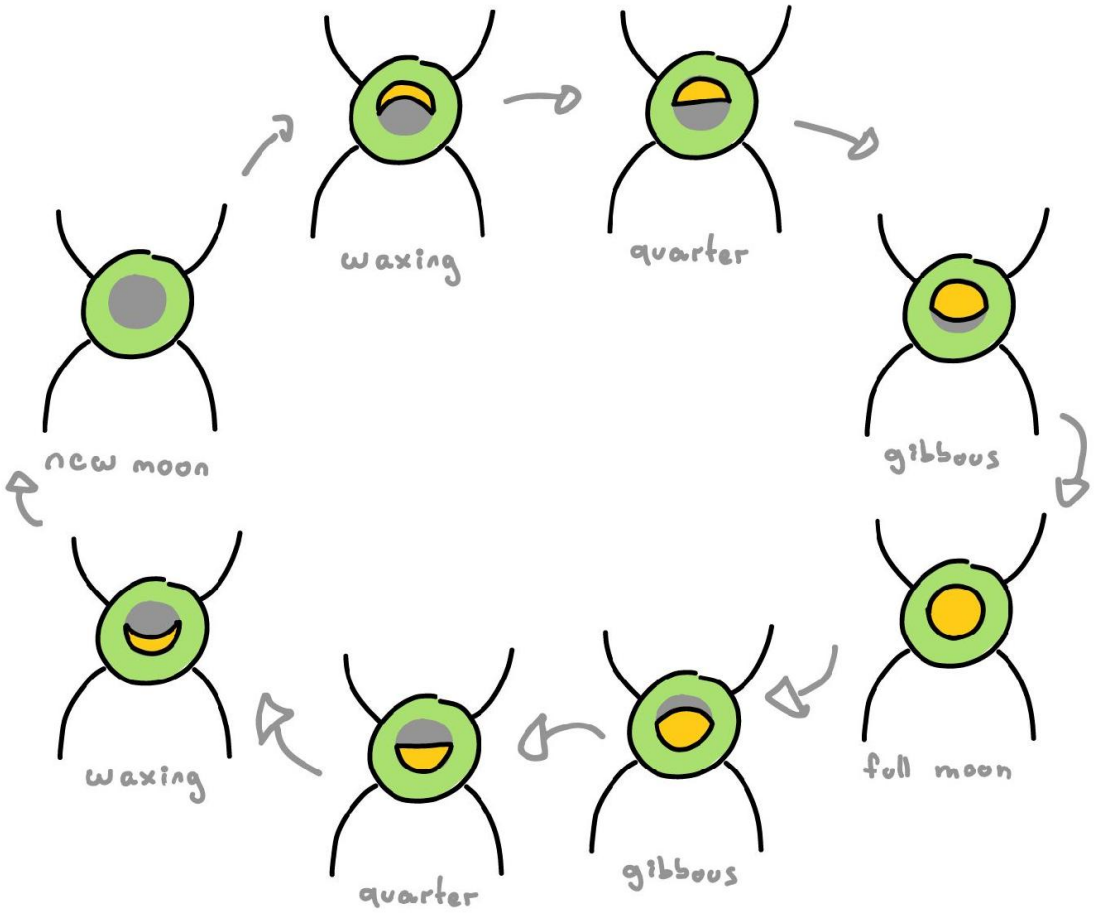
Particles can enter through any end, and travel through the spider to any other end:



Spiders by themselves are still a little bland. To spice them up, we throw in a new splash of colour and let them carry additional markings, making them a lot more interesting than wires:

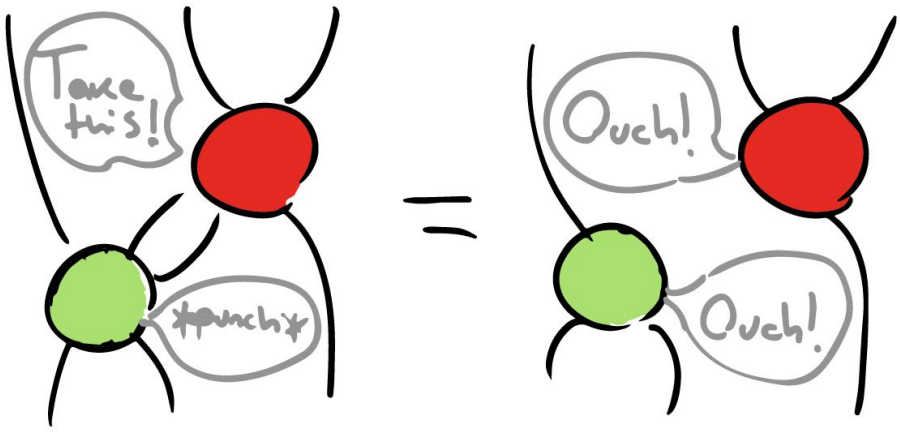


If the markings on the spiders look like a moon, that is because they work, in fact... exactly like the phases of the moon:

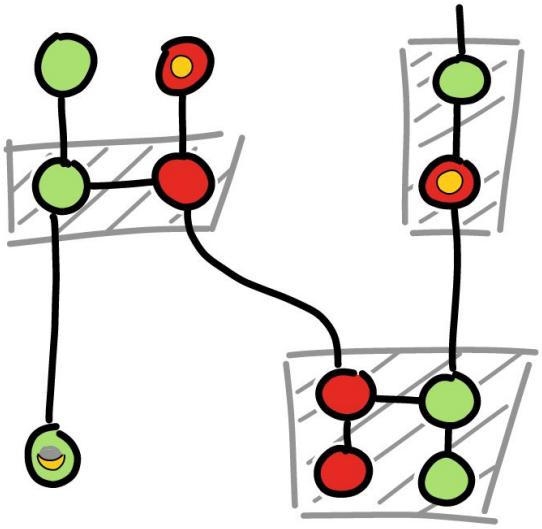


(If the moons look tilted to you, it's because those are the phases of the moon as seen from places around the equator. Spiders like some warmth!)

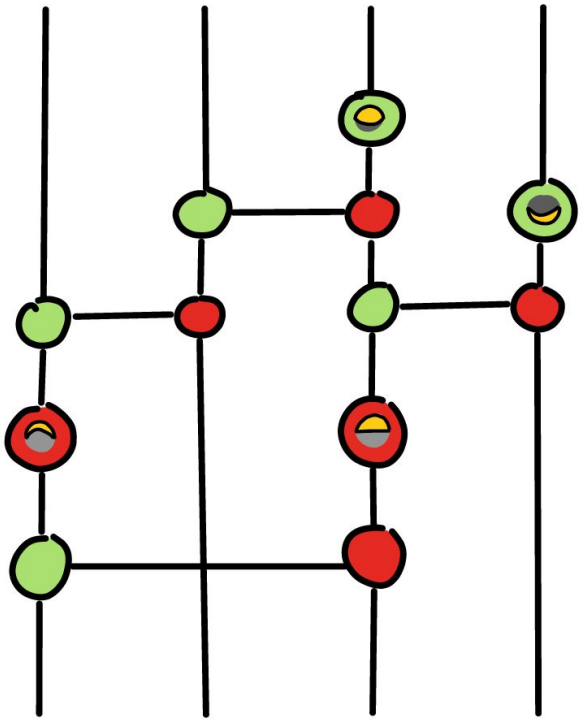
The way to think of the two different colours of spiders is to think of two different ways of looking at the world, like having two pairs of tinted glasses. We will explain more in detail how this works in the following chapters. For now, the important thing to know is that green spiders and red spiders don't like each other very much, and when they wrestle they lose some of their legs:



So, what can we do with our newly acquired, colourful spiders? Everything! That is, all quantum boxes used by physicists are made up of these spiders.



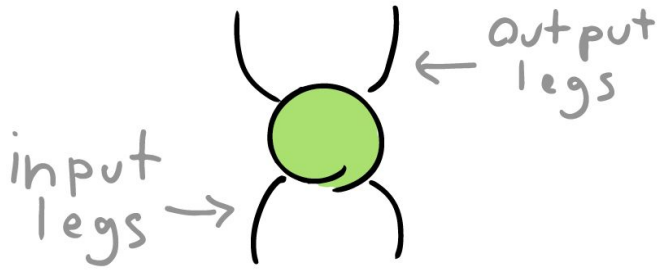
What's more, we will learn in the following chapters that spiders also tell us when two quantum boxes are equal or not, even if they look very different at first. So we truly live in a world made up of spiders. One particularly interesting thing that we can do with spiders is building quantum computers. Quantum computers are the stuff of the future, and we will teach you how they work. Here's an example of how spiders look in a quantum computer:



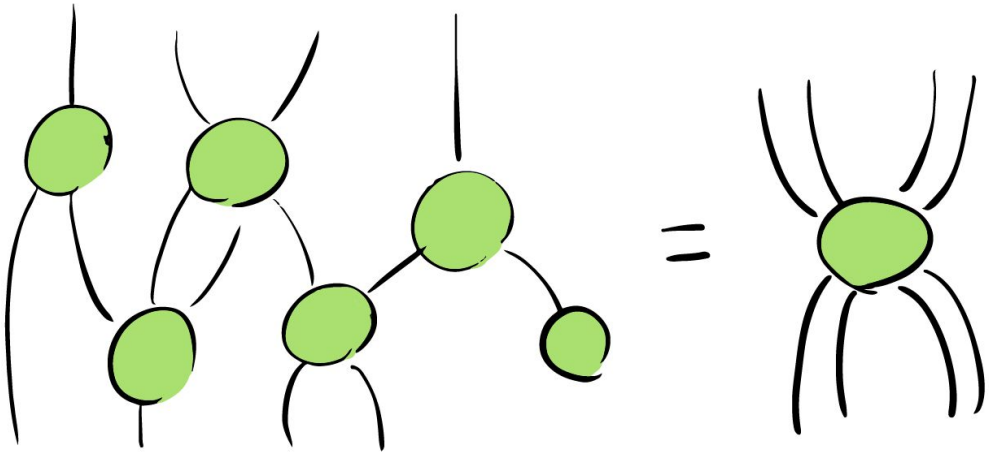
The smallest and simplest quantum particles around, the ones used in quantum computers, are called qubits, which is short for “quantum bits”. For the rest of this chapter, all wires will be qubit wires. Green and red spiders exist for all kinds of quantum particles, not just for qubits. Things get a wee bit more complicated for the other quantum particles, so we won’t discuss them.

2.1
The mathematics of spiders

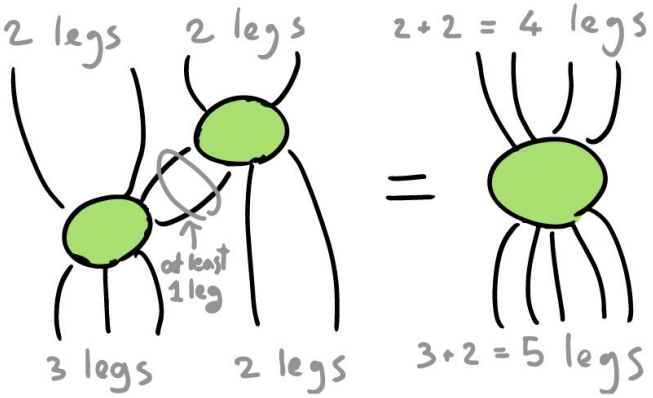
Spiders are special boxes which look like a spider:



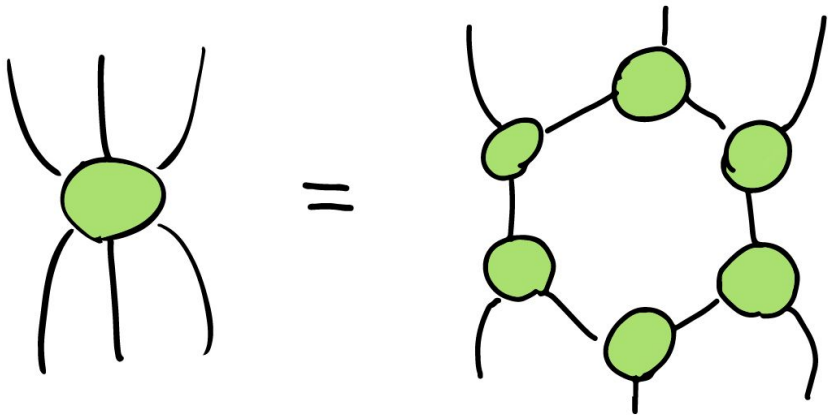
where all legs are the same kind of wire. What makes spiders special is that they fuse together to make new spiders:



In order to fuse, two spiders need to share at least one leg. The legs they share disappear as they fuse, and all the other legs are collected together:



Given a spider, we can also un-fuse it in any way we like:



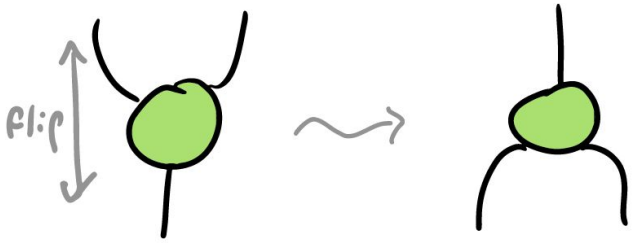
and if you bend the legs of a spider you again get a spider:



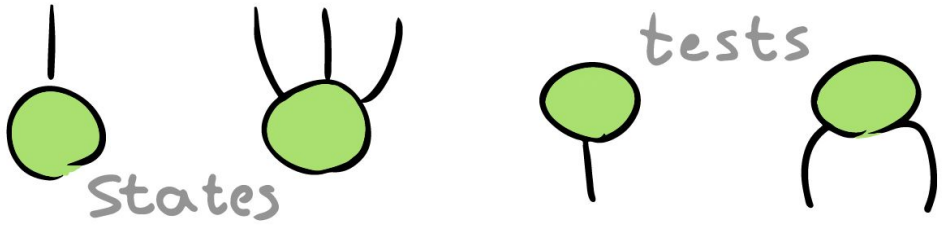
If you rotate it by 180°, you also get a spider:



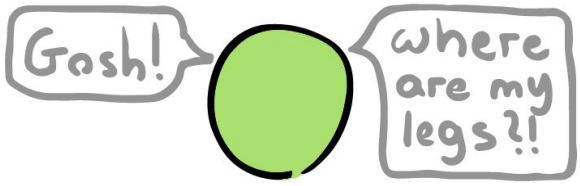
Vertical flipping again gives a spider:



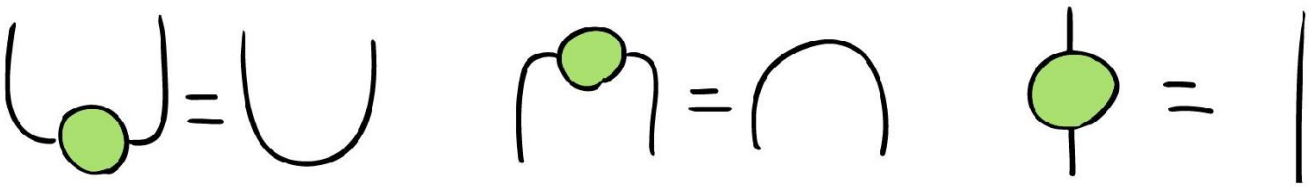
Spiders can also be states or tests:



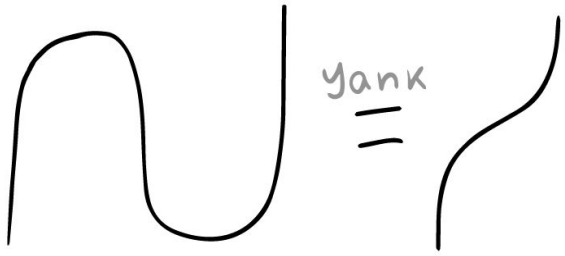
and really unfortunate spiders have no legs at all:



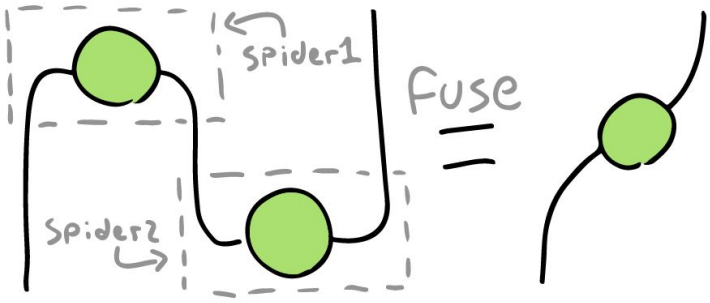
Also cup states, cap test and plain wires are examples of spiders. A spider with two legs up is the cup state, one with two legs down is the cap test, and one with one leg up and one leg down is a plain wire:



In this way, somewhat magically, the yanking of wires that we saw in the previous chapter:

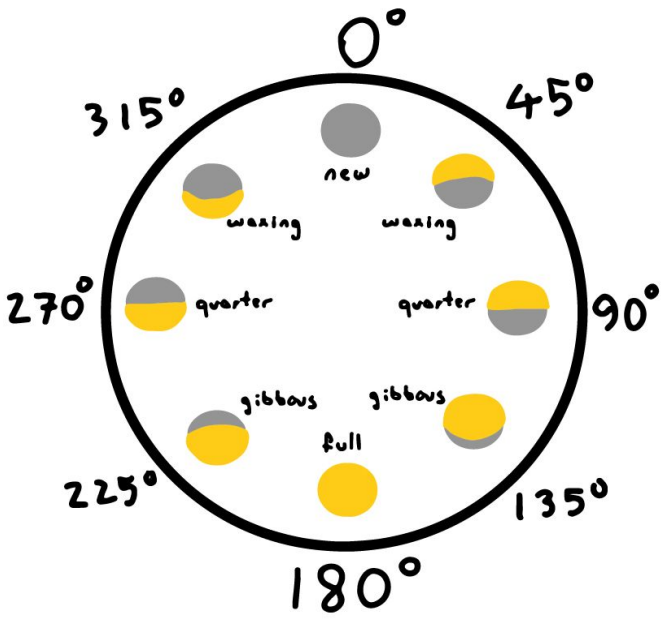


becomes an instance of fusing spiders:

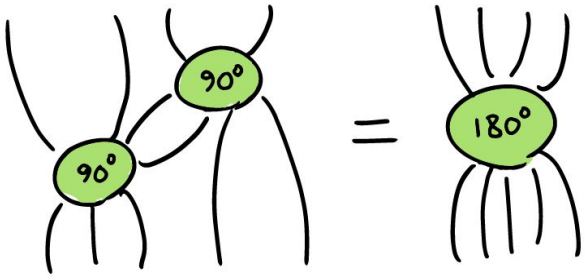


Phases

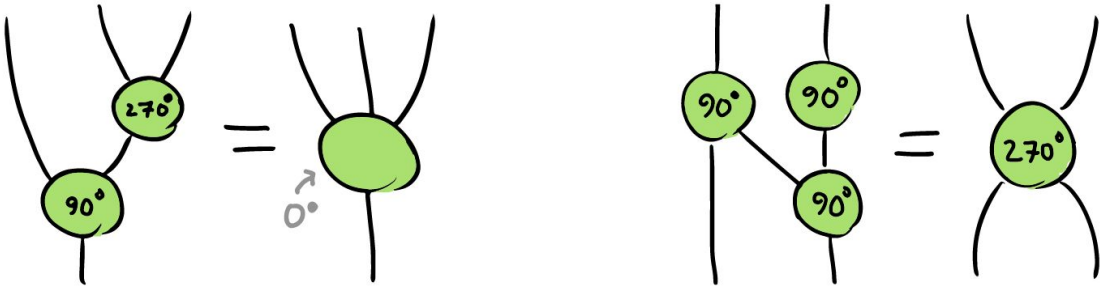
Phases are markings carried by spiders. So far we have represented them by moons, but from now on we will write them as angles, as a matter of convenience:



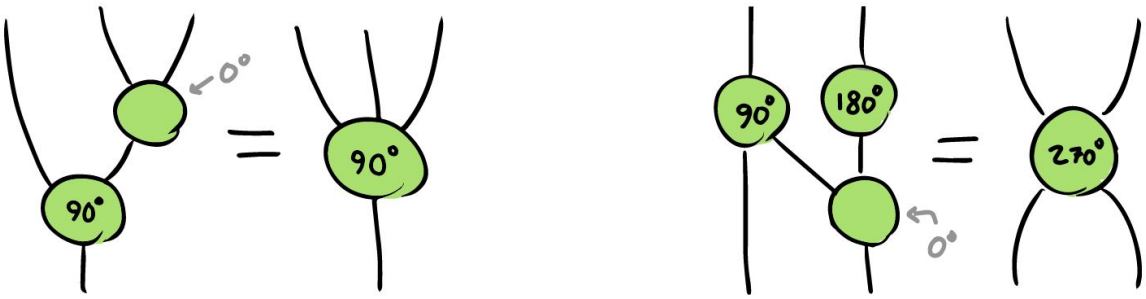
Spiders decorated with phases fuse as ordinary spiders would, and their phases add up as angles. For example, we have:



In another example, we have:



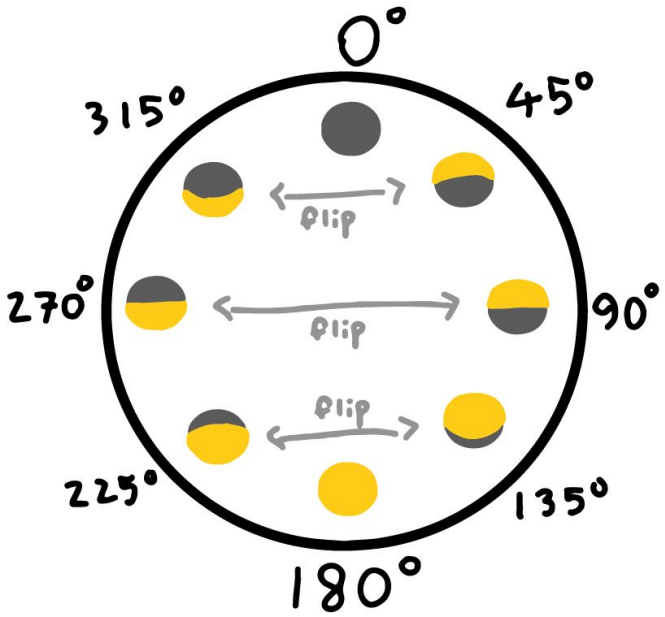
Ordinary spiders correspond to the 0° phase, so we have:



When a spider with phase is flipped vertically, the angle on it changes, and becomes 360° minus the original angle. For example, flipping the spider below changes 90° to 360° - 90° = 270°:

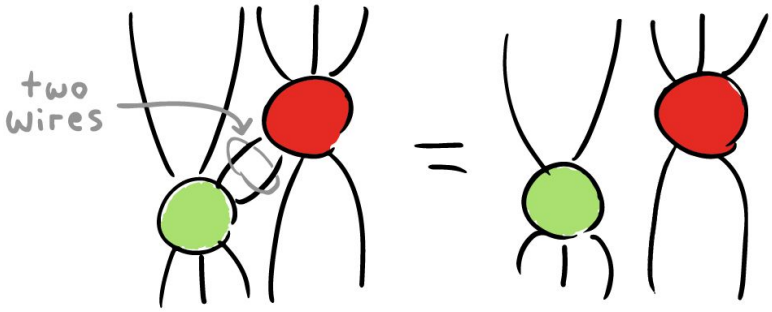


Why? Because what we're actually doing is flipping the phases of the moon upside down:

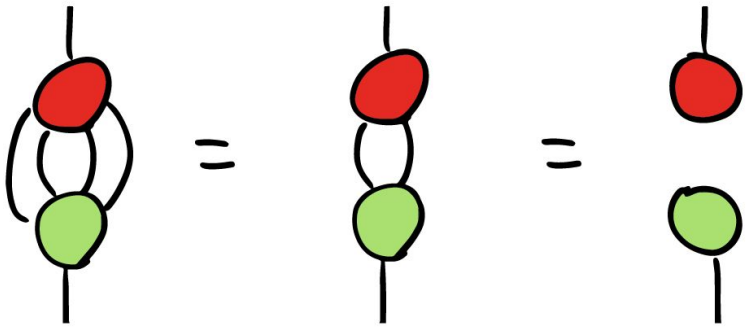


Green spiders and red spiders

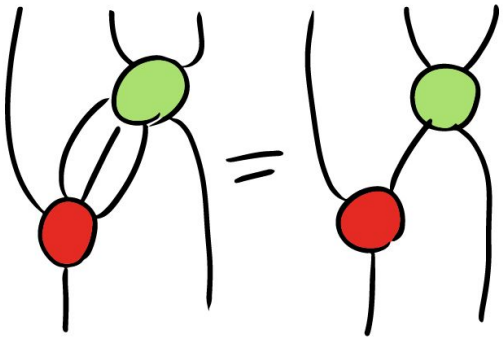
Spiders can be either green or red. When a green spider and a red spider are connected by two legs, the two legs just fall off:



This means that if there are four connecting legs, then these also fall off, first two and then the other two:



If there are three connecting legs, then two fall off and one remains:

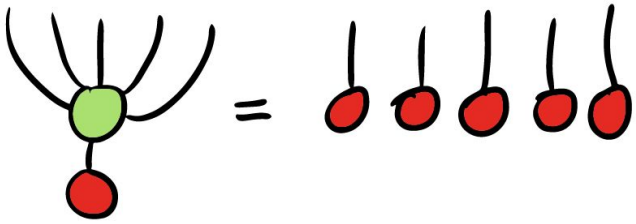


By the same reasoning, we are always left with either zero legs
(if the original number of legs was even) or one leg (if the original number of legs was odd). We call all of this leg-chopping.

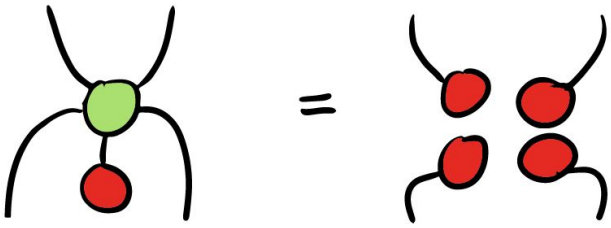
There are some other things that spiders do. A very sneaky thing is that spiders with only one leg can have themselves copied by spiders of the other colour:



Here we used three-legged spiders as copiers, but if the little ones can catch a bigger spider of the other colour, then they can even copy themselves many times at once:

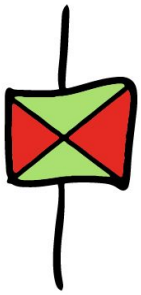


Also, copies don't just have to go up, they can go in all directions:

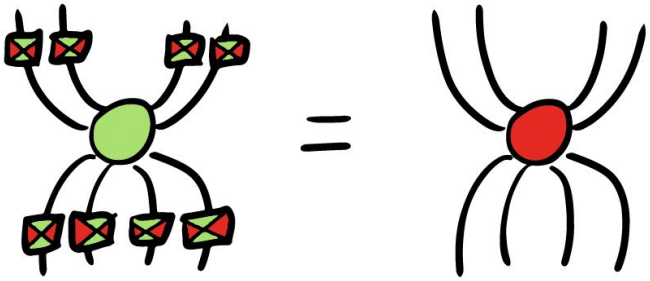


Colour change box

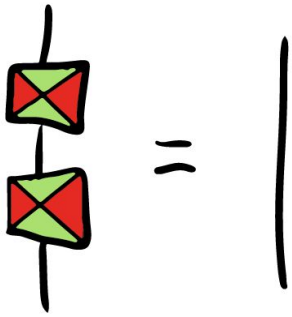
Besides the spiders themselves, there are also special boxes that act on spiders. In particular, there is a special colour change box:



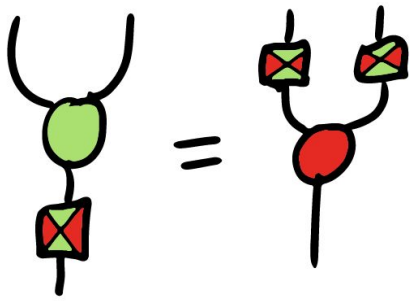
that changes the colour of spiders. It does so when we stick it to all the legs of a spider:



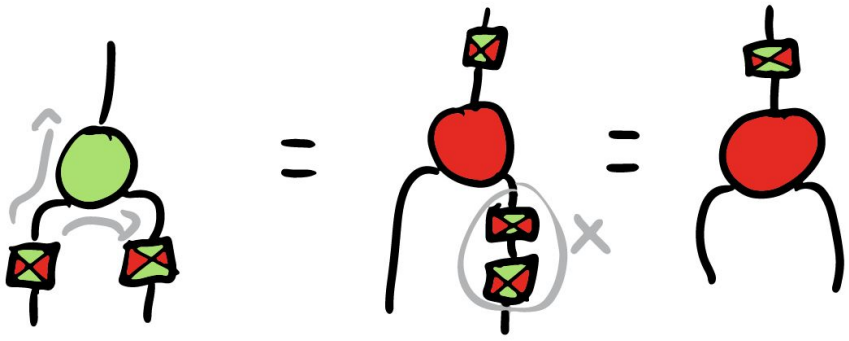
Of course, changing the colour twice leaves colours unchanged, so it is doubly-flip-undoable:



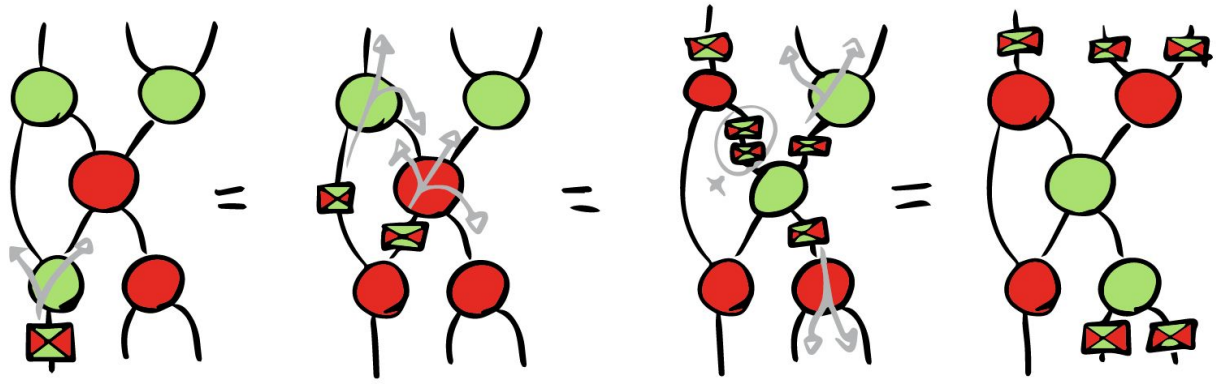
Another manner in which the colour-change box can change the colour of a spider is by pushing it through:



If you want to push two of them on different legs through a spider, just start with one and it all works out as desired:



We can even push one through a whole network of spiders:



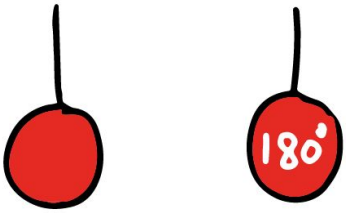
In the end, all the spiders have changed colours, and all the loose legs where there wasn't a colour-change box now have one.

2.2
A spider bit me!

Ordinary computers, mobile phones, and other electronic devices work with bits, which can be either 0 or 1. For example, if you type “hi” and hit send, the computer sees this as “0110100001101001”.

Quantum computers instead work with qubits, which are a lot more powerful than bits. Sometimes, however, we also want to use bits in our quantum computers, for example because we want to connect them to ordinary computers. In order to make qubits be like bits, we only allow certain qubit states and certain qubit boxes. These states and boxes make it look as if we were working with bits on an ordinary computer.

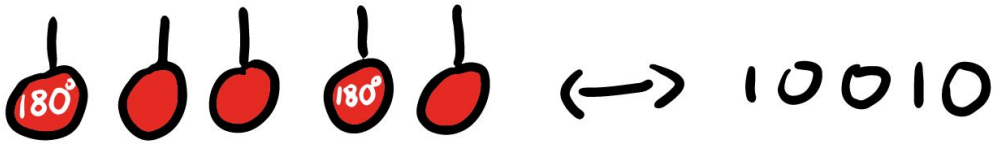
For single qubits we only allow two “bit” states:



We can think of the 0° red spider state on the left as a 0 bit and of the 180° red spider state on the right as a 1 bit:



So there’s our bit. Multiple qubits can each be in one of two states, so we obtain a string of 0’s and 1’s:



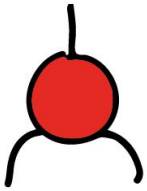
just like the ones that occupy ordinary computers.

And now some boxes. We already know that the 0° red spider states are copied by green spiders: as it turns out, the same is true for 180° red spider states.

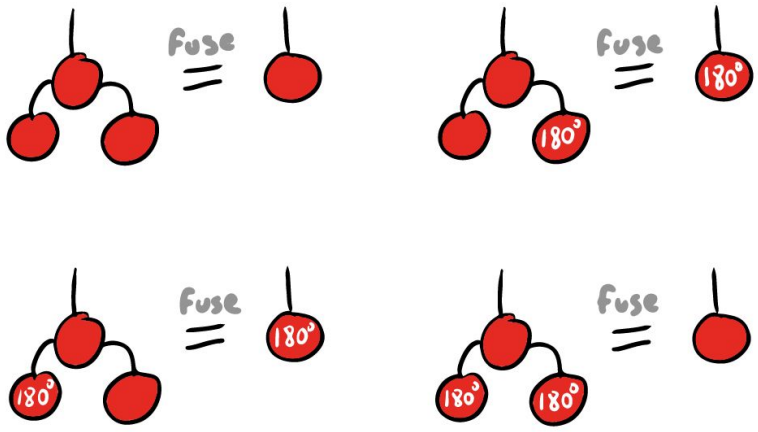


Since copying is something that ordinary computers can do very well, this green spider is a box that makes perfect sense for bits.

What about red spiders? Obviously they don't copy the 0° and 180° red spider states, since they fuse with them instead. But when we apply the following red spider:



to two bit states we get the following four cases:



It's similar to addition between 0's and 1's, but a little funny: there is no 2 available in bits, so we set 1 + 1 = 0 instead. We call this box for bits the ADD box, and write it with a funny plus sign:

$$0 \oplus 0 = 0$$
$$0 \oplus 1 = 1$$
$$1 \oplus 0 = 1$$
$$1 \oplus 1 = 0$$

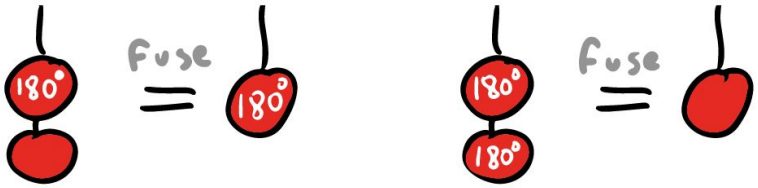
Another important box for bits is the NOT box, which flips the value of a bit from 0 to 1 and vice versa:

$$\text{NOT } 0 = 1 \qquad \text{NOT } 1 = 0$$

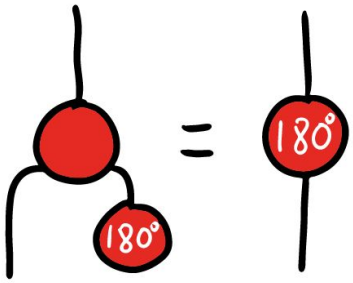
This is exactly what the following red spider:



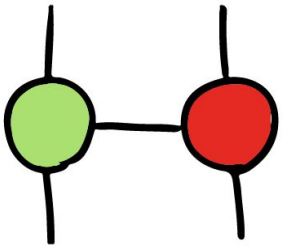
does to the 0° and the 180° red spider states:



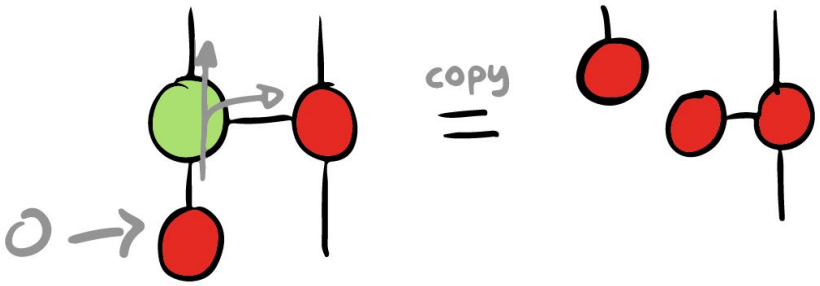
Spiders are also really useful for figuring out how boxes for bits like ADD and NOT relate to each other. For example, we learn that sticking a 1 bit into one leg of an ADD box turns it into a NOT box:



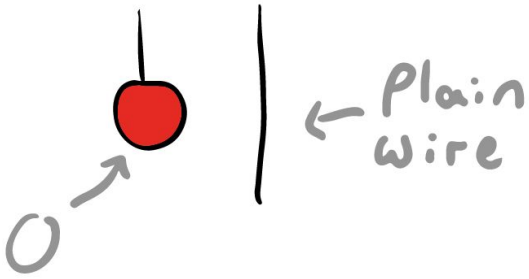
Another important box for bits is the CNOT, which has two inputs and two outputs and is made up of a three-legged green spider shaking a hand with a three-legged red spider:



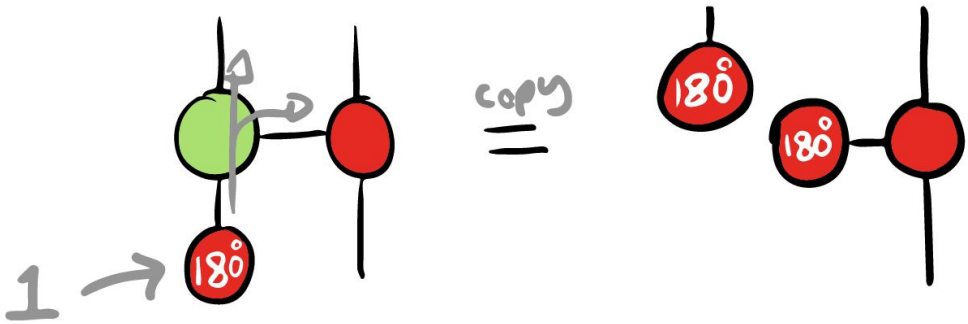
CNOT stands for Controlled NOT, since the left bit is used to "control" whether the right bit is left untouched or whether a NOT box is applied to it. Indeed, if we stick a 0 in the left input, it is first copied by the green spider:



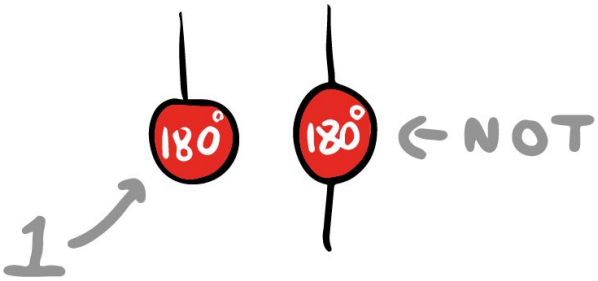
and after fusing the red spiders we get:



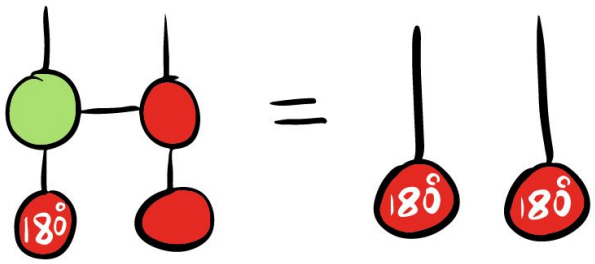
We therefore get the 0 back at the 1st output, and on the right we have a plain wire that would do nothing to the 2nd input. If we instead stick a 1 in the left input, it is also copied:



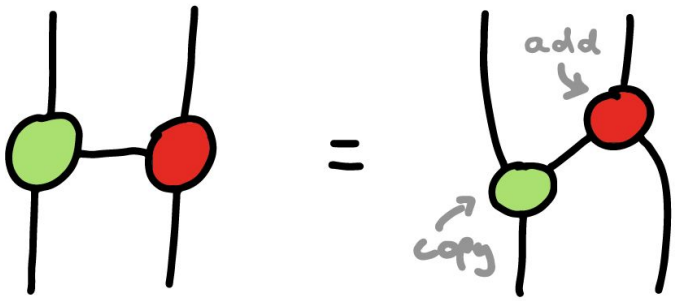
and after fusing the red spiders we obtain:



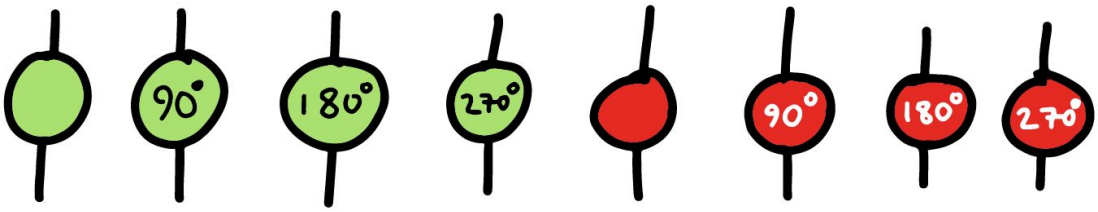
the 1 back at the 1st output, and on the right we now have a NOT box. For example, if we stick a 1 at the 1st input and a 0 at the 2nd, that 2nd bit now becomes 1:



When used in quantum computers, boxes such as the one above are called gates. One more thing that we now know, simply by looking at the pictures, is that we can make a CNOT from a copy and an ADD box as follows:



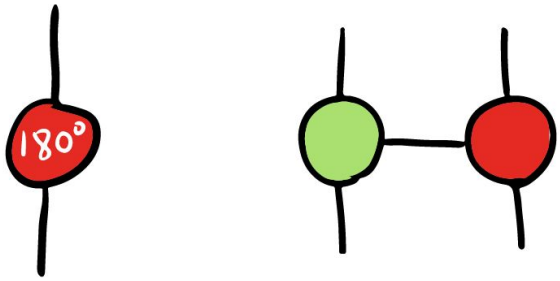
There are many more boxes for qubits, known as quantum gates, which can be used to build quantum computers. There are also many more states for qubits:



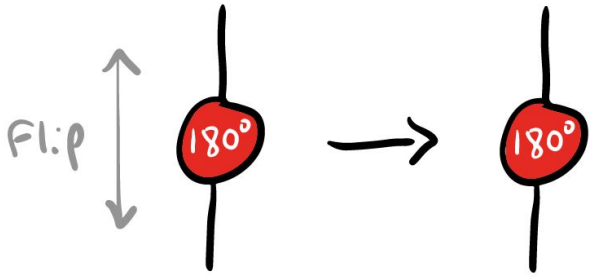
You will not find quantum computers in the shops yet: they are currently being invented! But we will now see a couple of ways in which people in science labs build them from spiders.

2.3
Quantum gate computing, Part I

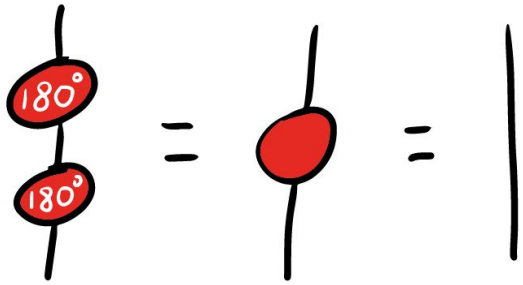
The first examples of qubit gates are some of the bit gates themselves:



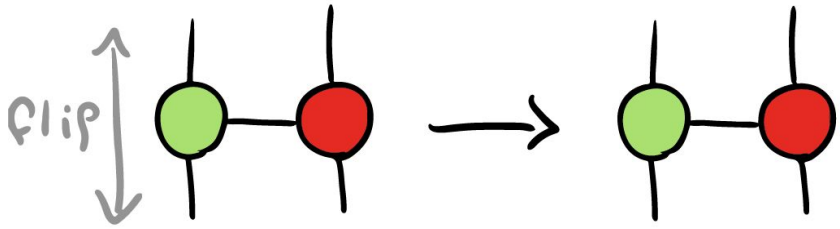
Some, and not all, because quantum gates need to be doubly-flip-undoable. Flipping the NOT gate vertically gives the NOT gate again:



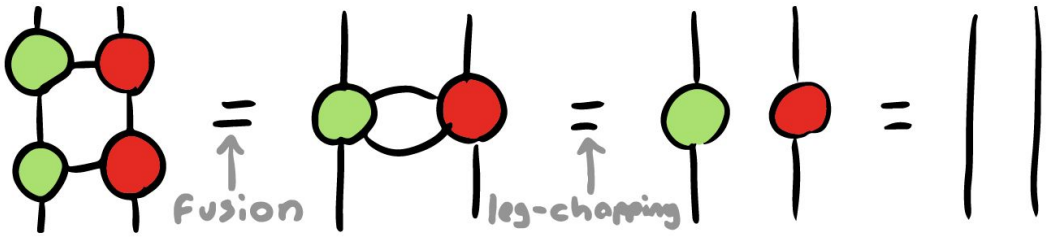
If we connect these we can fuse the spiders:



So the NOT gate is doubly-flip-undoable. Flipping the CNOT gate vertically also gives the CNOT gate again:

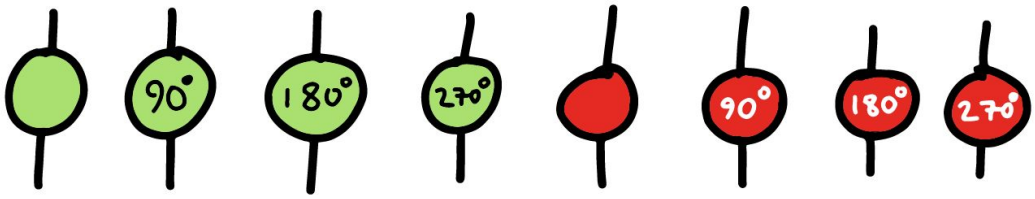


If we connect these we can again fuse the spiders and the middle legs fall off:

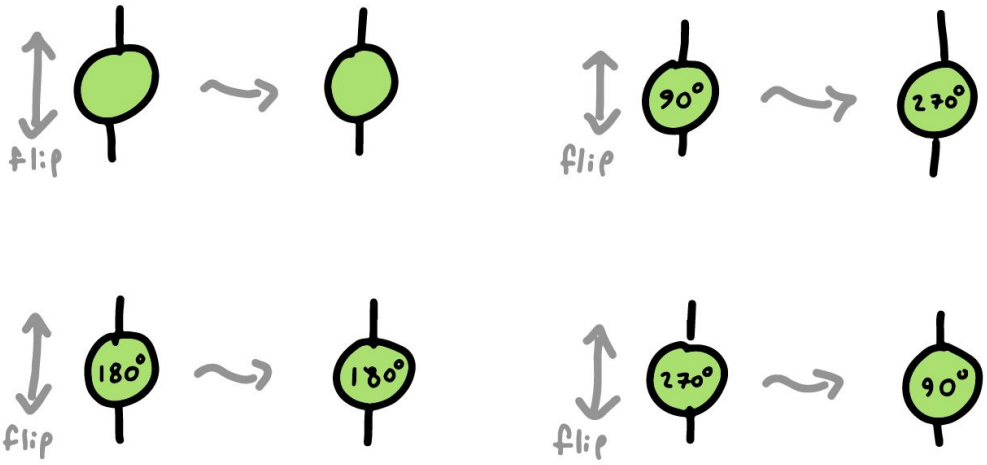


So the CNOT gate is also doubly-flip-undoable.

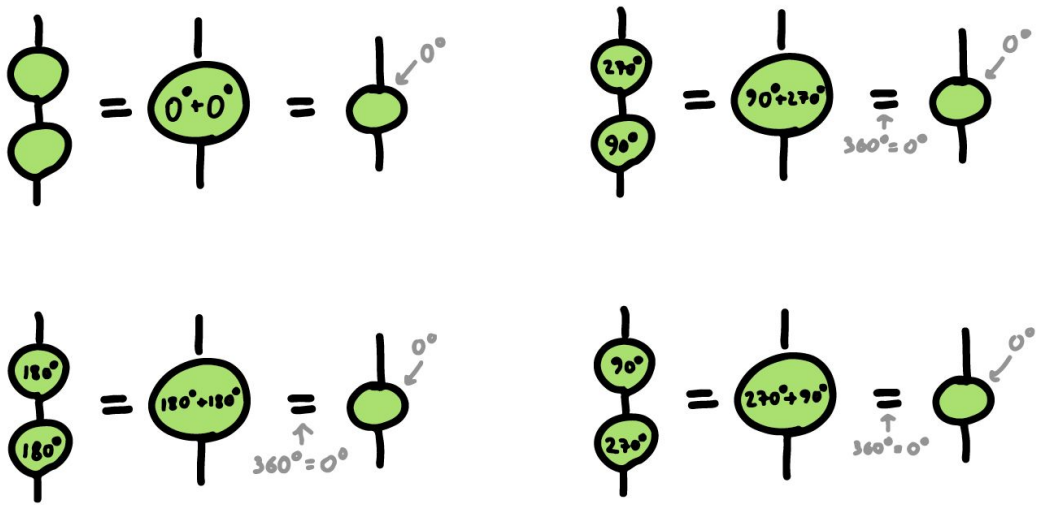
What makes a quantum computer much more powerful than an ordinary computer is that there are many more quantum gates than the NOT and CNOT. In particular, not only is the 180° red two-legged spider a qubit gate, but all spiders with one input leg and one output leg of any colour are as well:



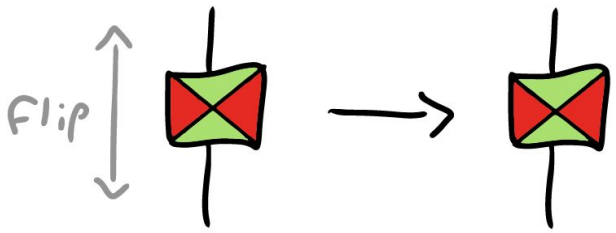
These are called phase gates. Flipping a spider with phase changes the angle of the spider (remember? that's 360° minus the original angle, because we're flipping the moons upside-down):



Above we have shown this for the green phase gates, but the same holds for the red phase gates. Phase gates are, of course, doubly-flip-undoable:

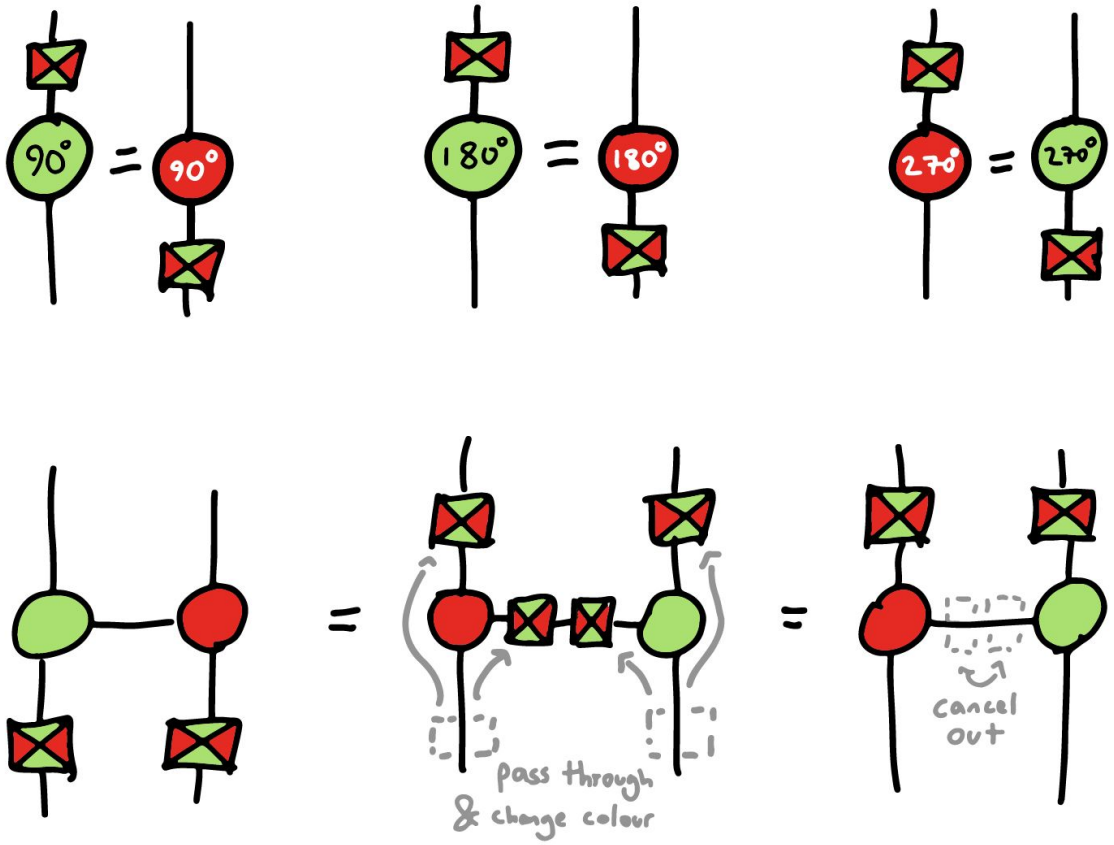


Flipping the colour change box gives the colour change box again:



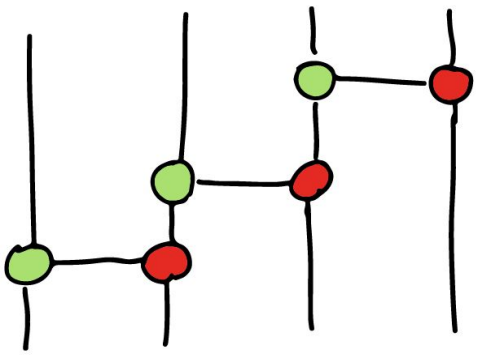
We already saw that the colour change box undoes itself, so it is also doubly-flip-undoable and hence a quantum gate. Almost all quantum gates that we saw are made of

spiders, and colour change boxes are a close friend of spiders. For example, we can push them through a phase gate or a CNOT:



In fact, colour change boxes are also (secretly!) made of spiders, but they are so useful that they get their own picture.

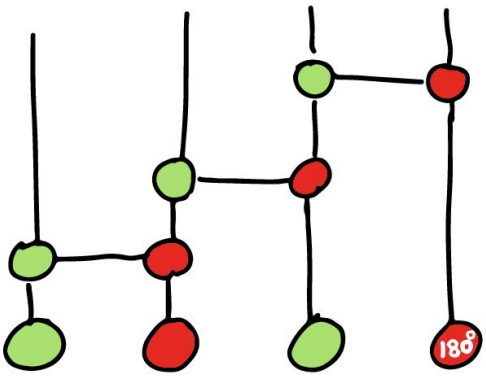
We can stick all these quantum gates together to form a quantum circuit, a diagram made of quantum gates:



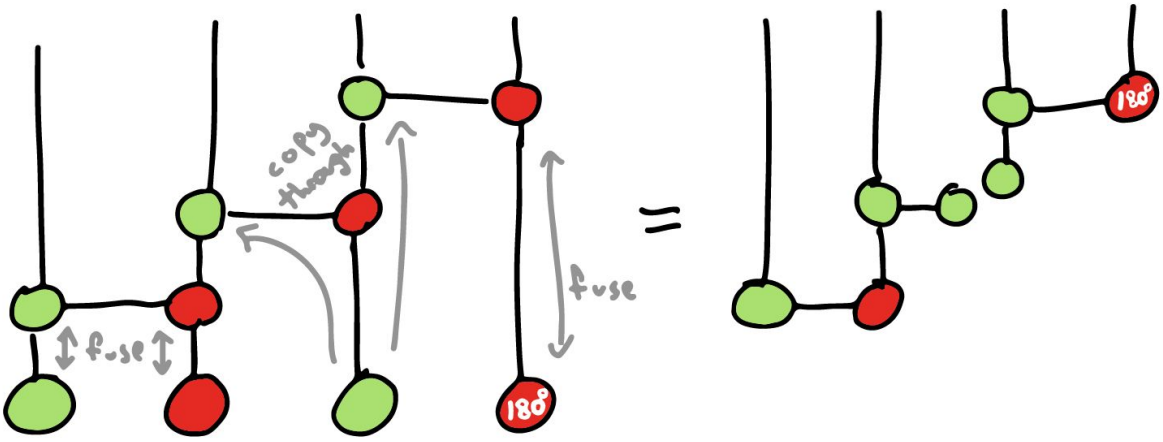
Such a circuit can then be applied to the states of qubits stored in the quantum computer. For example, if we apply it to the following states:



we get the following new states:



We can use spiders to figure out what these new states are after the circuit is applied. We see that the states of the 1st, 2nd and 4th qubits fuse with the spiders in the circuit, while the state of the 3rd qubit gets copied through:



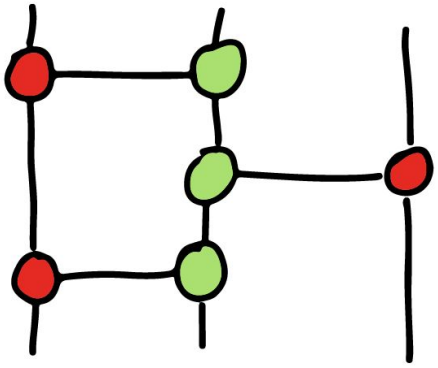
After some more fusing we get the following:



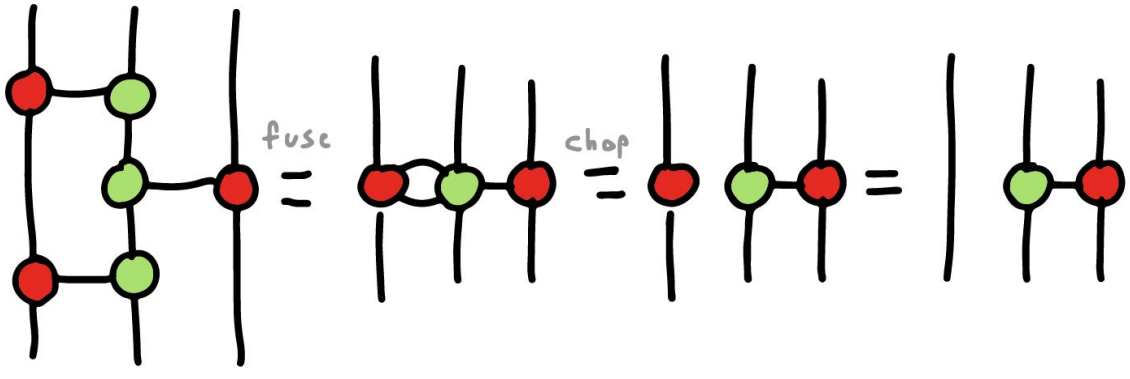
and removing all undecorated two-legged spiders yields a normal cup state and a cup state formed by a 180° red spider:



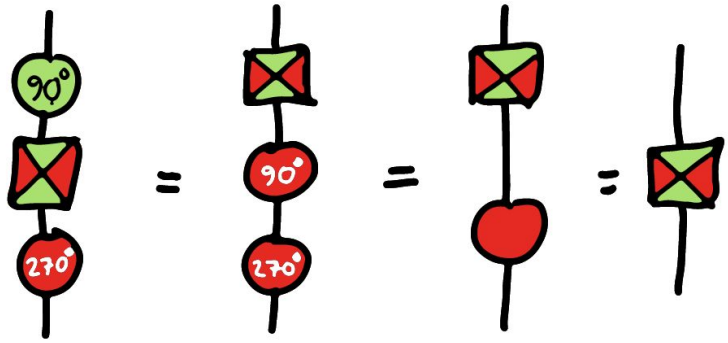
Quantum gates are quite expensive, so it is important to simplify our circuits as much as possible before we actually execute them on a quantum computer. For example, can the following circuit be simplified?



Of course it can! Spiders fuse, legs fall off and we get a much simpler circuit which does the exact same thing:



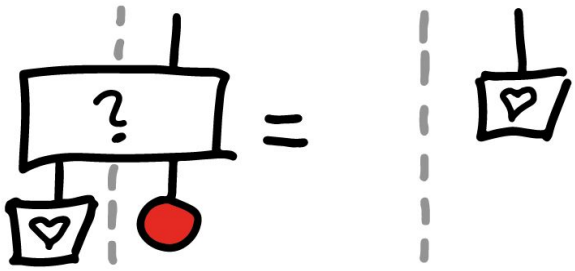
Here is another example, where three phase gates in a row are simplified to a single colour-changing box:



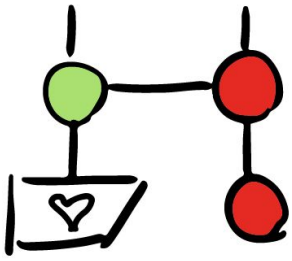
2.4
Quantum test computing

Suppose we want to move a quantum state  from one qubit to another in a quantum computer, just as Andrea and Blaise did in teleportation. Unfortunately, we have no cup state available, so teleportation won't work for us. However, if we can use any quantum gate we wish and perform any test, then it turns out that we can still make this happen!

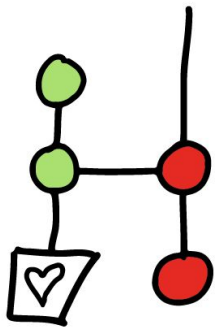
Instead of a cup, we now use a red spider state, so this is what we intend to achieve:



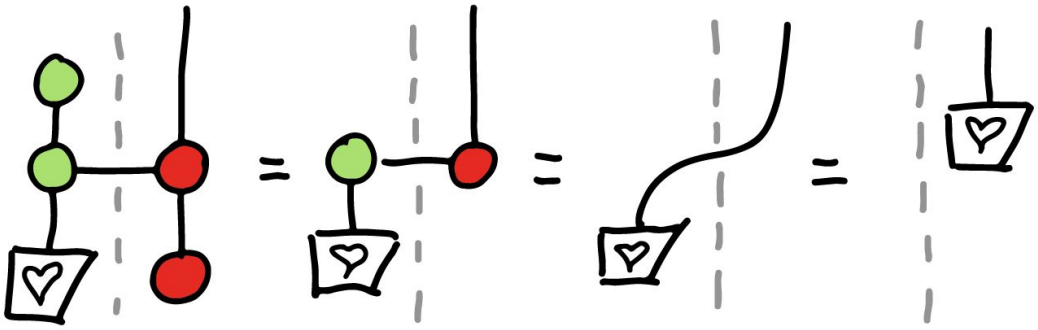
How could we do this? First we do a CNOT gate:



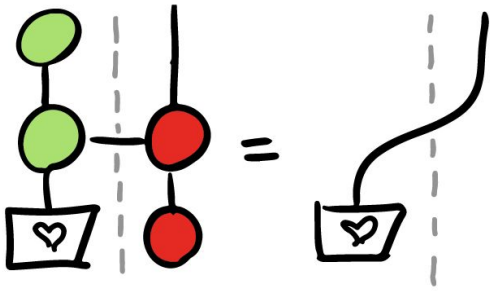
Then we apply a green 0°-test to the 1st qubit:



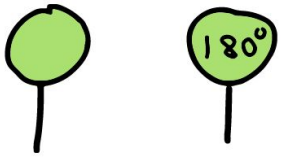
And then...we're done! Indeed, this is what we get:



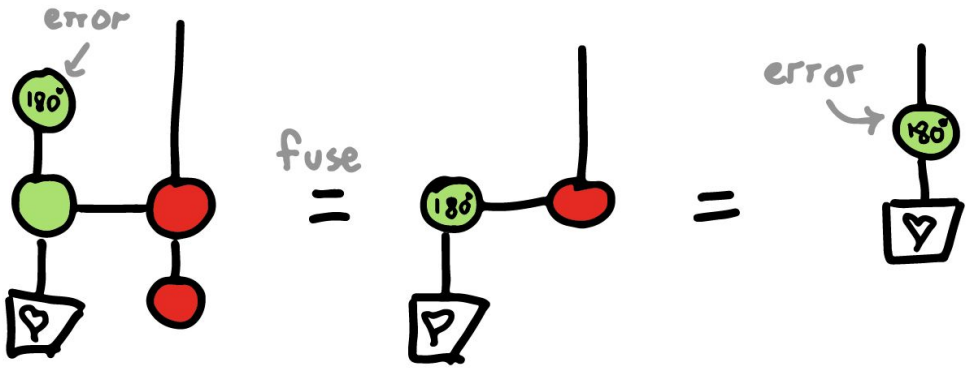
If the green 0°-test is applied to the 1st qubit, our state gets moved just as we wanted:



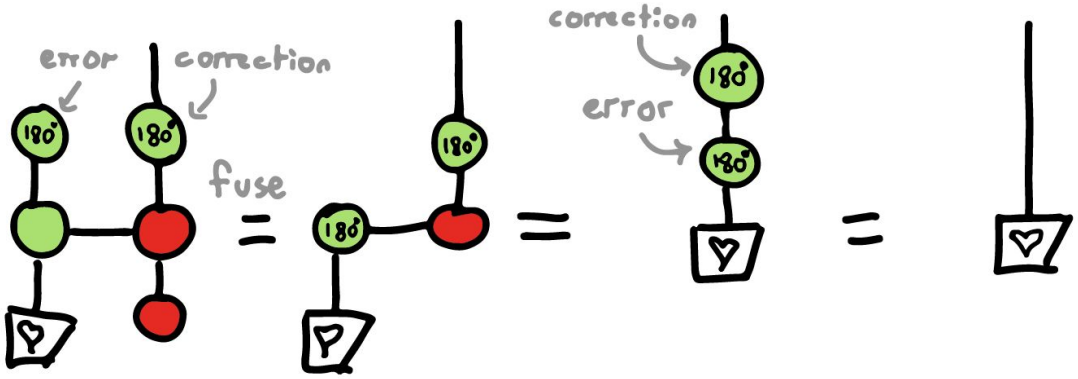
Unfortunately, the green 0°-test is a maybe-box, and it is applied about 50% of the time. The other 50% of the time, the green 180°-test is applied instead, so two tests are possible:



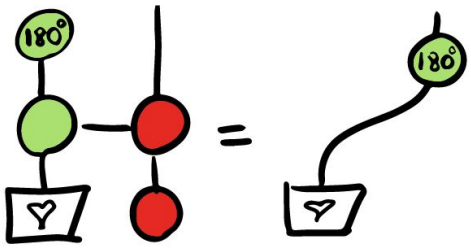
We don’t have control over which test is applied, so we may end up with an error, just as Andrea and Blaise did in teleportation:



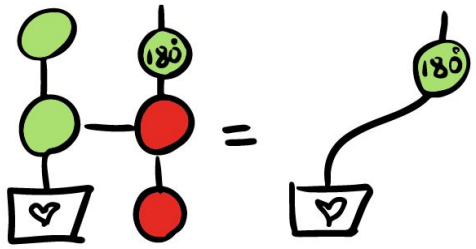
To fix this error, we simply apply a green 180° phase gate:



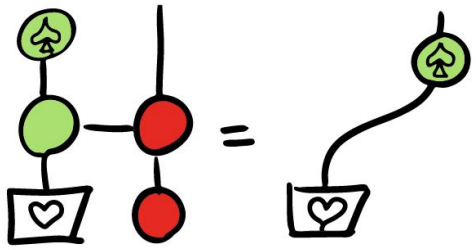
Instead of doing so, we could just leave the error there:



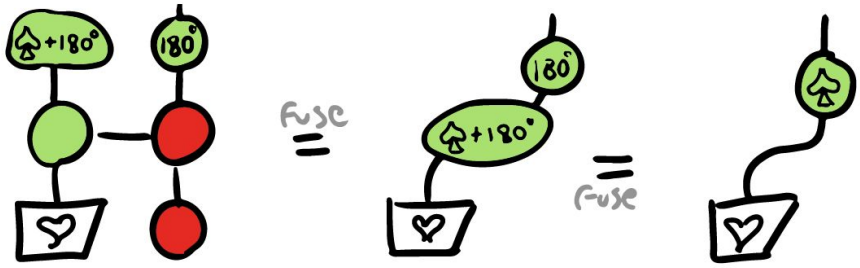
Why? Maybe we wanted to apply the green 180° phase gate anyway, and now that has happened automatically. If that is really what we wanted, then we instead have to apply a 180° phase gate to correct the case where the 0°-test was applied:



More generally, if we want to apply any green phase gate, we have to do the green test with the same phase:



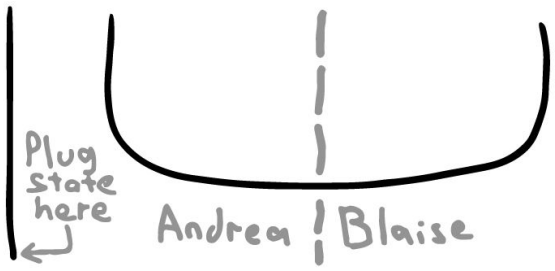
As before, the test we want is only applied 50% of the time: the other 50% of the time, the wrong test is applied. If the wrong test is applied we simply use a 180° green phase gate to fix the error:



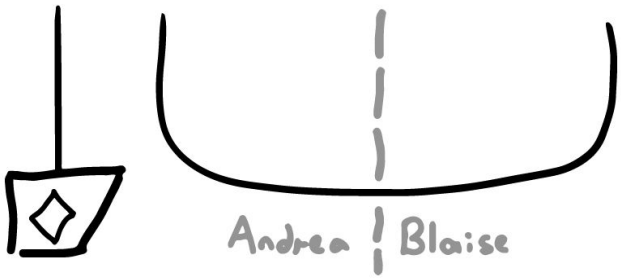
This gives us a “crazy” way of computing, by testing only! There are “crazy” people, all over the world, who are are building such “crazy” quantum computers.

2.5
Quantum teleportation, Part III

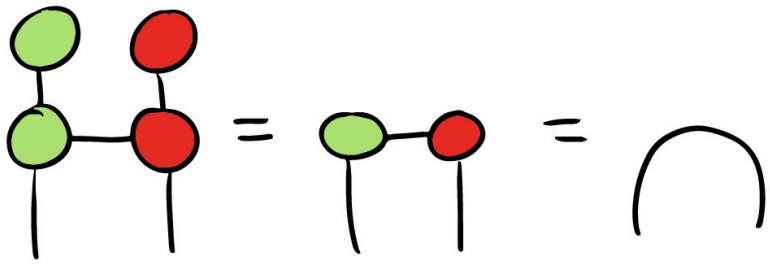
We can do the teleportation protocol from the previous Chapter using nothing but spiders. Recall that Andrea and Blaise share a cup state and that Andrea has some other qubit which she wants to send to Blaise. We won't bother to write its state, and simply draw an open input:



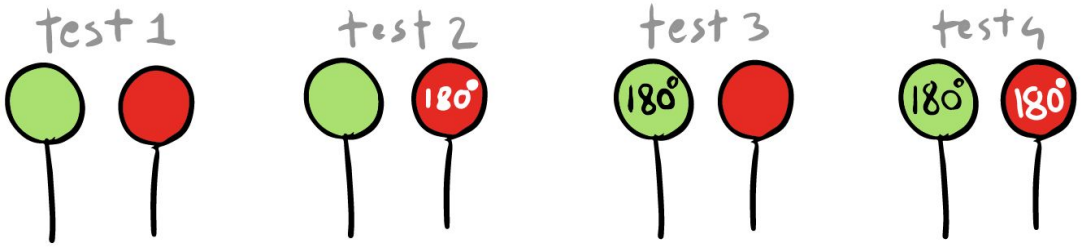
into which any state can be plugged:



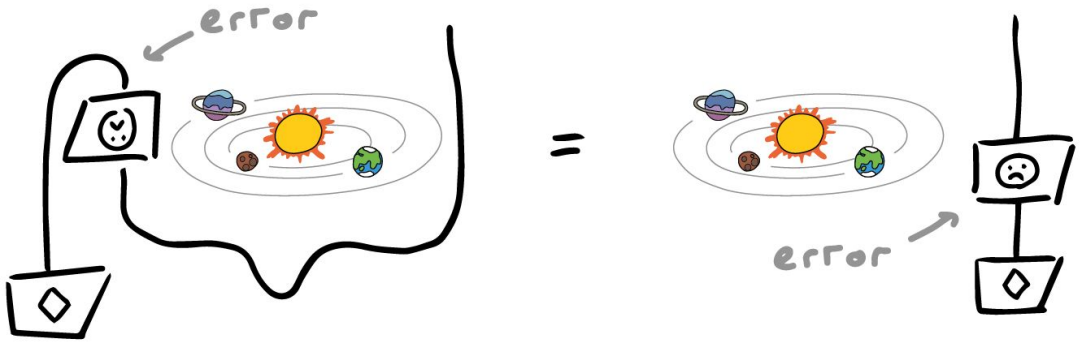
This will make our pictures a bit cleaner. Next, Andrea needs to apply a cap test. Since the last time we saw teleportation, we have gotten acquainted with spider tests, in particular with the 0° red/green tests. A cap test can be built from these together with a CNOT gate:



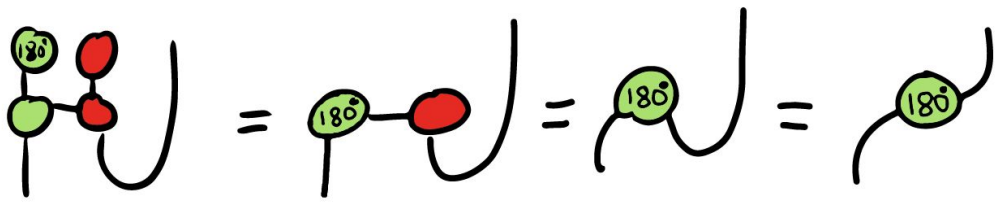
Because they are maybe-boxes, we know that either or both of the green 0°-test and red 0°-test may not actually happen, with the green and/or red 180°-tests happening instead. As a consequence, we get four possible combinations of tests, all equally likely:



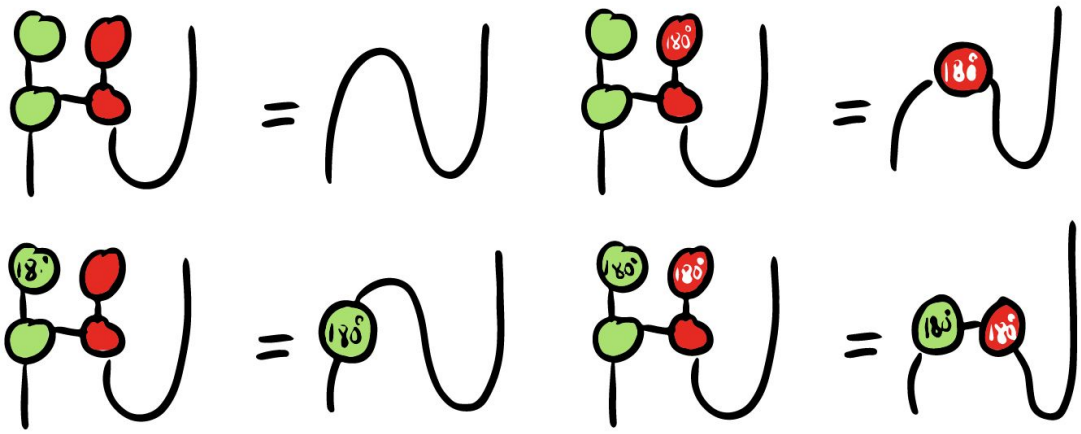
We saw that tests different from the cap test introduce errors:



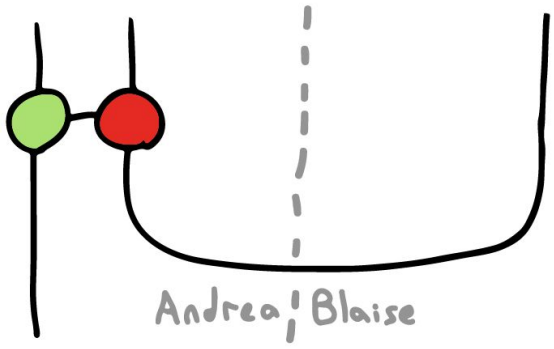
Now we can finally see what those errors really look like. For example:



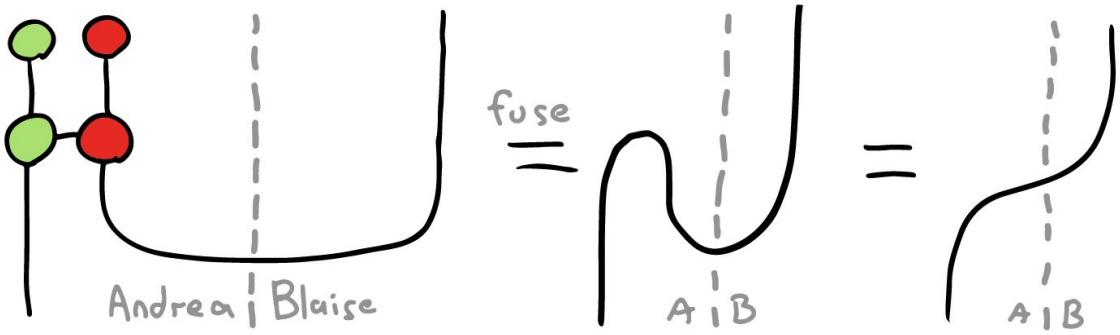
Here are all the cases together:



They all take the form of one or two phase gates, so Blaise can correct them by only using phase gates himself. Let's put all of this together and give the full description of teleportation using spiders. First, Andrea applies a CNOT gate to her two particles:

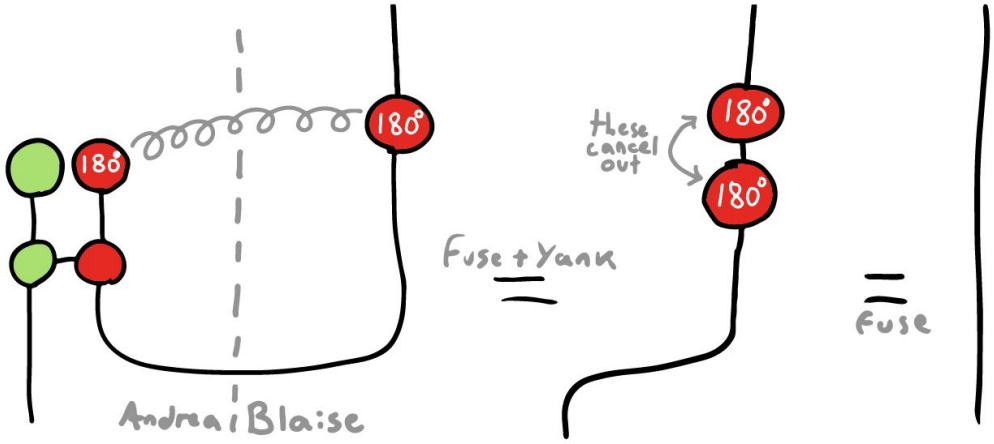


Andrea then tests her particles. There are four possible cases as detailed above, the simplest being the cap test:

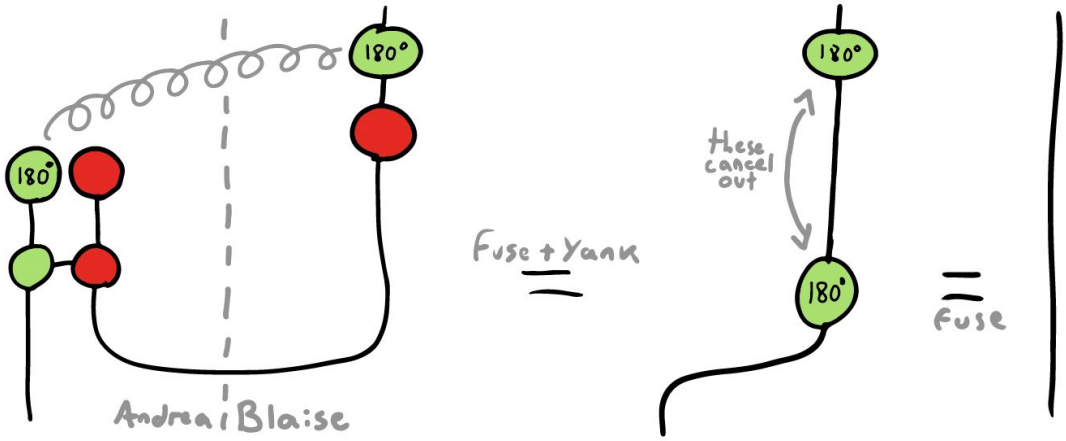


More generally, Blaise has to perform a correction using one or two phase gates, depending on which tests Andrea tells him she witnessed. This is the whole scenario

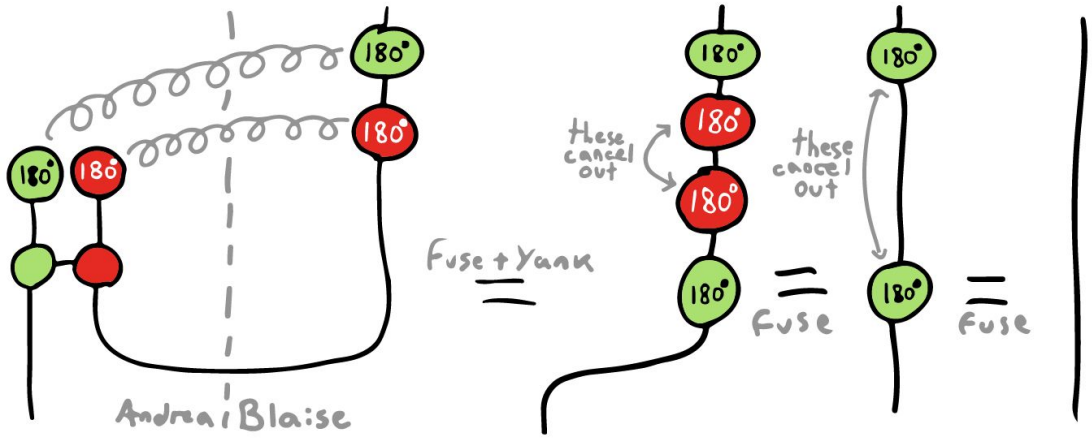
when Andrea sees that the 2nd of the four tests has happened:



This is the whole scenario when Andrea sees that the 3rd of the four tests has happened:



Finally, this is the whole scenario when Andrea sees that the 4th of the four tests has happened:

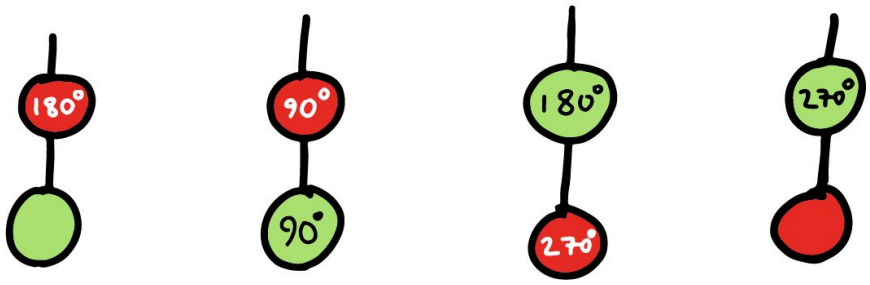


In this last case, Blaise has to use two phase gates, while in the previous two cases he only had to use one.

All this using nothing but spiders... so cool! But this is not just cool: it allows us to understand something very important about what teleportation actually does. For Blaise to know which phase to apply, he just needs to know which one of the four tests has happened, i.e. he needs to be told two bits. However, the particle that Andrea wishes to teleport could be in any state. For example, it could be a green (or red) spider state with any phase:



or something more complicated involving a combination of spider states and phase gates:



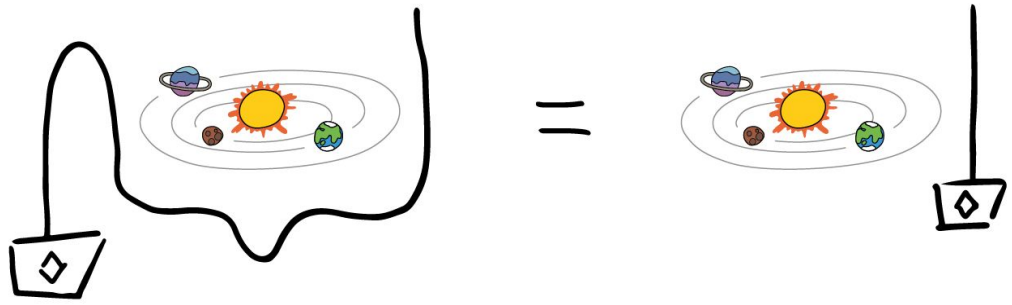
So that's the real magic of quantum teleportation: there are infinitely many states that we can teleport, and Andrea only needs to communicate two bits to Blaise to teleport any of them!

3.
KEEPING EINSTEIN HAPPY

You may have heard about Einstein's relativity theory: it trashed many things physicists believed for centuries. In particular, it puts a limit to how fast light—and everything else—can travel.

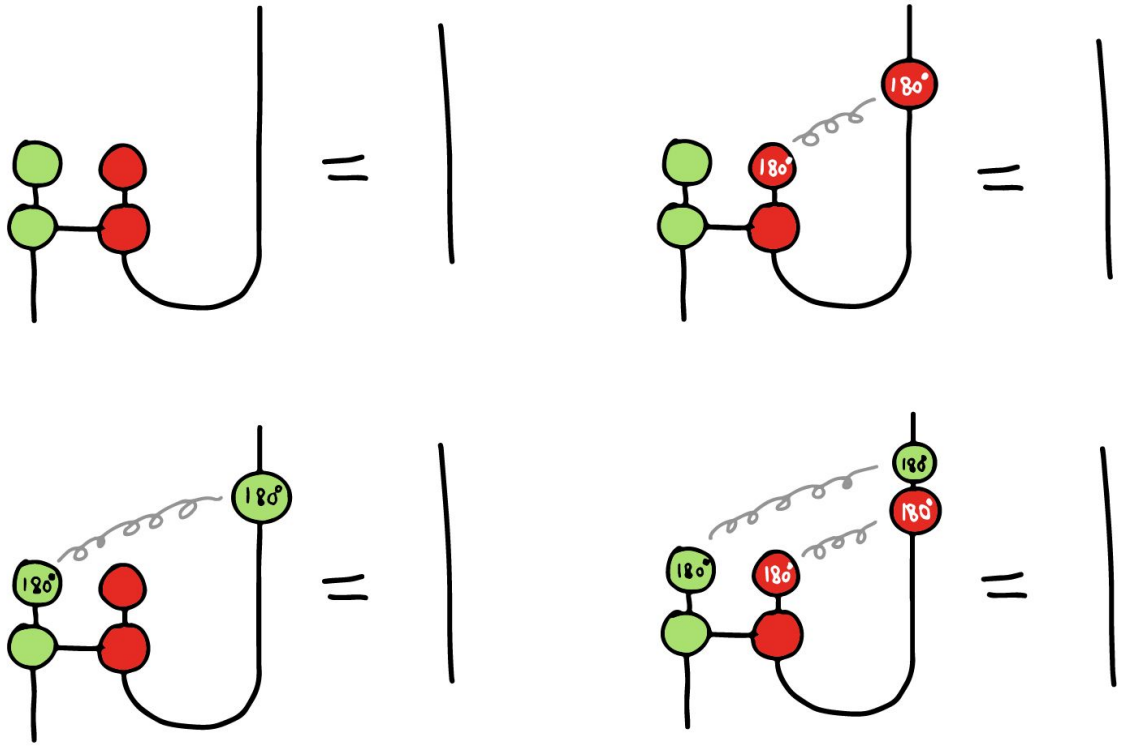


You may ask: Why is this relevant to us? It is relevant because it tells us that tests cannot be sure-boxes. If the cap test were a sure-box, then we could perform quantum teleportation as we originally described it:



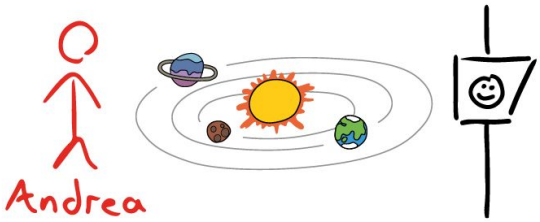
Andrea's state would appear at Blaise's end of the solar system instantly, while it would take hours for light to make the same trip. This breaks the rules of Einstein's relativity theory.

Luckily for us, the quantum lottery saves the day! The cap test is not a sure-box but a maybe-box. Using the spider representation of teleportation from the previous chapter, we know that whenever we try to do a 0°-test we may instead see that a 180°-test has happened. This means that Andrea must make a long-distance phone call to tell Blaise when and how to correct the error:

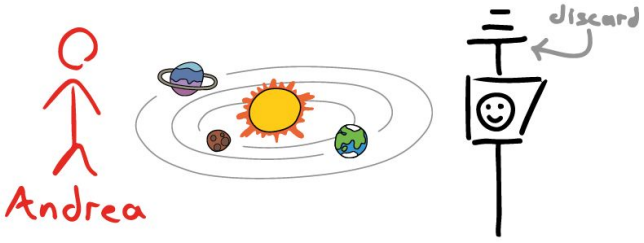


As Andrea's voice does not travel faster than light, all is now OK.

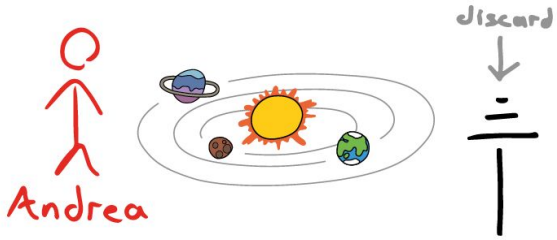
But how do we know which boxes are sure-boxes and which ones are maybe-boxes? This is what we will learn in this chapter. What makes a box a sure-box is exactly that it obeys the theory of relativity: things that are very, very far away must not immediately affect us, because what influences us cannot travel faster than the speed of light. Assume that Andrea is at one end of the solar system and that some box  is happening at the other end:



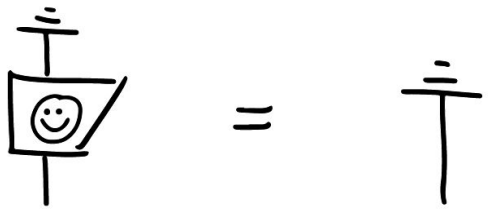
Due to the extreme distance, Andrea doesn't have to care about the output of  : it cannot reach her right now, so it is irrelevant to whatever she is doing at present. We can indicate this by letting Andrea <u>discard</u> whatever comes out of  :



Because the output of  has not yet reached her, whatever particle the box  has been applied to should also be irrelevant, so that Andrea might as well discard that altogether:



Taking the two pictures together, we get the following equation:



Reading this directly, it says that if the output of a box doesn't affect us, than we may as well discard what goes in it. For example, if someone starts with some ingredients, cooks a chocolate cake with them and throws the cake into the bin, then they might as well have thrown the ingredients into the bin from the start, as in both cases they are left with an empty stomach.

Not all boxes satisfy the equation above: sure-boxes are exactly the ones that do. Just looking at this equation on its own, it may seem like it has very little to do with Einstein's idea of things not travelling faster than light, but that impression is incorrect. We will see in this chapter that the equation above ensures that whatever happens at one end of the solar system cannot immediately affect what lies at the other end. We will also show that the same equation ensures that whatever happens in the future cannot affect the present and the past.

Of course, all this doesn't mean that we will throw our maybe-boxes out of the window! They are very useful, as we have seen previously. But we will learn how to use them in a way which keeps Einstein happy.

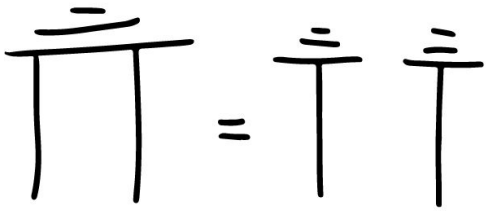
3.1
The mathematics of sure-boxes

Discarding

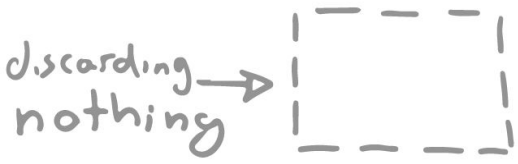
For each kind of particle, there is a special discarding test:



Of course, discarding many wires together is the same as discarding each individual wire:



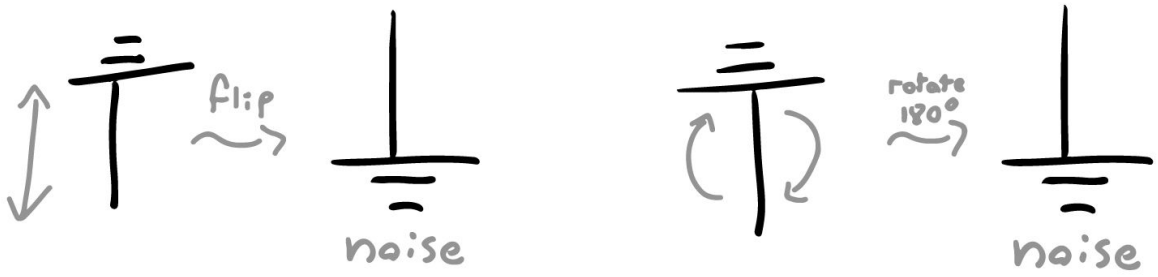
If we don't have any wire then there is nothing to discard:



This last one may sound rather silly, but we will need it below when we discard the output of a test.

Discarding remains the same if we flip it horizontally, but flipping it vertically or

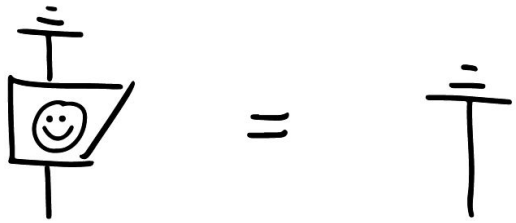
rotating it by 180° results in a state—the same in both cases—which we call noise (or randomness):



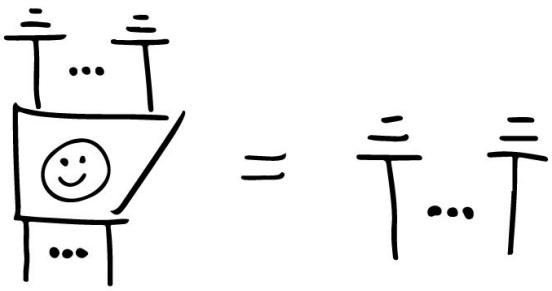
If discarding means we ignore what comes out of a box, noise means we somehow ignore what goes into it.

Sure-boxes

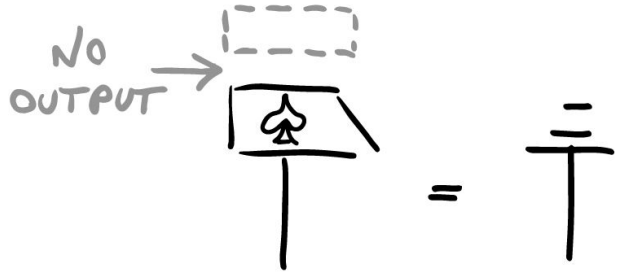
A sure-box is a box for which we have:



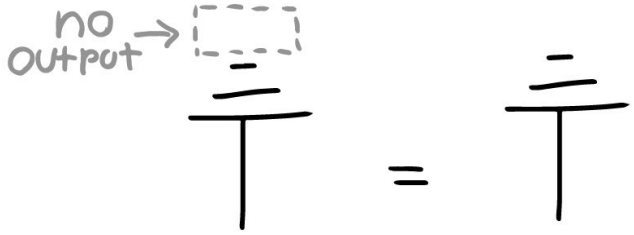
that is, discarding the output of a box is the same as discarding its input. It is important that all outputs are discarded, so that a box with multiple input and output wires is a sure-box if it satisfies the following equation:



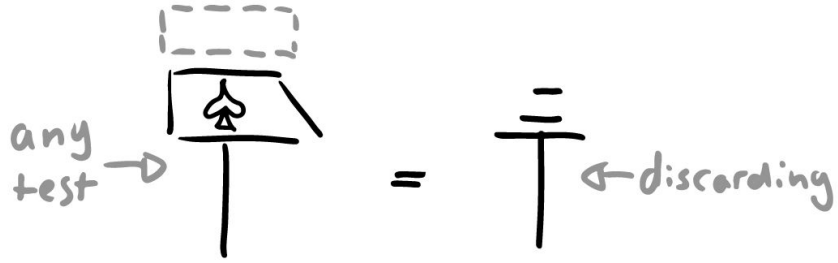
On the other hand, if a box has no output, nothing needs to be discarded, so a test is a sure-test if we have:



In particular, discarding is a sure-test:



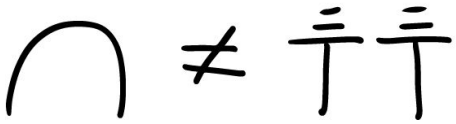
since it is of course equal to itself. In fact, discarding tests are the only sure-tests! Therefore, a test  is a sure-test if it satisfies:



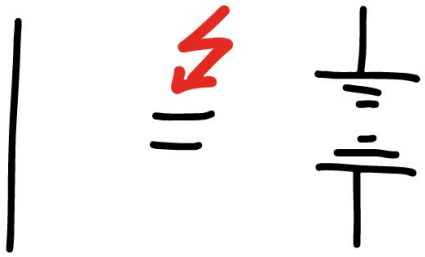
which plainly says that  must be equal to discarding. In particular, the green 0°-test and red 0°-test cannot be sure-tests, since they are not the same as discarding:



Similarly, the cap test is not a sure-test, since it is not the same as the discarding of two wires:



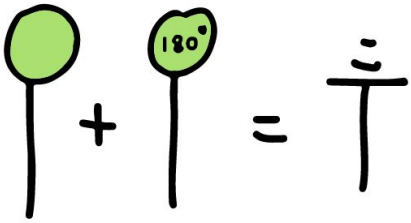
You might now wonder: what would happen if the cap test actually were the same as the discarding of two wires? Well... that would mean that every wire is broken! We can see this by bending one of the two wires upwards:

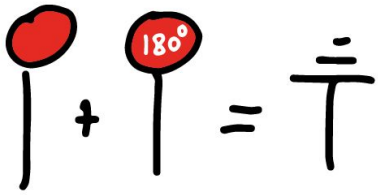


This, of course, would make no sense. So cap tests cannot be the same as discarding.

Sure-boxes from maybe-boxes

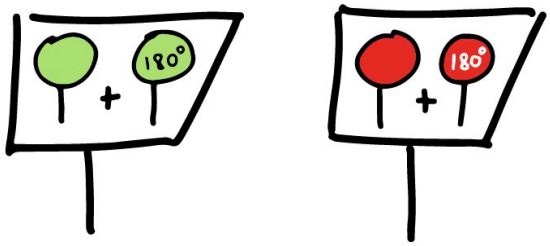
It is possible to make a sure-box from maybe-boxes by taking a few of them together. Indeed, we have already seen that neither a 0°-test nor the corresponding 180°-test can be made to happen with certainty, but that one of the two always happens. This means that, together, a 0°-test and the corresponding 180°-test form a sure-test, i.e. a discarding test:





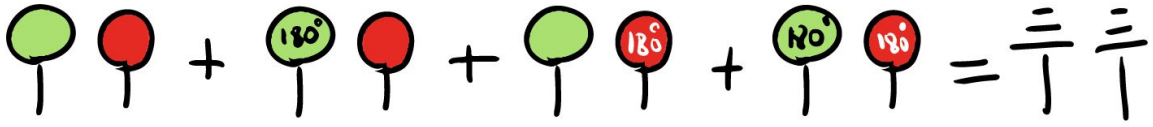
The + sign is a way of indicating that there are two cases: one of them will certainly happen, you just don't know which one. For example, when looking at a cake we can check whether it either has sprinkles or not: one of the two will certainly be true. Think of the 0°-test as 'sprinkles' and the 180°-test as 'no sprinkles'.

We didn't yet explain why either the 0°-test or the 180°-test will certainly happen: don't worry, we will do so in the coming chapters. For now, just remember that one should think of this sum of maybe-boxes as a single sure-box:

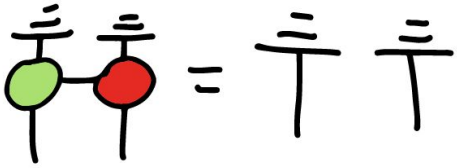


which stands for either the 0°-test or the 180°-test happening.

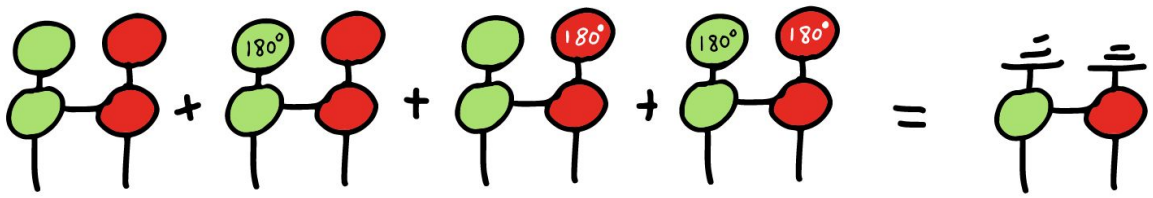
When we apply these sure-tests to two systems, we get a more complicated sure-box with four cases. We used this more complicated sure-box in the previous chapter when dealing with teleportation, applying the green tests to one particle and the red tests to the other:



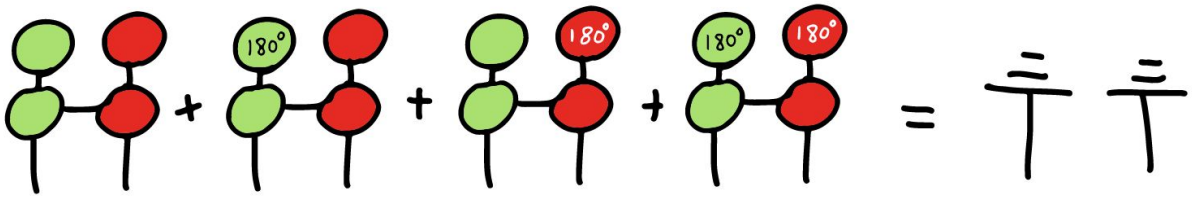
Now, the CNOT is also a sure-box (we shall see why this is in the next chapter):



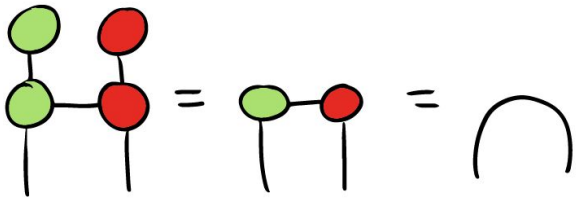
By inserting a CNOT at the inputs of the four tests in the big equation above we therefore get:



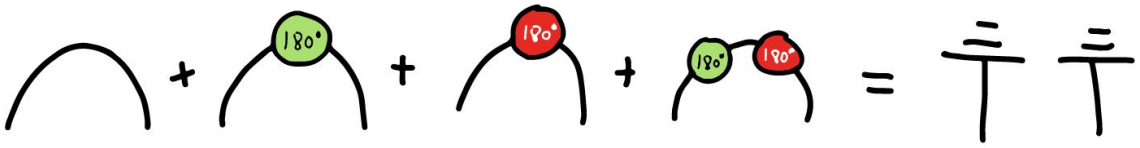
Because the CNOT is a sure-box, this becomes:



The first of these tests is the cap test:



and the other three can be added to the cap test to form a sure-test:

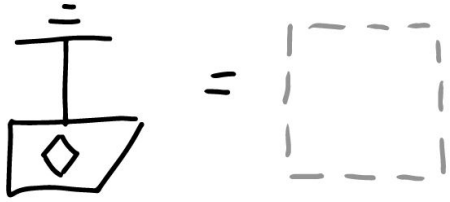


In fact, it turns out that—one way or another—every maybe-test is part of a sure-test.

Sure-states

There may only be a single sure-test for each wire, but there are plenty of sure-states,

namely those such that:



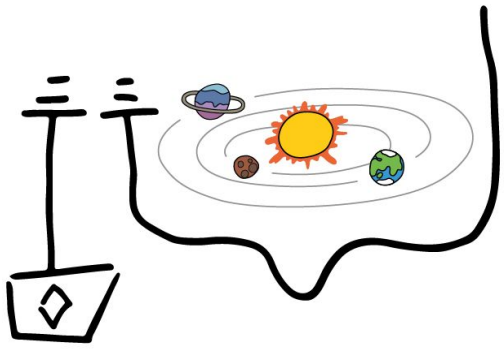
Because states don’t have input wires, the empty diagram appears on the right. However, be aware: while you can flip every state into a test, a sure-state is not going to be a sure-test if you flip it! Flipping does not respect “sureness”: there are many kinds of sure-states, but only one kind of sure-test.

3.2
No teleportation without a telephone

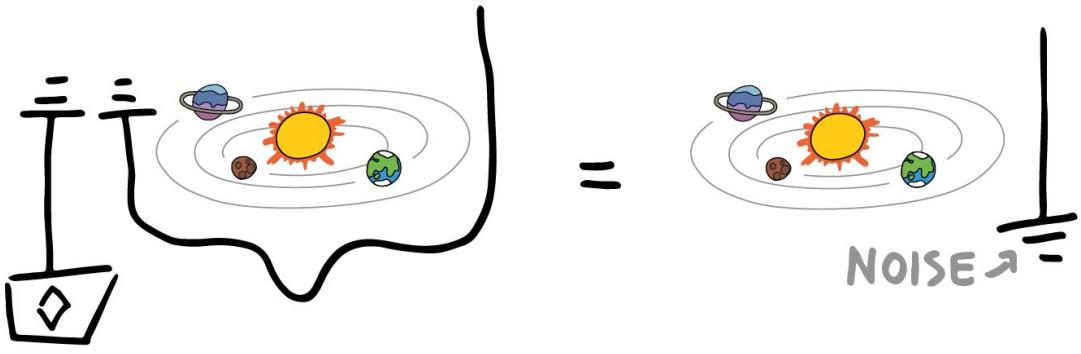
The mantra of this Chapter is: if you want to make sure that something happens for sure, only use sure-boxes in your diagrams!

For teleportation to succeed we had to consider four cases in total, corresponding to four possible maybe-tests that might happen. This means that Blaise has to hear from Andrea before applying the correction on his side. Now suppose that he never hears from Andrea: What is he left with?

If Blaise doesn't hear from Andrea, he can discard the two wires at Andrea's end, since he doesn't know which of the four maybe-tests happened:



Because ◇ must be a sure-state, we obtain:

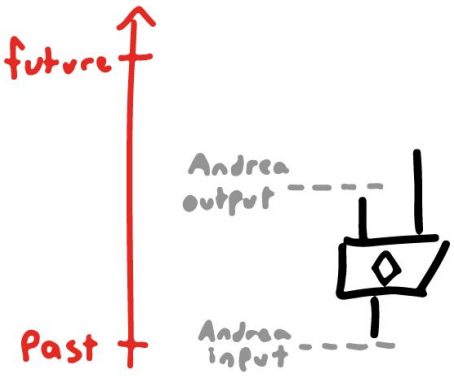


That’s it: without a telephone call, Blaise receives nothing but random noise, and nothing is teleported.

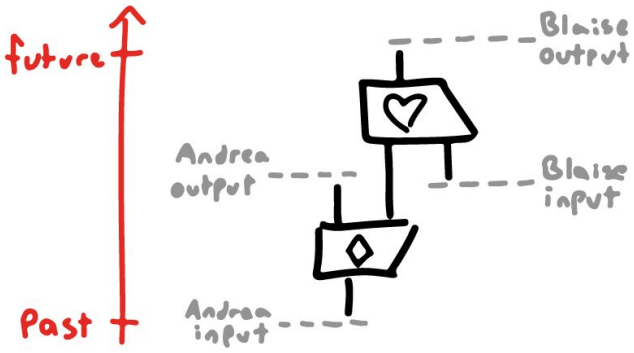
3.3
No texting from the future

With quantum boxes around and the crazy stuff they already allow us to do, it could well be that some even crazier stuff becomes possible, such as the future affecting the past. Luckily, sure-boxes come to the rescue: this cannot happen. What a relief!

Assume that, in the present, Andrea has access to a sure-box ◇. She can stick particles into the input of the box and she can receive particles from the output of the box:

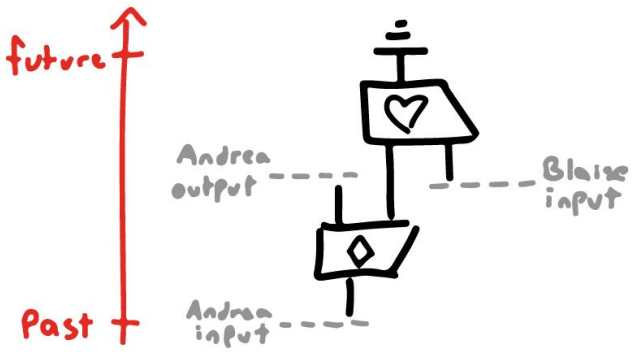


In the future, Blaise has access to another sure-box ♡ and the two boxes are connected by a wire:

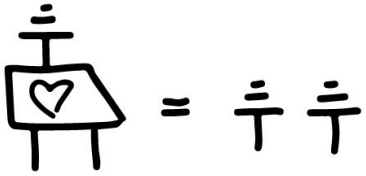


The question now is: Can Blaise use his quantum box ♡ to send a love letter to Andrea from the future? Sadly, the answer is no. Evidently, Andrea has no access to Blaise's

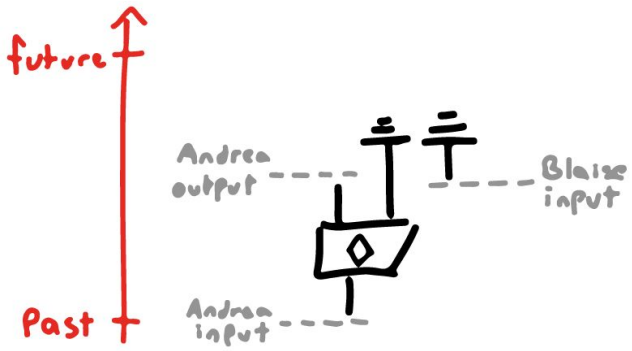
output, since it lives in the future. That means that, in the present, she must discard it:



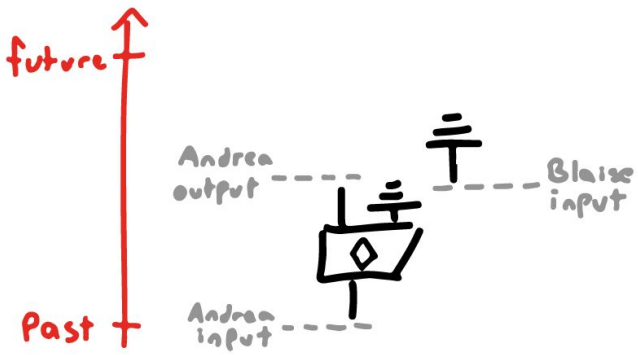
Because the box  is a sure-box, discarding Blaise's output is the same as discarding his box altogether:



so we obtain:



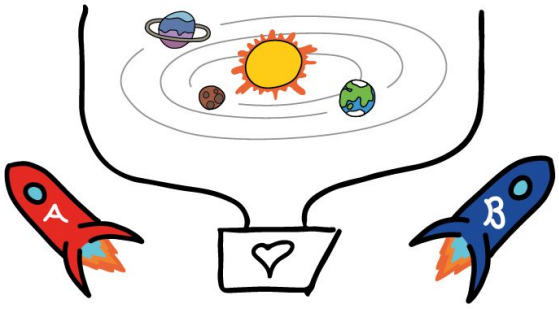
Sliding the discarding down to Andrea:



we see that there is no longer a connection between Blaise's input and Andrea's output. He may declare his love in the future, but it can never make its way to her in the past.

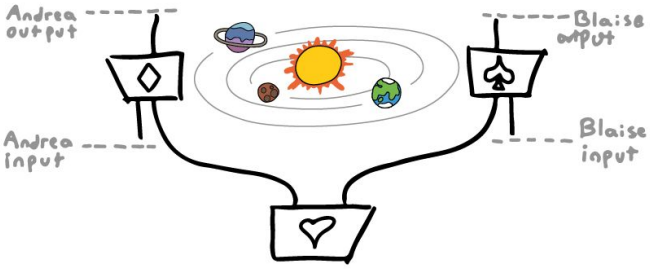
3.4
No real-time texting from far away

Assume now that Andrea and Blaise are at opposite ends of the solar system. Blaise wants once again to declare his love to Andrea, and he (understandably!) doesn't want to wait thousands of years for the message to reach her. As they are both in the present, there can't be a wire that instantaneously sends particles from Blaise to Andrea. However, imagine that, some time before this, Chi sent them one particle each, in some two-particle state 🂱:

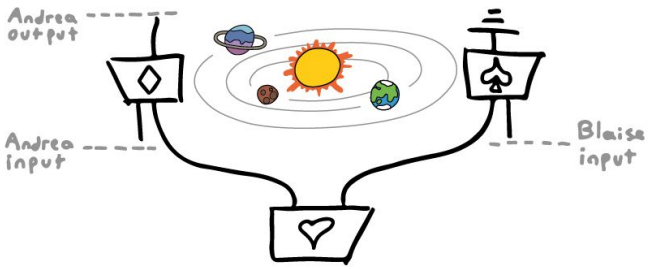


For example, you could imagine 🂱 to be the cup state. Can Blaise use this shared state to send an instantaneous love letter to Andrea from so far away? The answer, unfortunately, is once again no.

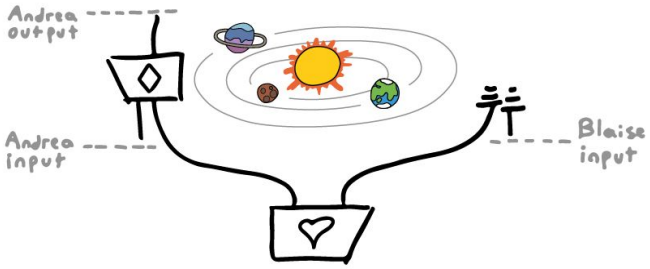
To see this, assume again that Andrea and Blaise each have access to a sure box, 🃁 and 🂡 respectively, into which they can stick particles and from which they can receive particles:



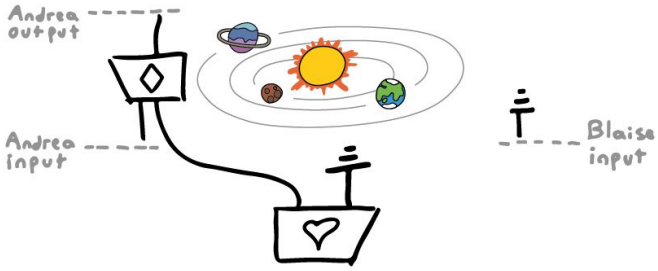
Andrea has no access to Blaise's output, since it is very far away, so she must discard it:



The box ♤ is a sure-box, so we once again obtain:



After some sliding this becomes:



There is no longer a wire connecting Blaise's input to Andrea's output, so there is no way for him to instantaneously declare his love from the other end of the solar system. Talk about star-crossed love...

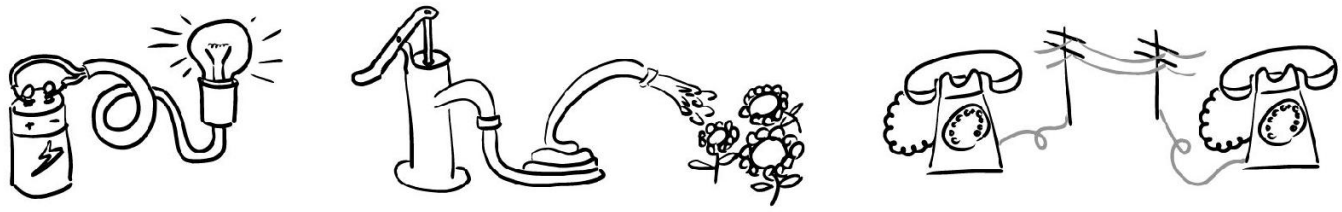
4.
QUANTUM VS. ORDINARY PARTICLES

One big problem with quantum particles is that we can't see them, at least not directly. We have to test them by means of fancy machines to know what they look like, and only then we might get to know something about them, e.g. on a computer screen. One such machine, located at CERN (in Switzerland), is bigger than many towns and is used to study the smallest quantum particles around.



The things which we can instead see and experience directly are called ordinary particles: these include billiard balls, dots on a computer screen, and pretty much anything else which makes up the everyday world we know and love. At the very beginning of this book, we saw that ordinary particles also travel in wires. Electricity travels in electrical wires, water travels in pipes, and phone calls travel in telephone

wires:



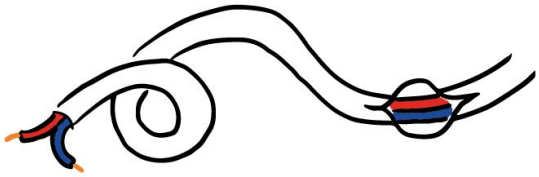
Sometimes, we want to draw diagrams that have both ordinary and quantum wires. For example, in teleportation Andrea has to make a long-distance call to Blaise, and we may want to draw the wire carrying her voice in the same diagram that already contains the wires carrying the quantum particles.

This means that we need two kinds of wires: ordinary wires, carrying ordinary particles, and quantum wires, carrying quantum particles. These wires should look different, since the things we can do with ordinary particles are very different from the things we can do with quantum particles. We already used a traditional telephone wire to depict Andrea's call to Blaise, but that's not very good: we still want to use spiders, and spiders don't have telephone wires coming out of them. We need to find a better solution.

How will we distinguish the two kinds of wires using only spiders? We go for the simplest possible option: we use a single wire (an ordinary wire) for ordinary particles, and two wires bundled together (a quantum wire) for quantum particles:



So, a quantum wire is secretly made up of two wires, just like an electrical wire is made up of a pair of smaller wires:

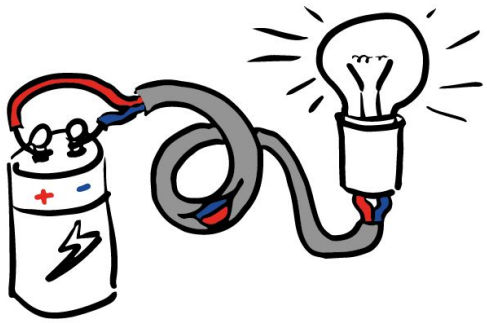


Why this way around? Well, we can do much crazier stuff with quantum particles than we can with ordinary ones, so it makes sense that we would need more wires to carry the craziness.

Similarly, we use single boxes (ordinary boxes) for ordinary particles and double boxes (quantum boxes) for quantum particles:

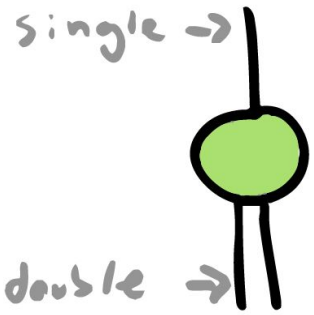


You notice that we horizontally flipped one of the two boxes. You can think of this exactly as in the case of electricity, where the outputs of a battery and the inputs of a light bulb are always in pairs of positive/negative charge:



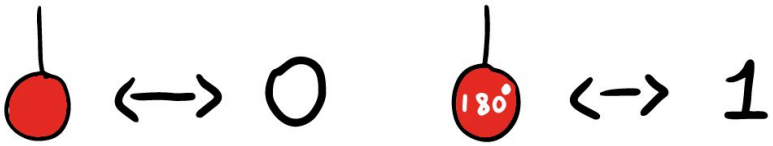
We said earlier that we need fancy machines to know what quantum particles look like, but how can we draw these machines with wires and boxes? Since they turn quantum

particles (i.e. double wires) into ordinary particles (i.e. single wires), we need to find boxes that do exactly that. Our mighty spiders come to the rescue:

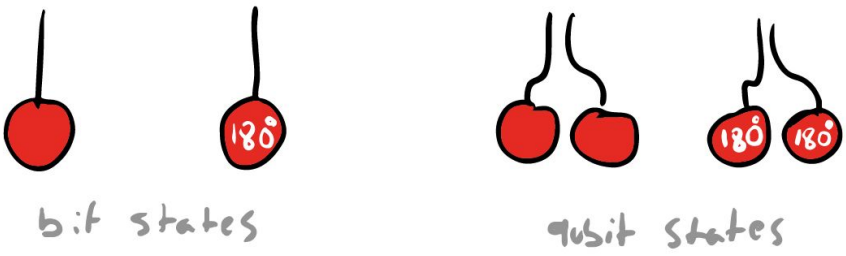


The two input legs of this spider are a quantum wire, while the single output leg is an ordinary wire. Since the fancy machines are usually called “measurement devices”, we will call such a spider a measurement.

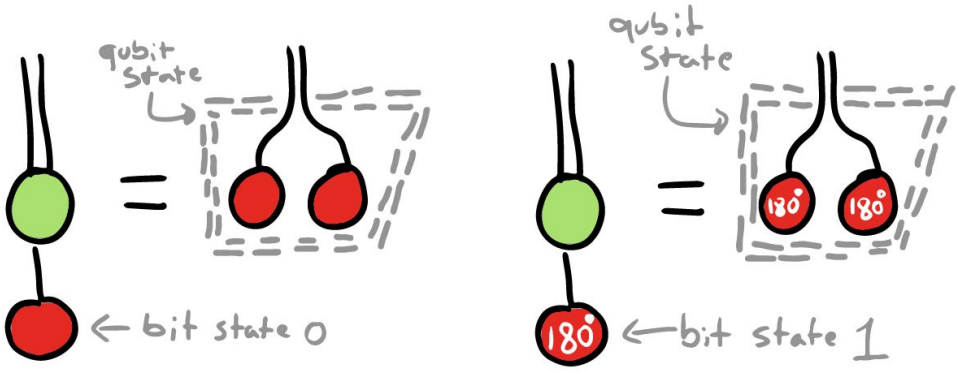
We can also do the opposite, using spiders to turn an ordinary particle into a quantum particle. Earlier on, we sneakily did this when using the 0° and 180° red spider states to make bits out of qubits:



Under our new rules for pictures—single wires for ordinary particles, double wires for quantum particles—it is completely clear when we mean the 0° and 180° red spiders to be bit states and when we mean them to be qubit states:



If we vertically flip the measurement spider, we can now truly encode a bit state as a qubit state:



We used the fact that both the 0° and the 180° red spider states are copied by the green spider, something that we learned some time ago. All in all, we now have a way to turn quantum states into ordinary states and vice versa.

Now that we can both talk about ordinary and quantum particles in the same diagram, we can also figure out how different they are. In our ordinary world, for example, we can copy classical particles whenever we like, but we will soon see that this is no longer possible for quantum particles. And if this weren't confusing already, we will also see that even looking at quantum particles to learn something about them is enough to completely change them!

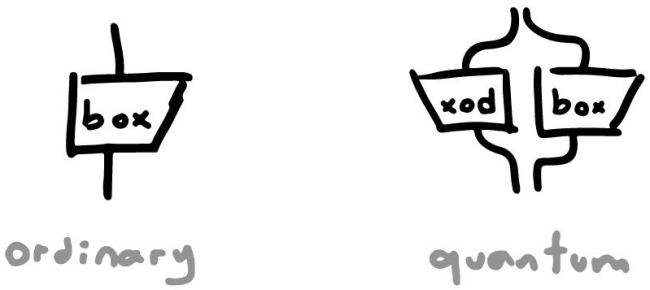
4.1
The mathematics of quantum vs. ordinary

Quantum vs. ordinary wires and boxes

There are two kinds of wires:



Ordinary boxes are single boxes and quantum boxes are double boxes, where the left box is horizontally flipped:

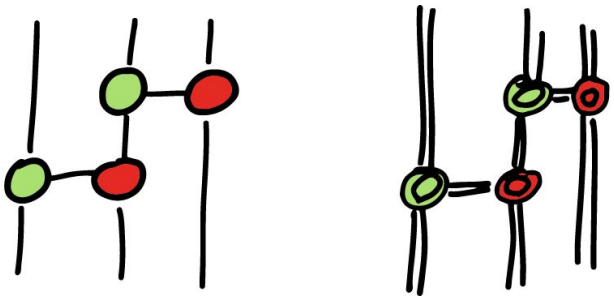


To make them look more like ordinary boxes, we will often draw quantum boxes as a single box with doubled wires and doubled borders:

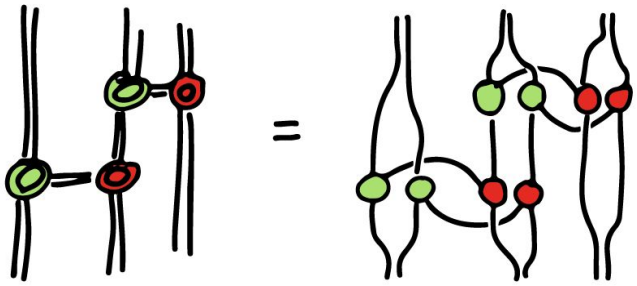


For example, we can now easily distinguish an ordinary circuit from the corresponding

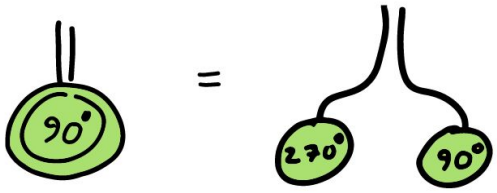
quantum circuit:



where the quantum circuit secretly is:



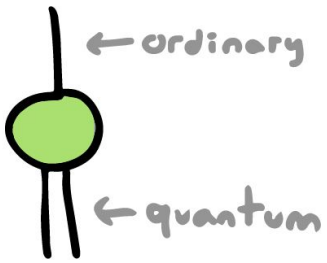
Because boxes on the left are horizontally flipped, for quantum spider states we should not forget to change the angle of the spider on the left:



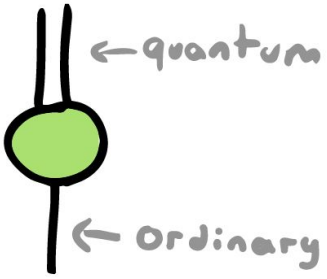
The same is true for every doubled box involving spiders!

Measurement and encoding

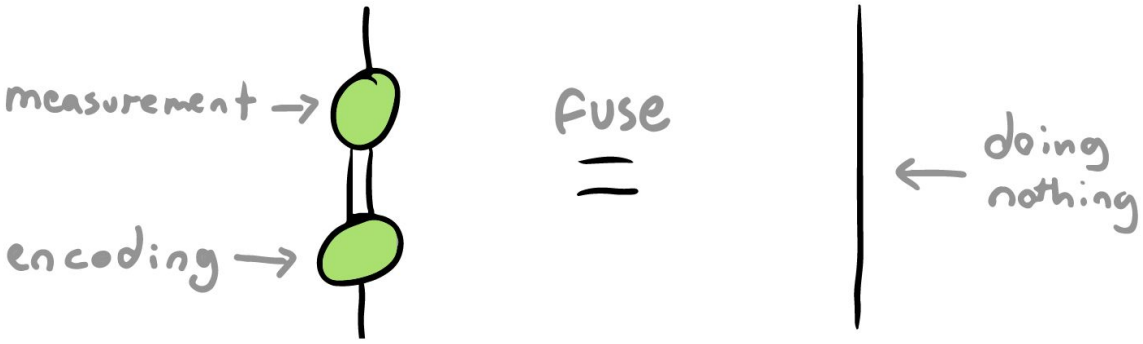
A measurement is a spider:



Where we think of the two input legs as a quantum wire and of the output leg as an ordinary wire. An encoding is the vertically flipped spider:



Due to spider-fusion, encoding and then measuring in the same colour amounts to doing nothing at all:

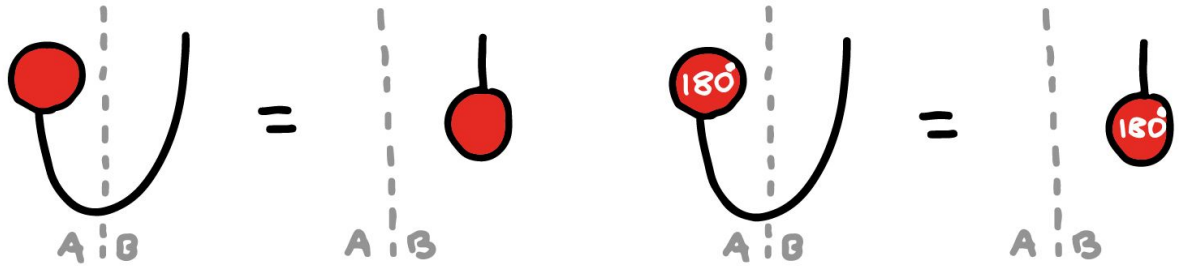


Ordinary and quantum cup states

We now have an ordinary cup state and a quantum cup state, and they are not the same:



You may ask yourself: What is an ordinary cup state? This is a state where Andrea and Blaise each have a bit and they don't know which bit they have. However, whichever bit Andrea has, Blaise always has the same: either they both have 0 or they both have 1. To see this, simply plug in a bit test at Andrea's end:



So whatever bit Andrea gets, Blaise gets the same on his end.

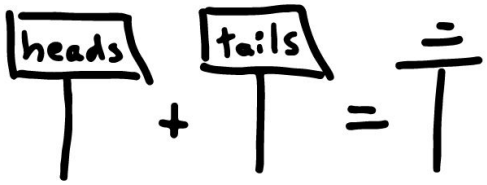
4.2
Tests forming a sure-box

As previously promised, we will now explain why the 0°-test and the 180°-test of the same colour make up a sure-test. We will do so using our new doubled notation:

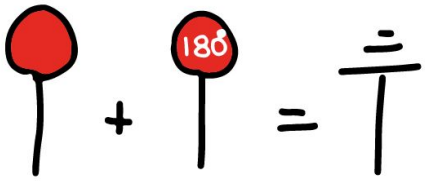


The reason will boil down to the fact that quantum sure-boxes arise from ordinary sure-boxes. So, what are ordinary sure-boxes?

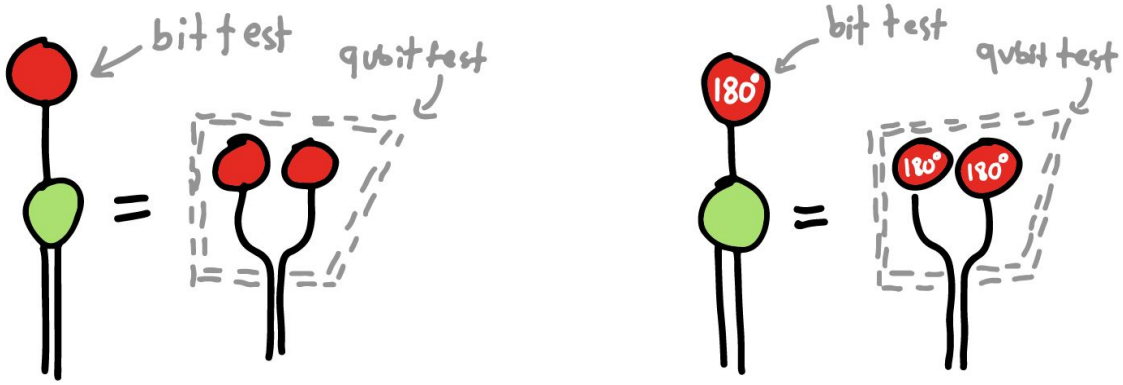
If you flip a coin, there are only two possible outcomes: heads or tails. This corresponds to two tests: “obtaining heads” and “obtaining tails”. Because one of the two has to happen, these two tests taken together make up a sure-test:



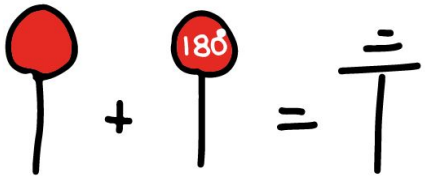
Instead of heads and tails, we could use 0 and 1, or equivalently the 0° and the 180° spider tests:



This is what we mean by ordinary sure-tests. We can now use our measurement spiders to turn ordinary tests into quantum tests: in particular, we can turn ordinary sure-tests into quantum sure-tests. To get a quantum test from an ordinary test, we simply plug the ordinary tests into the output wire of a measurement:



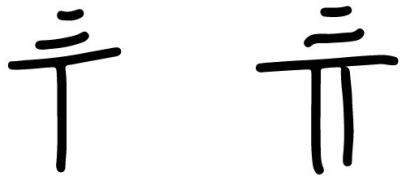
To get a quantum sure-test, we bunch together all the quantum tests arising this way from the tests in an ordinary sure-test:



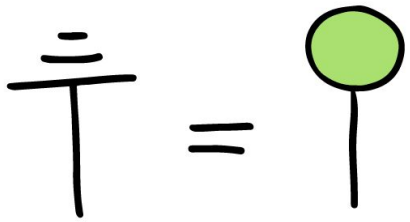
That's our mystery solved: we can think of a quantum sure-test as a measurement followed by an ordinary-sure test, and we can figure out what makes an ordinary sure-test by referring to our everyday experience.

Ordinary vs. quantum discarding

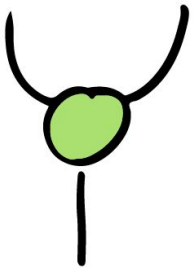
We previously claimed that spiders were enough to do everything we wanted. If that's the case: What's up with these not-very-spidery-looking-boxes??



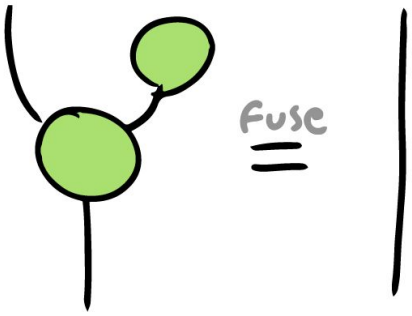
Are they spiders? If so, which spiders? Let's start with ordinary discarding. We claim that it is this spider:



Why? If we take an ordinary particle, we can copy it as follows:

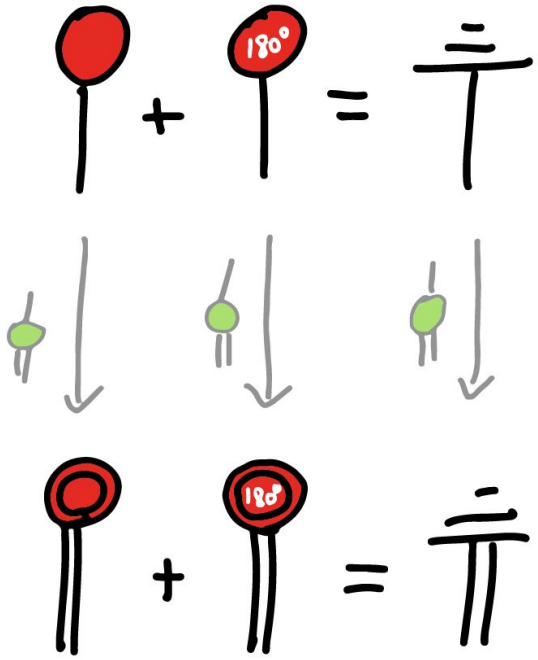


Then we can discard one of the resulting copies, and this should be the same as doing nothing at all. Bingo:

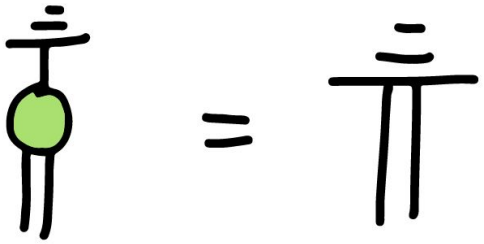


What about quantum discarding? We just saw that by plugging ordinary tests into a

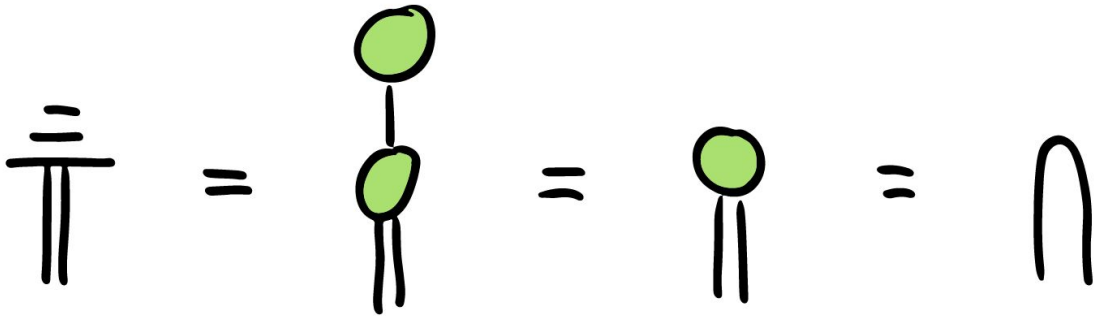
measurement we get quantum tests, and our notation seemed to indicate that plugging ordinary discarding into a measurement gives quantum discarding:



A measurement can be performed with certainty in a lab, so it had better be a sure-box. This means that we must have:



That is, discarding the (ordinary) output of a measurement is the same as discarding its (quantum) input. We can use this to figure out what quantum discarding should look like:

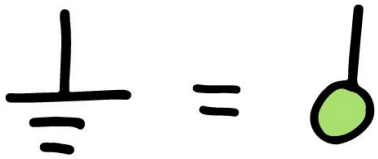


From the above, we get the following:



That's it! Everything is a spider!

We also said in the previous chapter that if we flip discarding vertically—or rotate it by 180°—then we get noise. Flipping ordinary discarding, this is what ordinary noise looks like:

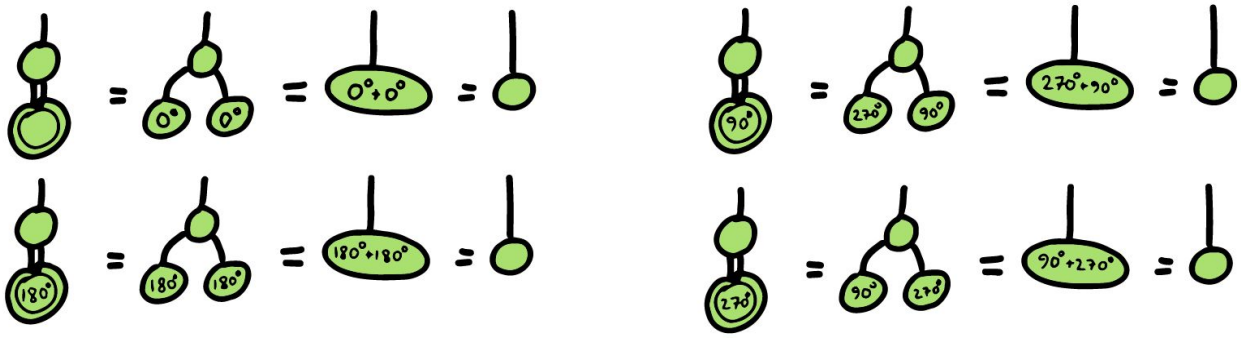


Flipping quantum discarding, this is what quantum noise looks like:

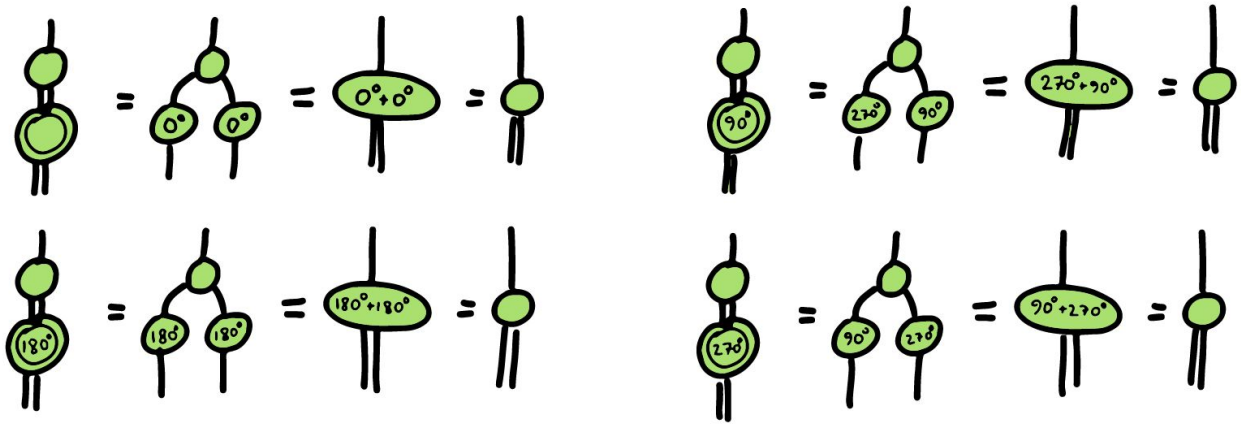


4.3
Phases are the face of quantum

This is what happens if we measure a phase state:



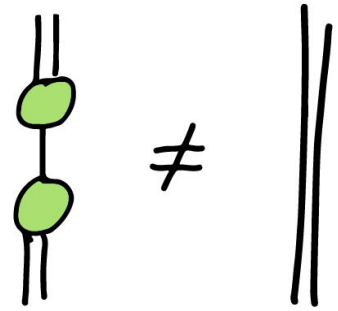
The phase is gone! The same is true if we measure a phase gate:



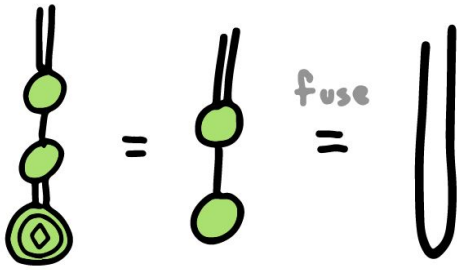
More generally, the same is true if we measure any leg of any spider with phases. What is happening here?! Well, phases live in the doubled quantum wires, but there isn't any space for them in the single ordinary wires, so they vanish. Phases are truly the face of the quantum world!

Earlier we saw that first encoding and then measuring does nothing. On the other hand, measuring a qubit and then encoding the ordinary outcome back into the qubit is very

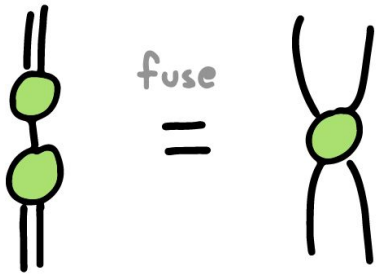
much not the same as doing nothing at all to it:



To see why this is, we simply stick an encoding on top of the first calculation we did in this section:



The phase on our quantum spider state is gone! We call this box phase-erasure. We can simplify it using spider fusion:

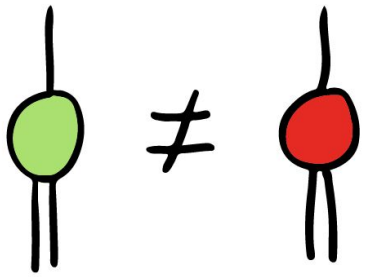


Phase-erasure is the reason building quantum computers is so hard. Why? Well, phase-erasure has a habit of happening whenever you bring a quantum particle in contact with our ordinary world, as indicated by the ordinary wire in the middle. Avoiding this contact turns out to be very hard, and a common way to do it involves keeping the quantum particles at very cold temperatures. Of course, we want quantum computers to be part of our world, which means not living in a giant fridge!

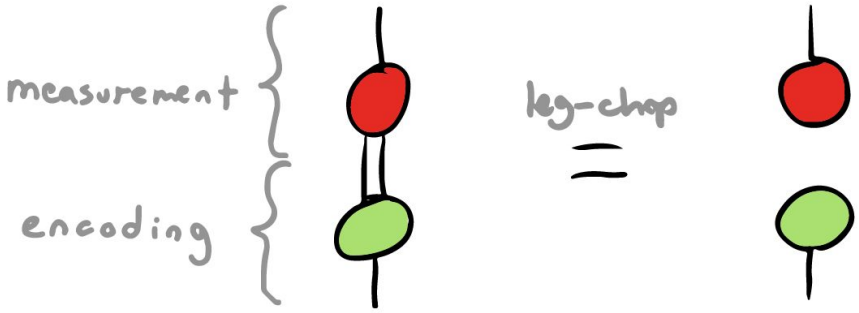
But it is these same phases that make quantum computers so much more interesting than ordinary computers: they can be used in a clever way to do things that ordinary computers can never do. This is what quantum algorithms are all about.

4.4
Quantum uncertainty, Part I

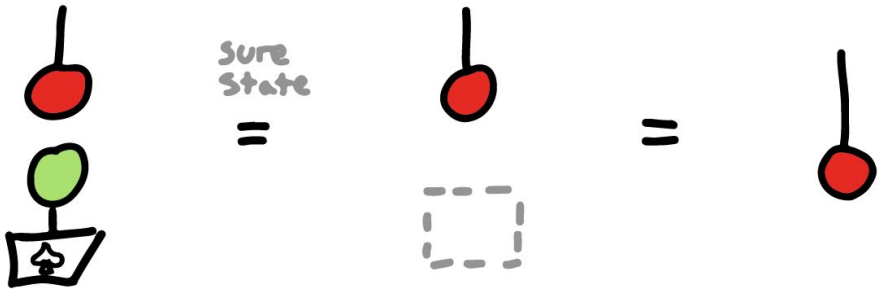
We said that the two different colours of spiders should be thought of as two different ways of looking at the world. The world we look at is the quantum world, and how we look at it is by means of measurements. Differently coloured spiders give different measurements:



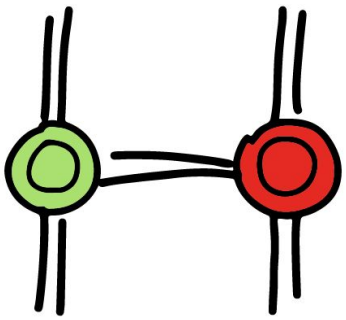
Each of the two measurements corresponds to a different way of looking at the quantum world. To see that they are different, we simply encode in one colour and then measure in the other colour:



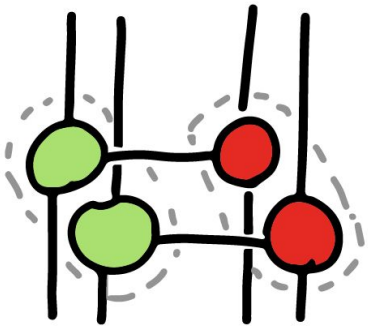
There are no wires connecting the input and the output: the spiders are now disconnected. To see what this means, we plug a state in the input:



The state gets deleted! If we encode a bit in one colour and measure it in the other colour, the bit is lost: we have total and complete uncertainty about what it was. Let's do a few more calculations, to get a feel for this. In this example:



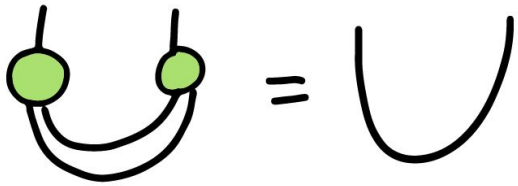
There is a red spider and a green one, with two wires between them. Do they vanish? No! If we draw this more carefully:



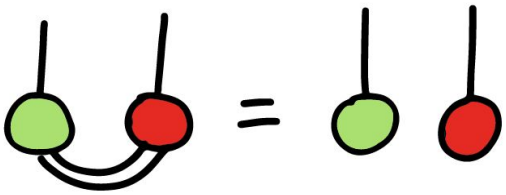
Then we see that there is only one wire between each of the green and red spiders, so no leg-chopping happens.

If we measure two particles in a quantum cup state with the same measurement on both

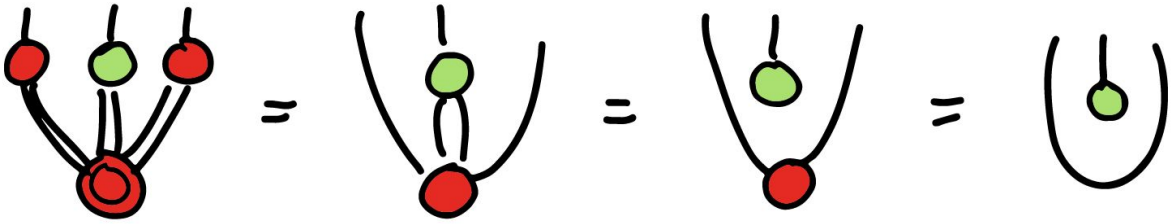
ends, we get a classical cup:



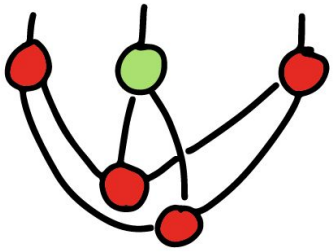
If we measure the two particles with different colour measurements, on the other hand, the spiders disconnect:



Here is a slightly more interesting example, involving three measurements on three particles in a red spider-state:



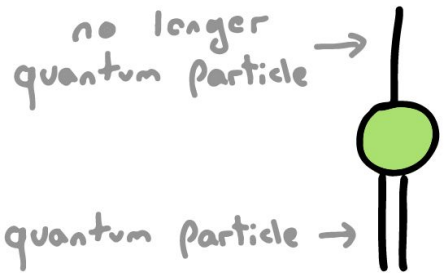
We end up with an ordinary cup state and noise in between. Note that, at first, it may seem that we can't get rid of the two wires from the green spider to the doubled red spider:



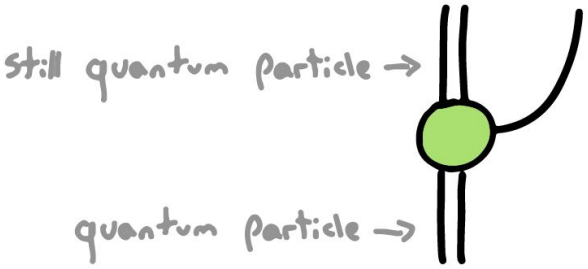
But then we fuse the red spiders, two wires appear, and the legs can be chopped.

4.5
Measurements disturb particles

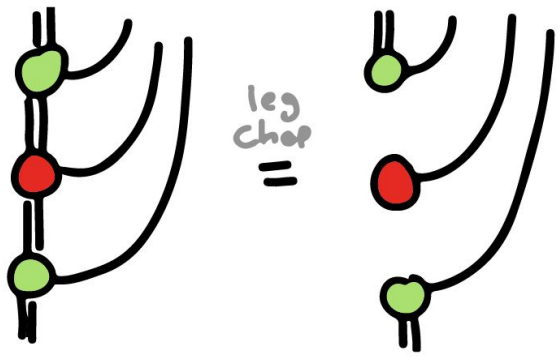
The measurements we've seen so far are such that after they have been done there is no quantum particle left:



It is however possible to do measurements where the quantum particle survives. In terms of spiders, they look like this:

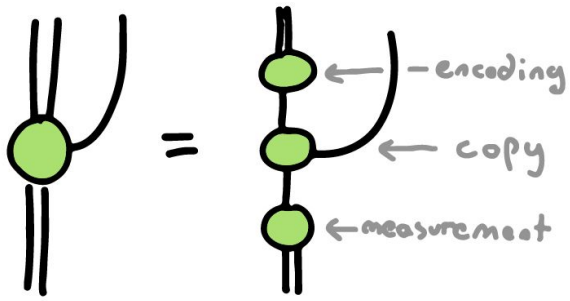


Now consider three such measurements of a quantum particle: first a green one, then a red one, and then again, a green one.

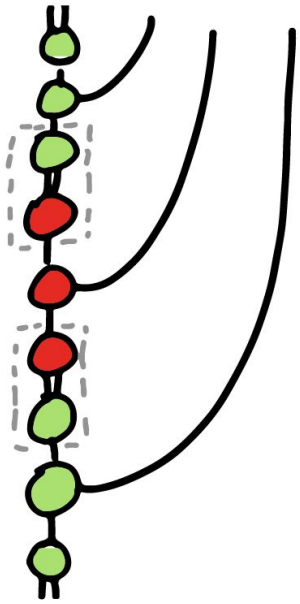


As we saw before, there are no wires connecting the spider at the input and the spider at the output.

This should—by now—no longer be a big surprise. A measurement where the quantum particle survives is—by means of some un-fusing—the same as first doing a measurement that destroys the particle, then doing an ordinary copy of the output, and finally encoding the output back into a quantum particle:



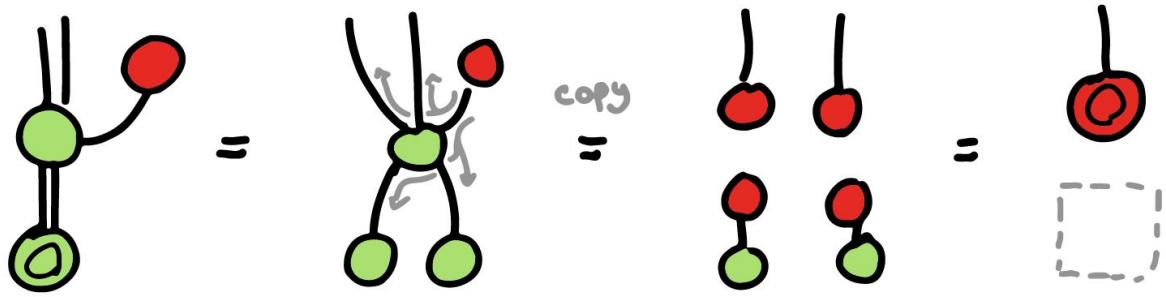
We can thus re-write the three measurements above as follows:



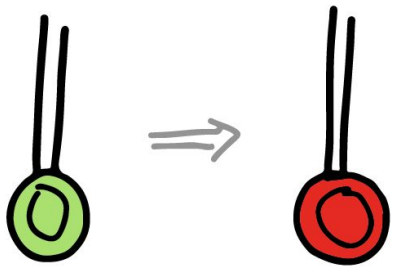
The disconnection happens in exactly the same way as detailed in the previous section.

In fact, it's not just the way we see the qubit that changes: the qubit itself has changed!

To see this, assume we do a green measurement on a green spider state, and we obtain a red quantum spider state as a result:

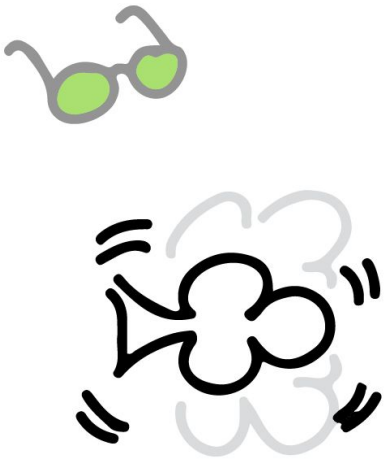


The two spider blobs floating around with no inputs or outputs are only there to tell us how likely we were to observe the 0 outcomes (50% of the time): we can ignore them for now and we will come back to them in the following chapters. We now see that the state of the qubit has changed:

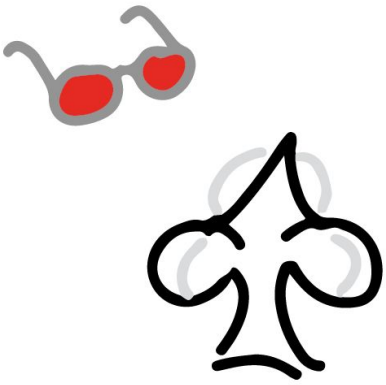


There is an important lesson to be learned here: quantum things change in a weird way when we look at them. To understand this, imagine that our quantum particle is a chocolate cake: green measurement means looking at the top of the chocolate cake to see whether there are sprinkles or not, while red measurement means looking at the sides to see whether there is frosting or not. If we first look at the top, then at the sides, then at the top again, what we see the second time on top may be entirely different from what we saw the first time, before looking at the frosting. The cake changed right before our eyes! How is this possible? Well, it may be that the only way to look at the sides was to take the cake out of its container and the sprinkles fell off. With a cake, we could try to be careful and avoid the sprinkles falling off.

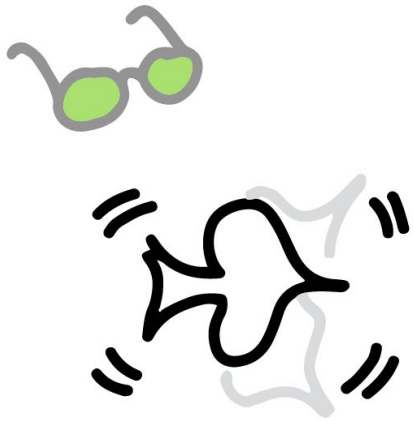
With quantum particles, on the other hand, there is nothing we can do to stop this. Imagine that our green glasses let us see whether a quantum particle is a club 🃑 or a spade 🂡 and that our red glasses let us see whether the same quantum particle is pointing up (🃑 or 🂡) or pointing down (🃍 or 🃝). We look at our quantum particle with our green glasses: it's a club.



Maybe pointing up, maybe pointing down: we can't quite see it with these glasses on. So, we put our red glasses on and look at our quantum particle again: it's definitely pointing up.



Great! We now know that our quantum particle must look like this: 🃑. We swap our red glasses out, casually put our green glasses on, and...



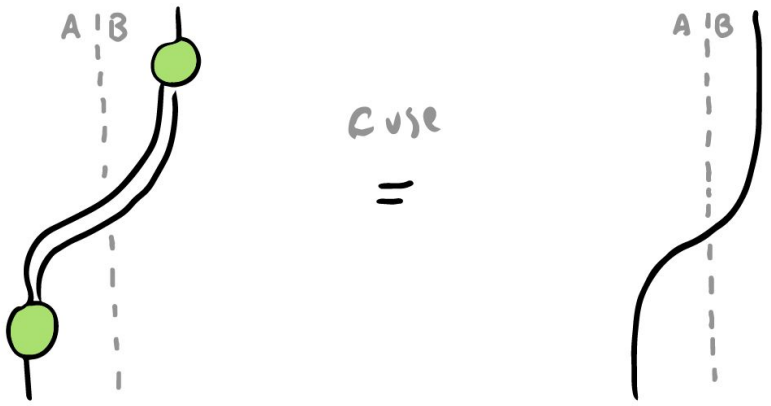
The particle is now a spade! What happened??? Well... Looking at our particle with the red glasses changed what the particle looks like with the green glasses, and vice versa. That’s quantum particles for you!

4.6
Inventing a secret language

Imagine that Andrea and Blaise are far away from each other but want to discuss their complicated love affair without anyone overhearing. They may want to use a secret language, but they forgot to agree on one before departing. They could agree on one over the phone, but Chi—who also has his eye on Andrea and is listening on the wire—would immediately learn it and snoop on their conversations.

Instead, Andrea and Blaise can let quantum particles make a secret language for them, in a way that lets them know whether Chi (or anyone else) is listening in on the conversation. In fact, Andrea and Blaise only need to share a secret string of bits: this string can, in turn, be used as a secret key to hide their messages, like your phones and computers do every day.

How can we use quantum particles to make such a secret string and make sure that Chi cannot learn it? We can do so by using a trick: initially it will look like a total waste of time, but it turns out to be extremely clever in the end. It starts simple: Andrea sends bits encoded as qubits, which Blaise then measures. However, Andrea randomly encodes each bit using either a green encoding or a red encoding. Similarly, Blaise randomly measures each qubit he receives using either a green measurement or a red measurement. If Andrea and Blaise chose the same colour, Blaise receives Andrea's bit:

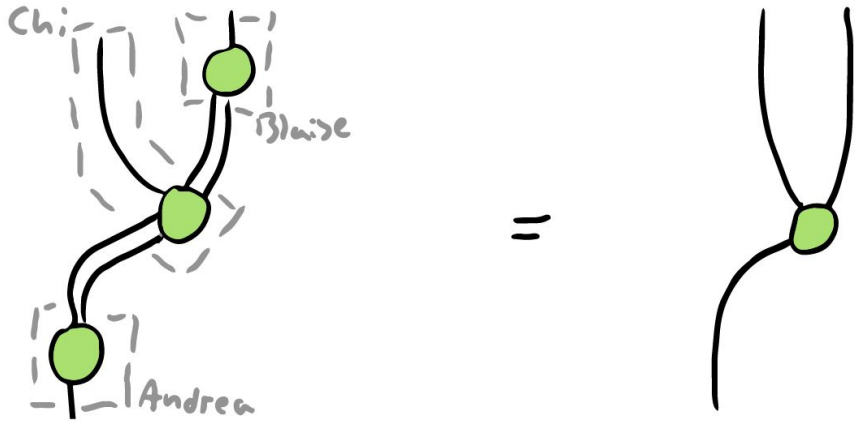


If Andrea and Blaise chose different colours, on the other hand, Andrea's bit gets discarded, and Blaise gets something completely unrelated:

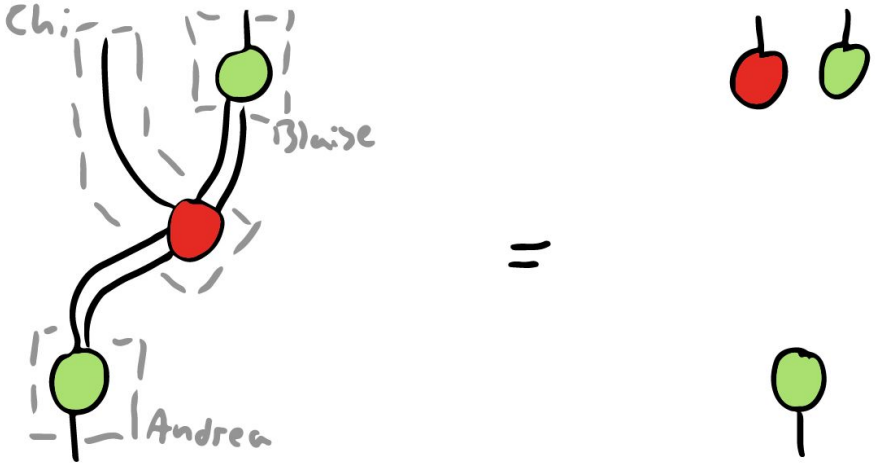


After they are done, Andrea and Blaise scream their choices of colour at each other, keep the bits for which the colours matched and throw away all the other ones.

Such a waste, right? Not at all! Doing this helps Andrea and Blaise learn whether Chi was snooping on the bits while they were on their way from one friend to the other. Indeed, if Chi wants to know what bit was sent, he must measure it. If he chooses the same colour as Andrea and Blaise, he will learn one bit in their secret string:



If the colours somehow mismatch, however, he ends up breaking the wire from Andrea to Blaise:



Because the wire is broken, the bit received by Blaise is now unrelated to the one sent by Andrea: 50% of the time it will be the same, 50% of the time it will be different. This is different from the situation where Chi doesn't snoop in: in that case, the bit received by Blaise is always the same as the one sent by Andrea. It is this difference that makes it possible for Andrea and Blaise to catch Chi in the act! After they have discarded the bits where they chose different colours, Andrea and Blaise also scream to each other some of the bits where they chose the same colour: if the bits match, Andrea and Blaise can be reasonably sure that Chi was not listening in on the quantum wire. They can then use the remaining bits as their secret and start talking in private.

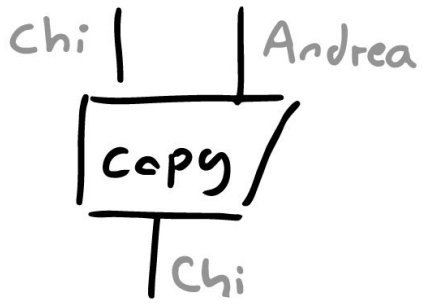
4.7
Quantum things cannot be copied

You may wonder why Chi used a measurement to listen in on Andrea and Blaise: could he not simply copy the message? This is exactly what you do when you have a nice picture on your laptop or tablet, and you send it to one of your friends: you send them an exact copy and keep the other copy to yourself. Shockingly, this is not something that you can do if your picture is quantum! You cannot make a copy of quantum particles, whether they are bits or pictures or love stories.

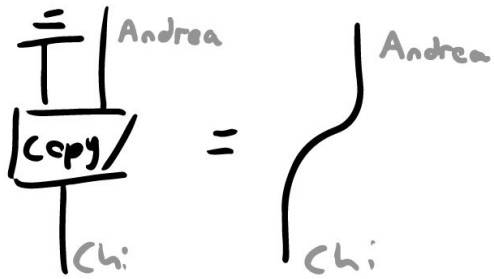
This is very annoying in many ways: you cannot make a backup of qubits, you cannot copy and share qubits, you cannot re-send qubits if they get lost in the post. However, it is also good in other ways, for example if you have something personal that you don't want other people to copy or share without your knowledge, just like Andrea's and Blaise's love letters to each other.

Quantum copy box

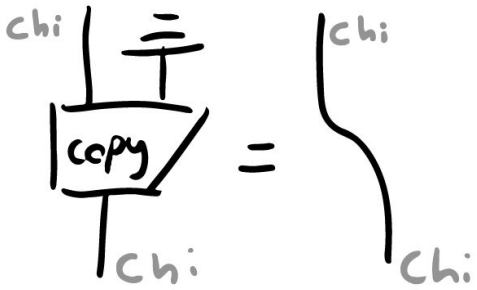
What would making a copy of a quantum particle look like in diagrams? Suppose that sneaky Chi has obtained the questions of the next exam in advance and wishes to send them to Andrea. All he needs to do is make a copy of the exam, and send it to her:



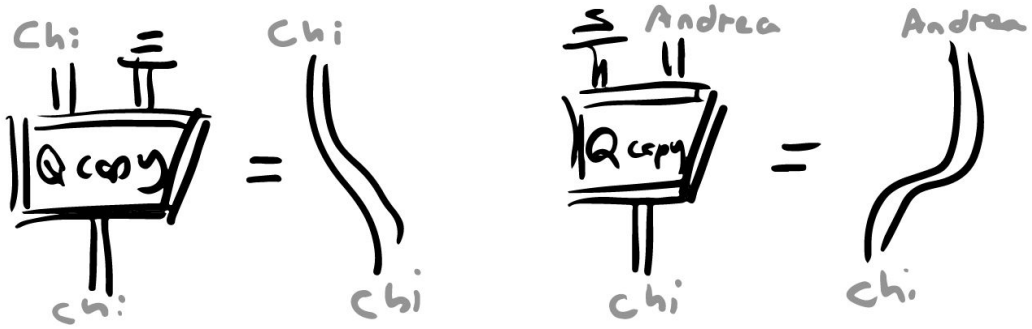
This is what it means for Andrea's copy to match Chi's original:



Above, we discarded Chi's copy: Andrea never gets it and doesn't really know what it contains. For all she knows, Chi could have sent her the only copy he had! This is what it means for Chi to retain his own copy of the original exam questions:



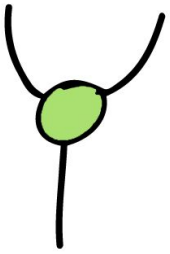
From the discussion above, we conclude that, were a quantum copy box to exist, it would at least do the following:



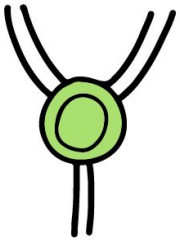
We will now see that such a box cannot exist: quantum stuff cannot be copied!

There exists no quantum copy box

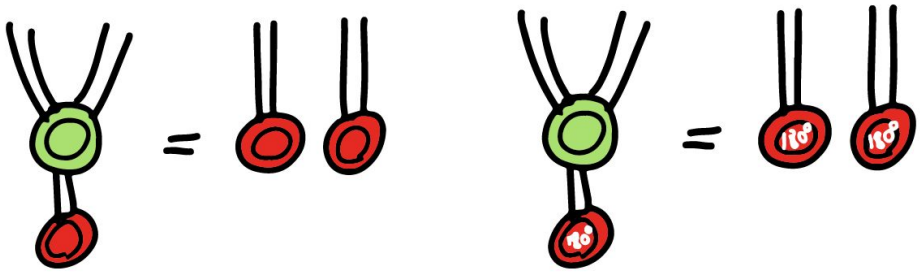
Wait a minute! We already have a box that copies bits:



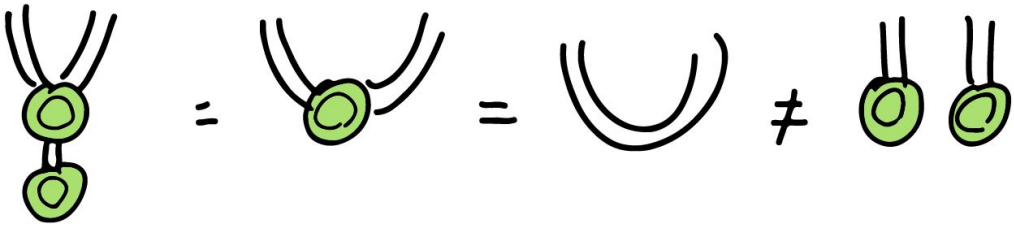
If we double it, don't we get a quantum copy box, like this?



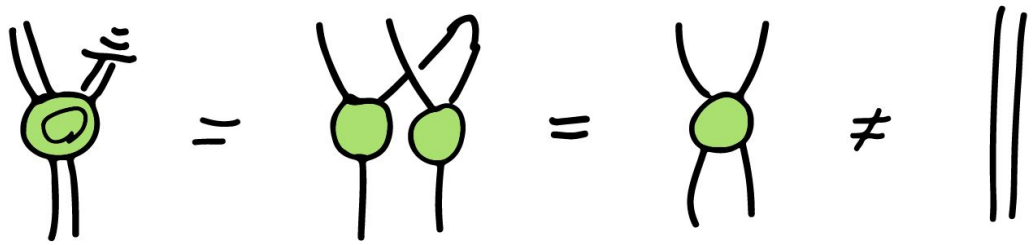
Well... no, we don't. The box above does indeed copy the bits encoded into qubits:



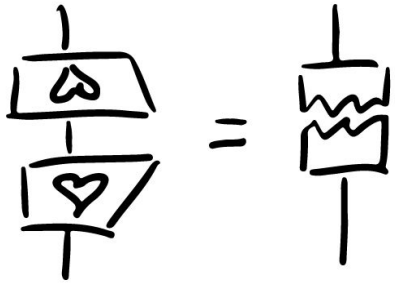
However, it does not copy all quantum states. For example, it doesn't copy the green spider states:



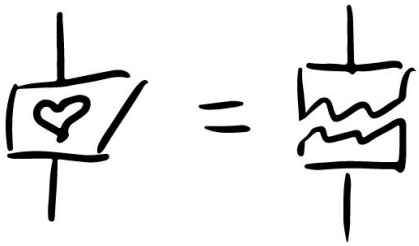
In fact, the spider above does not satisfy the equations we expect a quantum copy box to satisfy:



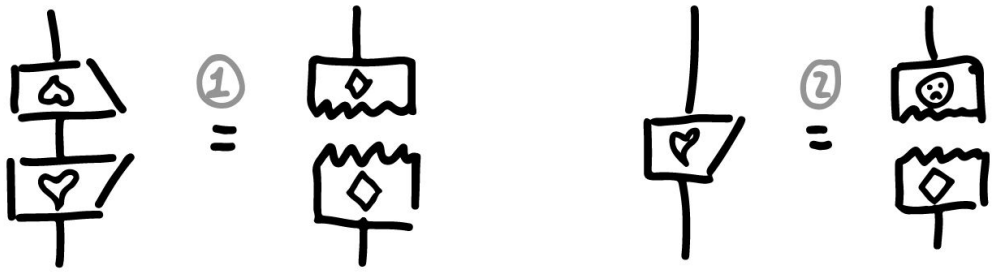
OK. Doubling the copy box for bits doesn't work. How do we know no quantum box will ever work? Suppose that when we connect a box to its flipped counterpart we get a disconnected box, which we indicate by a crack as follows:



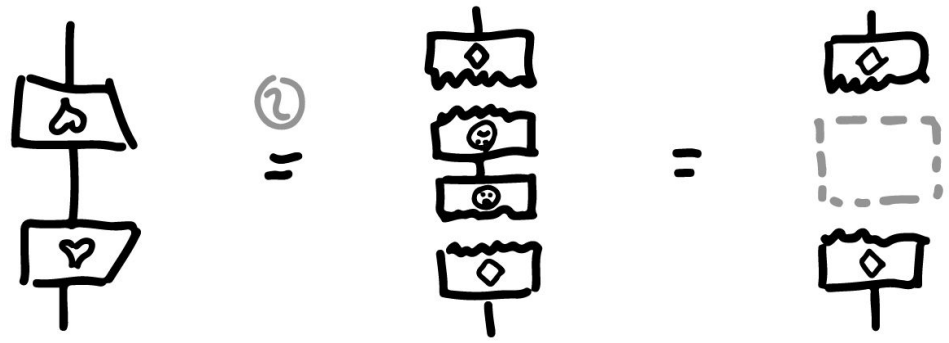
A wire has no cracks, and the two boxes are perfect reflections of each other, so the crack must actually come from the original box:



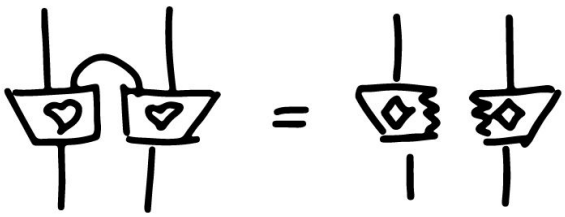
Moreover, we can see that the same cracked bits ◇ appear in both the original box and the box connected to its flipped counterpart:



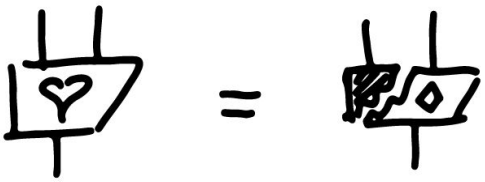
If we connect [♡] from the second equation above to its flipped counterpart [♤] we get the first equation above:



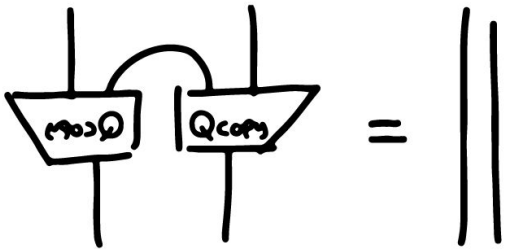
A similar reasoning holds for quantum boxes. If we have a crack on the right:



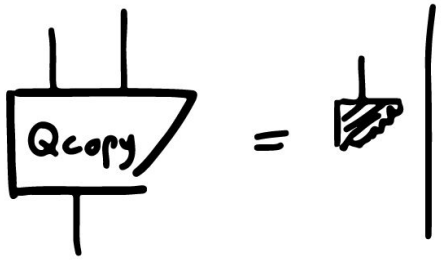
Then the box [♡] must have had a crack in the first place:



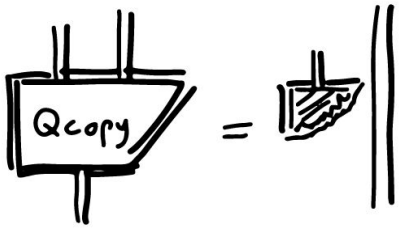
Since quantum discarding is a cap, we can now write the first of the equations of a quantum copy box as follows:



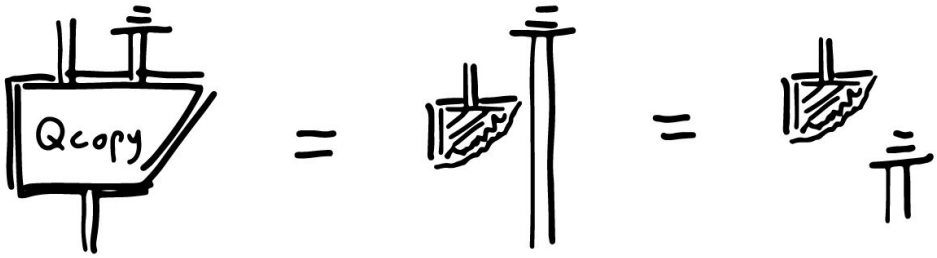
There is clearly a crack on the right, so there must have been one in our box on the left:



If we double this, we get the following picture:



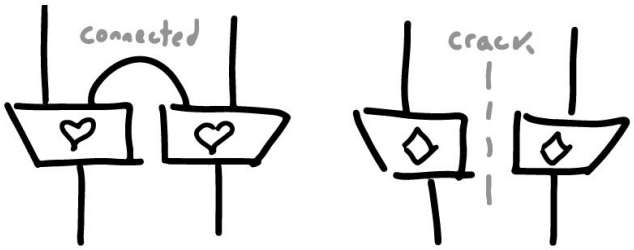
Discarding Andrea's output, we see that Chi always receives something unrelated to what he put in:



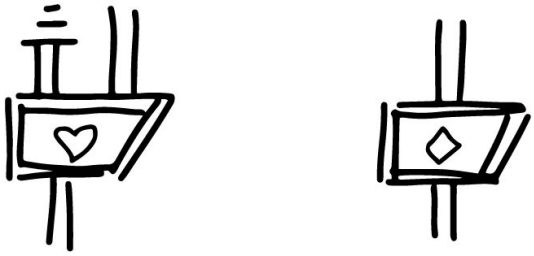
If he wants to send a leaked quantum exam to Andrea, he cannot keep a copy to himself! This shows that one of the equations required by a quantum copy box can never be satisfied, no matter what box we use in the process. Quantum copy boxes don't exist.

Leaky vs. non-leaky boxes

We have so far encountered two kinds of quantum boxes:

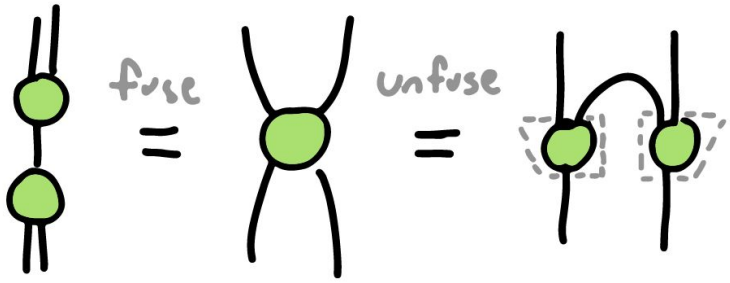


or, written with doubled wires:

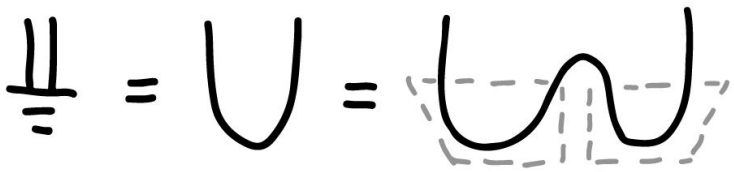


We can think of the first kind as "leaking" some quantum particles at the point where discarding takes place, so we call them leaky. The other boxes are called non-leaky.

We already saw some examples of leaky boxes. Discarding is itself a leaky box, of course, but so is phase-erasure:



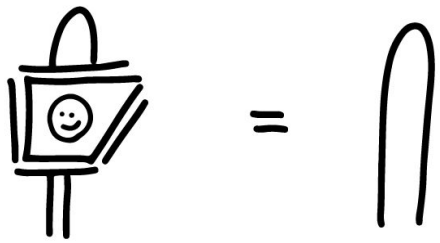
Quantum noise is also a leaky box, since we can re-write it as follows:



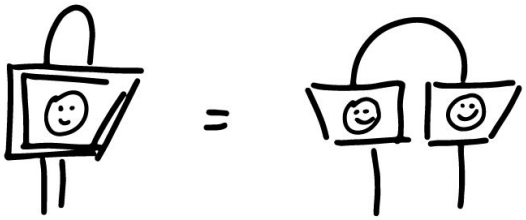
All non-leaky sure-boxes are flip-undoable

We have already seen that leaky boxes earn their name: some quantum stuff is lost forever when we apply them. But what about non-leaky boxes? Are they truly not leaking any quantumness around? We now show that this is indeed the case: non-leaky sure-boxes can be undone, so nothing can get lost in them!

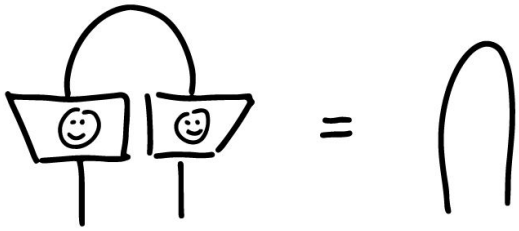
This is what it means to be a quantum sure-box:



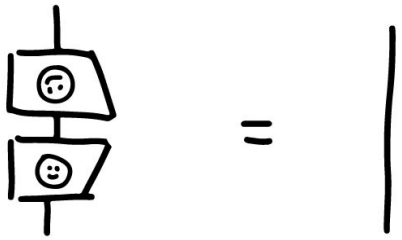
For a non-leaky quantum box, the left diagram is also:



Putting the two together we get:



Bending the wires on the left we then get:

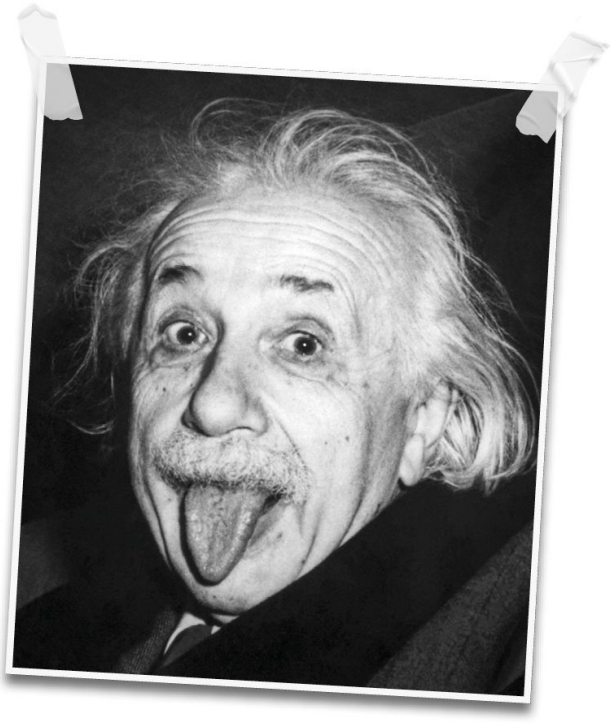


This is exactly what a flip-undoable box is! The non-leaky quantum sure-boxes are exactly the flip-undoable quantum sure-boxes.

5.
EVERYTHING JUST FROM PICTURES!

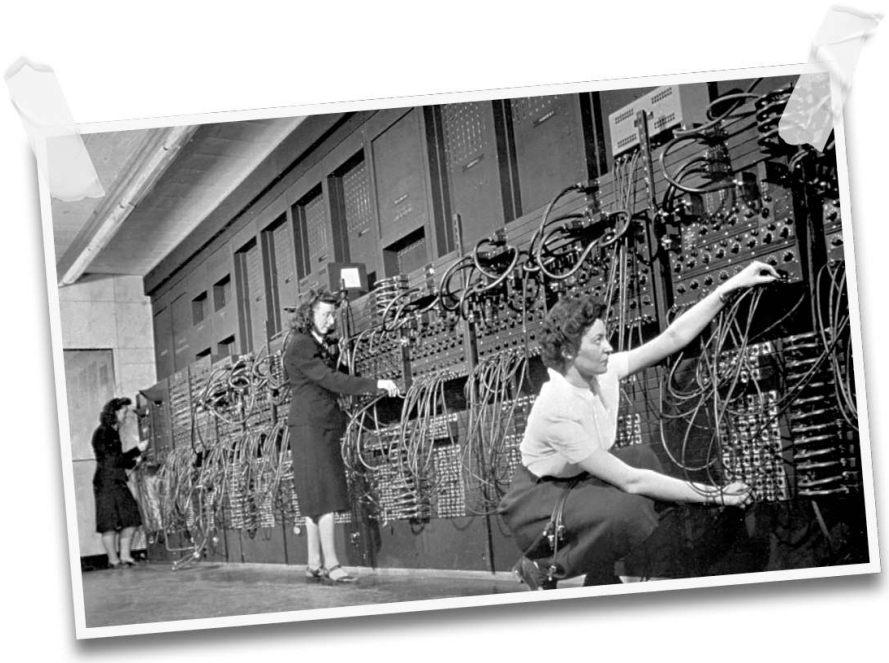
We've learned lots of stuff about quantum theory just by playing around with pictures of spiders. Specifically, we derived lots of diagram equations just by doing things like spider fusion and leg-chopping. How much of quantum theory can we learn in this way? As it turns out, all of it! We just need to understand spiders a bit better in order to get there.

What this will let us prove the most striking fact about the quantum world: quantum particles are non-local. This confused Einstein till the end of his days and physicists are still pulling their hair out about this.



The other thing that this will allow us to do is to make quantum computers faster and

smaller. Currently, quantum computers are huge machines, and running a quantum box is difficult and expensive. We are pretty much in the same situation as when running a computer program looked something like this:



If we can get rid of some of the quantum boxes in our circuits, it is a very big win. Quantum computer engineers doing this today use the spiders you learned about in this book: you'll be learning a bit about the state-of-the-art!

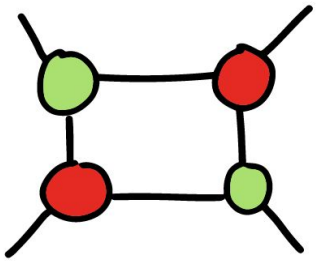
With spiders in them, quantum computers will tackle certain problems much faster than classical computers. For example, quantum computers will be good at understanding the behaviour of complex molecules, helping us find new cures for diseases, discover new materials and figure out new ways to help our planet. They may also be used to improve the way we manufacture complex objects—such as planes and skyscrapers—and to optimise transport and delivery times on intricate routes. Bob, who is also a musician, has already been making music with quantum computers. In the coming years, quantum computers will be able to solve, in mere seconds, problems that would take thousands of years on even the most powerful of classical computers. Aren't spiders great?!

5.1
Square-popping

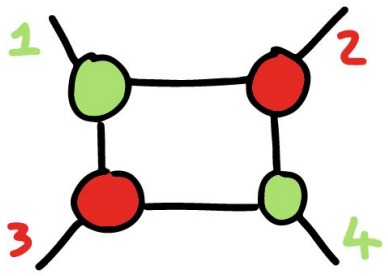
One thing that spiders do and that we haven't yet seen is "square-popping". We have seen that if two spiders of different colours share two legs:



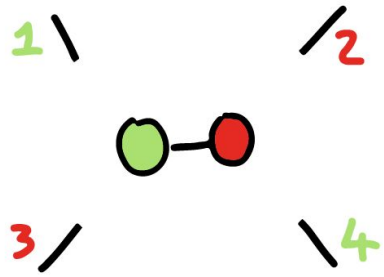
the two legs get chopped. Suppose instead that we have four spiders forming a square, like this:



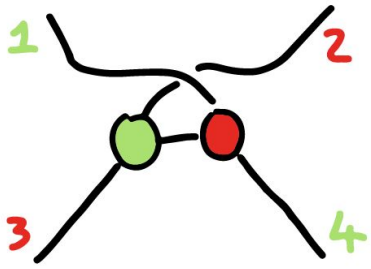
We can't use leg-chopping, because any two spiders of different colours share only one leg, and we cannot use spider fusion, because no two spiders of the same colour share any legs. But we can do something else, which makes the four spiders become two in a somewhat bizarre manner. To see how this works, we label each of the four outgoing wires using the colour of the spider it is attached to:



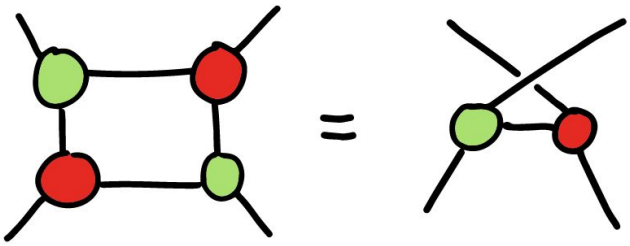
Then we cut the square and replace it with two spiders sharing a leg:



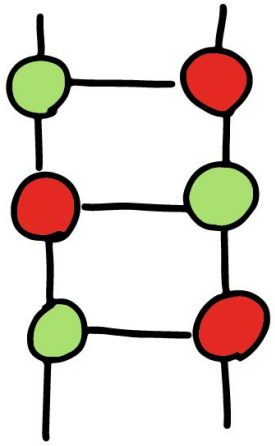
We then reconnect the legs, but to the spider of the opposite colour from the label:



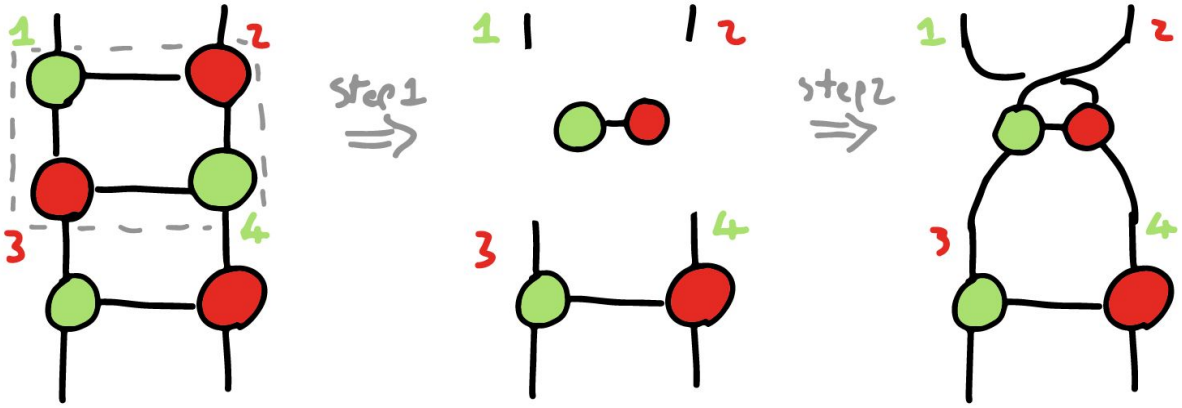
All in all, we get the following equation, called square-popping:



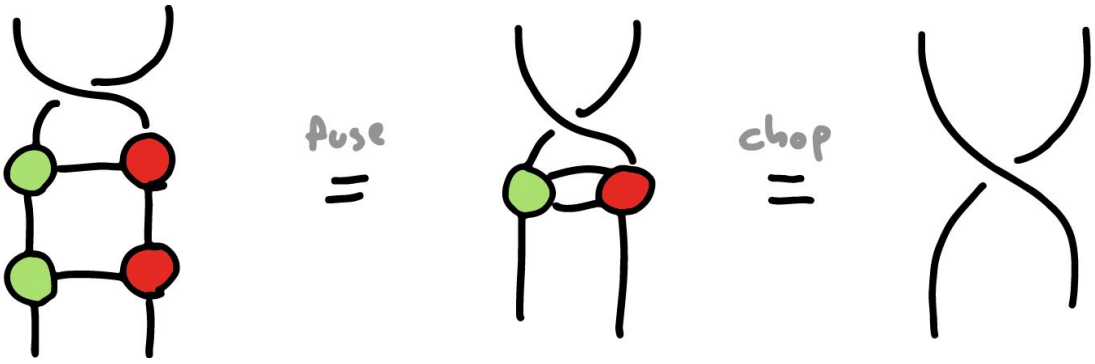
We can now do things we couldn't do before. For example, we can further simplify this circuit:



Indeed, we see four spiders that form a square, so we can use square-popping:



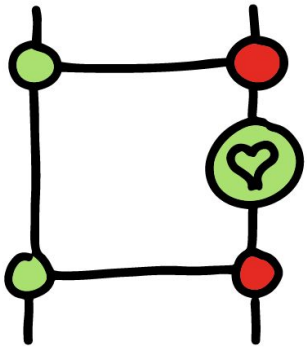
We now find two CNOT gates, which can be made to disappear by spider fusion and leg-chopping:



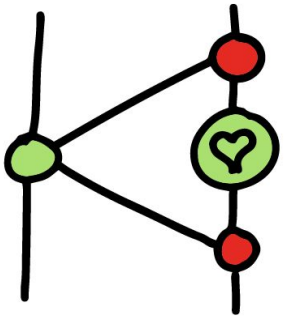
We learned something new: applying three CNOTs of alternating colours is the same as swapping the two wires.

A square doesn’t always look like a square!

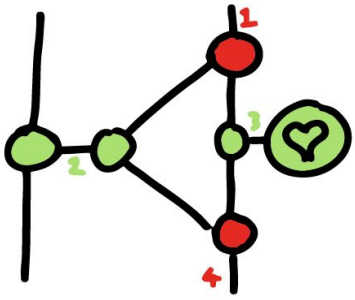
Sometimes a square is a bit camouflaged, like many animals in the wild. This circuit is a very good example of this:



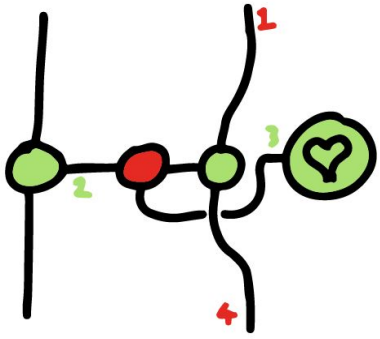
Doesn’t look like a square, right? But it is! To see this, first fuse the two green spiders on the left:



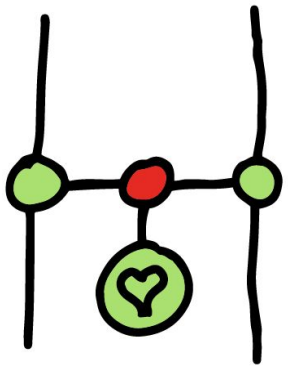
and then un-fuse a bit:



We got ourselves a square! Now we can use square popping:



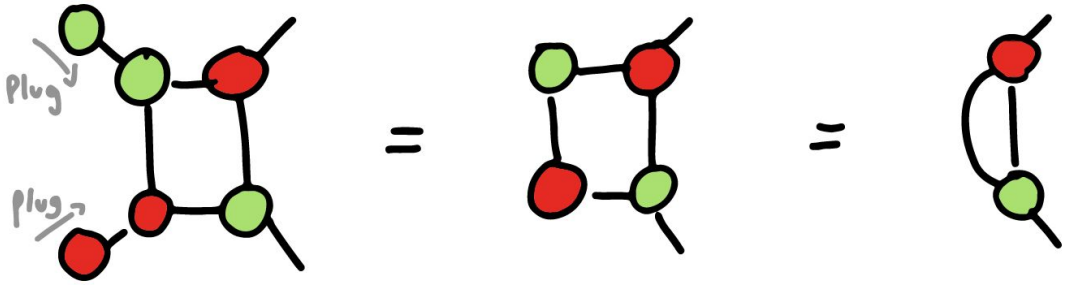
and we get this funky box:



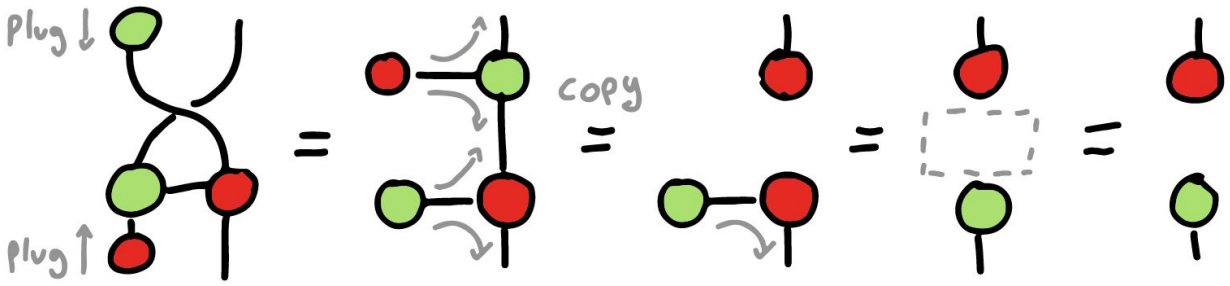
This box is called a phase-gadget and it is much easier to play with than the circuit we started with.

Square-popping does leg-chopping too!

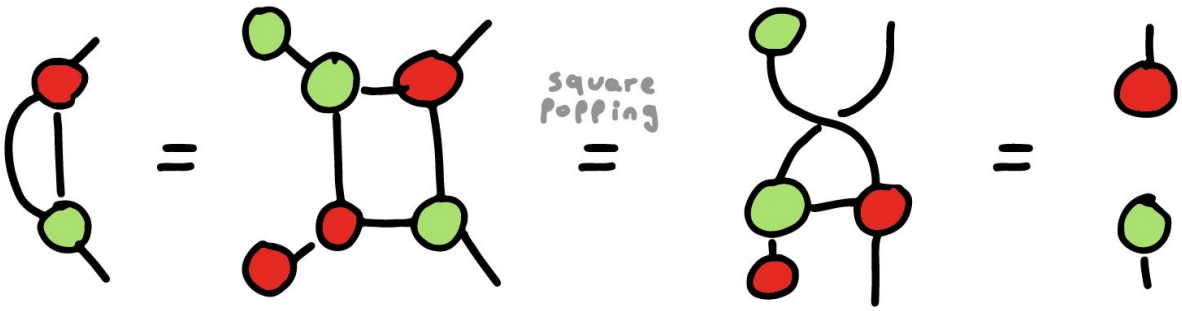
The bizarre square-popping behaviour of spiders may not seem related to leg-chopping in any way, but it is. To see why this is the case, we start by turning a spider square into two spiders sharing two legs, as on the left of the leg-chopping equation:



We now take the same spider square, apply square-popping to it and turn it into two spiders sharing no legs, as on the right of the leg-chopping equation:



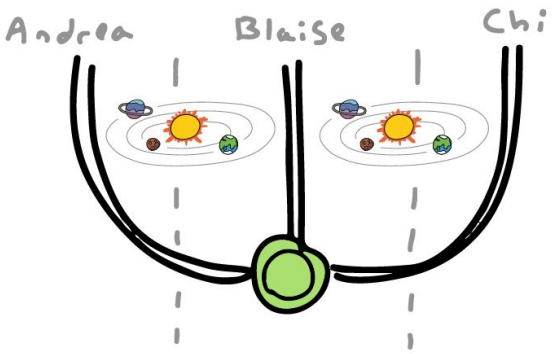
Because in both cases we started with the same diagram on the left, the diagrams on the right are equal, and we conclude that square-popping can be used to do leg-chopping:



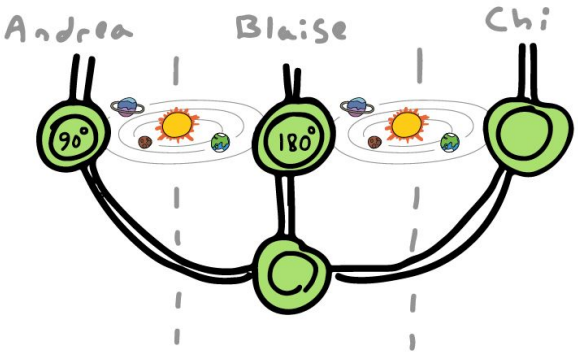
Neat, huh!

5.2
Quantum non-locality

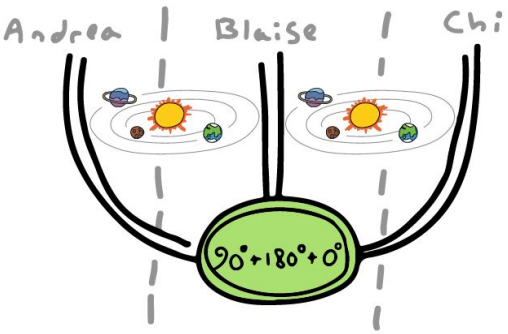
Imagine that Andrea, Blaise and Chi are all separated by solar systems, and they are sharing a spider state:



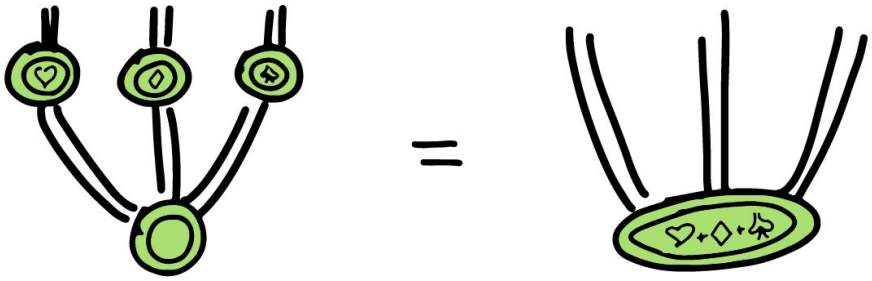
They each apply a green phase gate of their choice:



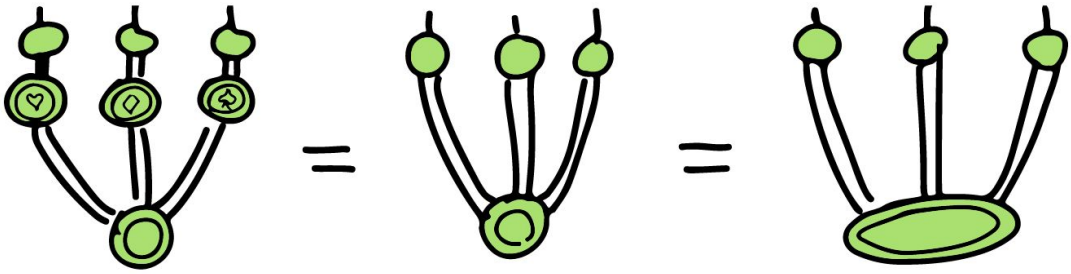
and something strange happens:



It is as if the phase gates “slid back in time” and fused together in the far past! Somehow, the state of the particle of each of them depends on the choices made by their friends completely independently and extremely far away. There is nothing special about the phases above, any other phases would have worked the same:



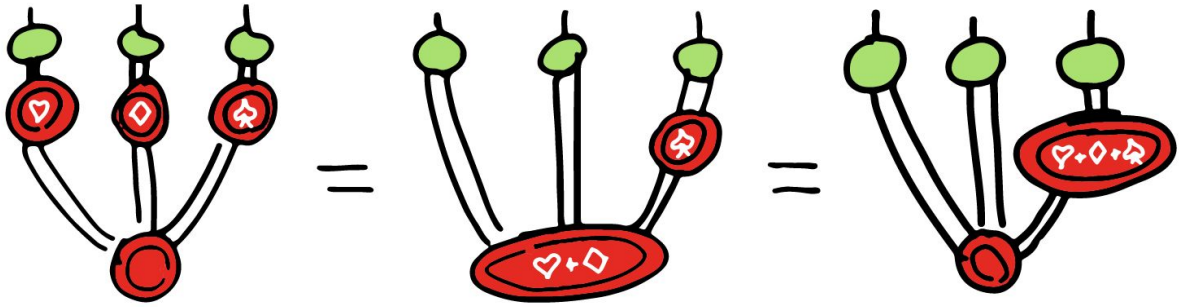
Something weird has surely happened, but how can the three friends know that? They could try measuring their qubits:



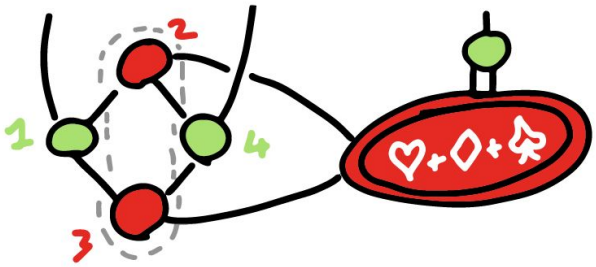
Oh dear! The phases vanished and the three friends learned nothing at all! There was magic, and now it’s gone.

What quantum theory tells us

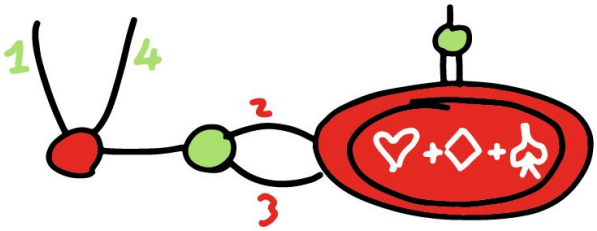
Can we fix this? Yes, we can. We can take phases of a different colour from the colour of the measurement:



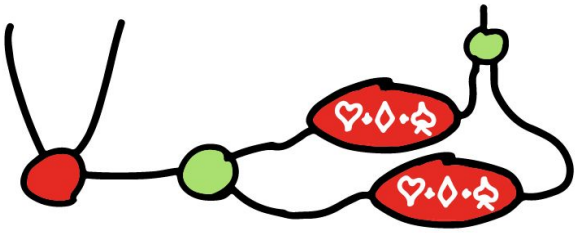
Why did we move all phases to the right? So that we can unleash the power of square-popping to simplify this diagram. But where is the spider square hiding? Can you see it? There it is:



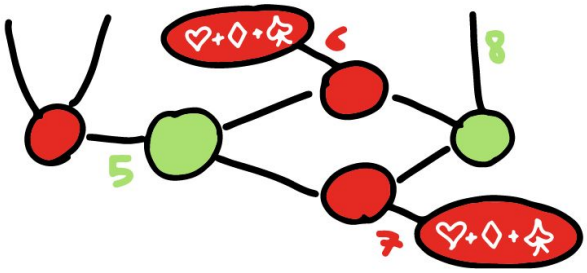
By square-popping we obtain:



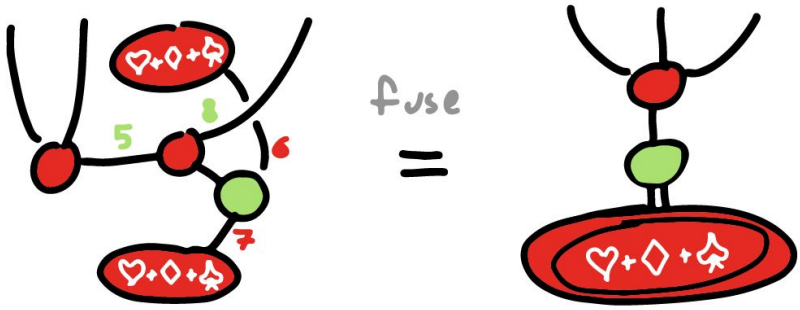
If we open up the doubled spider a bit:



and we apply some spider fusion, we see another square:



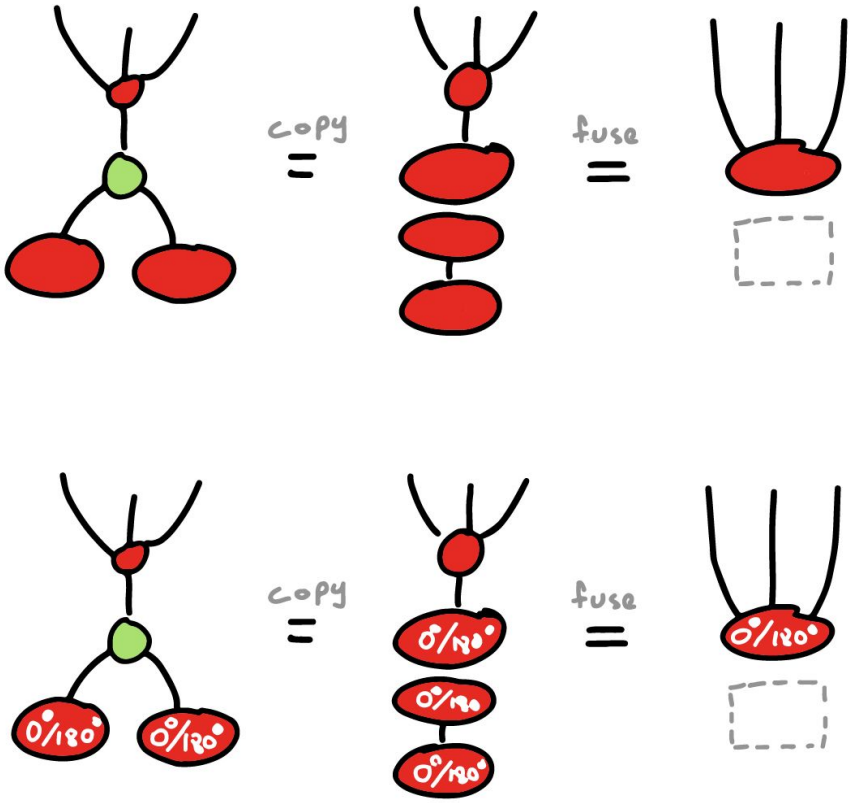
By square-popping again we obtain:



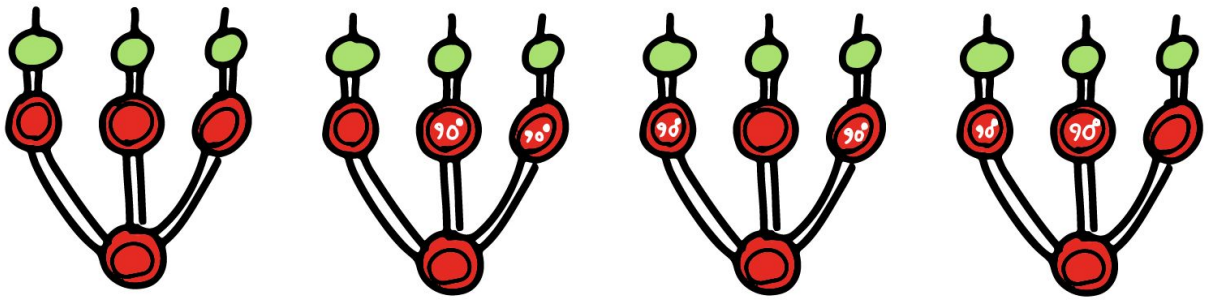
But wait, shouldn't we have changed one of the phases above into 360° minus that phase? Not if our angles add up to either 0° or 180°:



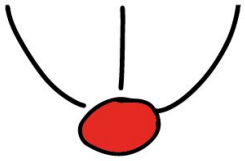
In that case, everything becomes even simpler! This is because red 0° and 180° spider states are copied by green spiders:



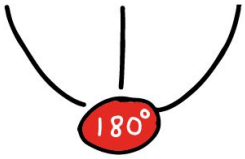
That's pretty simple. But, most importantly, the sum of the phases is still there, and the three friends can now observe it! Here are four examples of phases that Andrea, Blaise and Chi could choose to get the desired 0° or 180° sums:



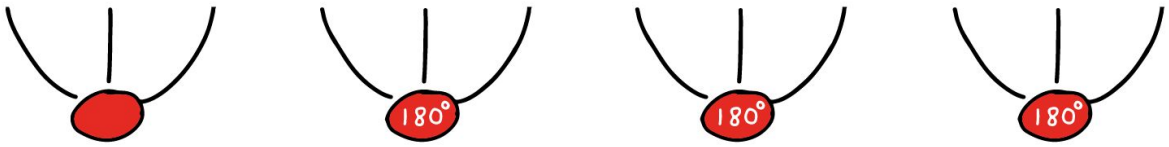
The first example gives:



The other three examples all give:



Putting all four together, this is what we get:



What if quantum particles were like cakes?

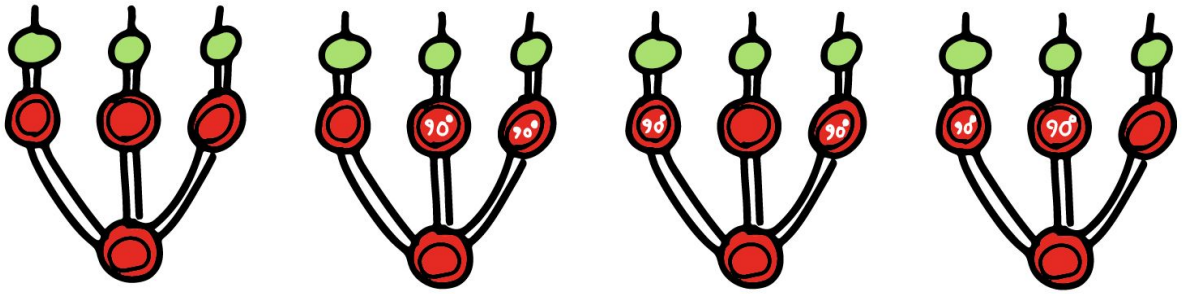
To understand what the three friends might observe if quantum particles were actually like cakes, let's imagine that the following phase gate means adding sprinkles to a cake:



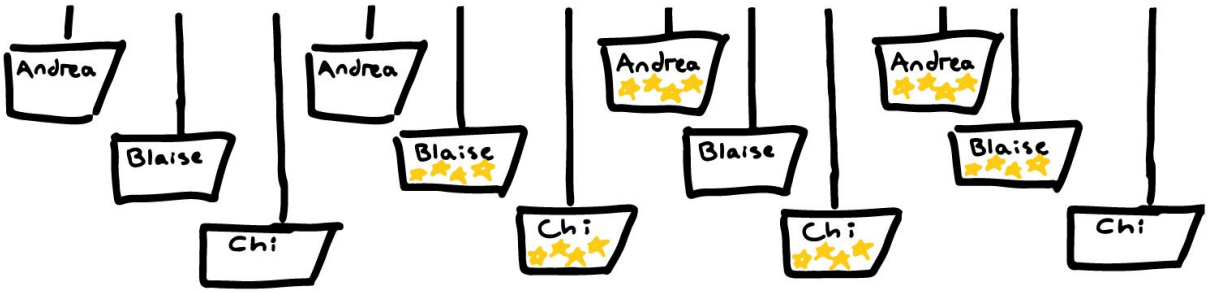
The following two "sprinkle states" will stand for Andrea seeing nothing on top of her cake (state on the left) or instead seeing sprinkles on top of her cake (state on the right):



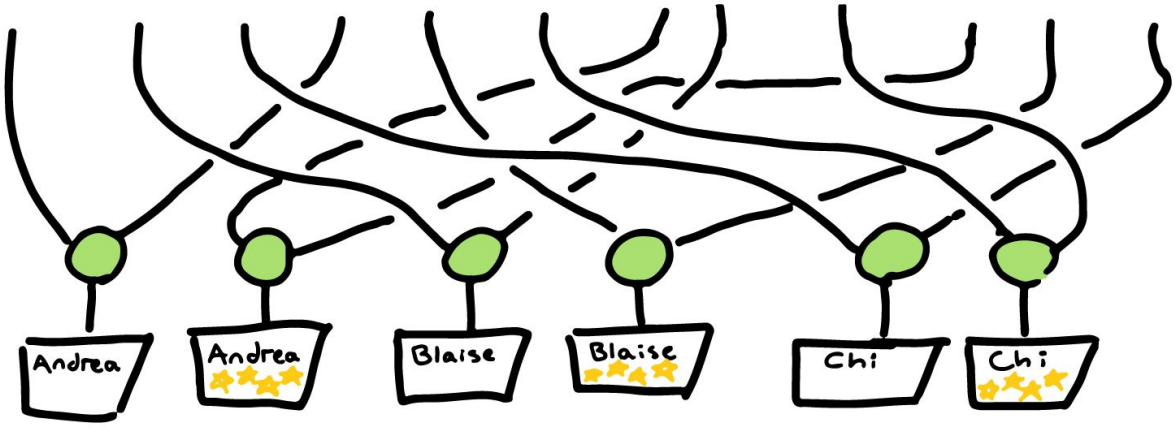
Going back to the four scenarios for Andrea, Blaise and Chi:



and interpreting the 90° phase gate as putting sprinkles on a cake, this is what we get:



We can also write the states above as follows, where each sprinkle state is written once and copied each time we need it:

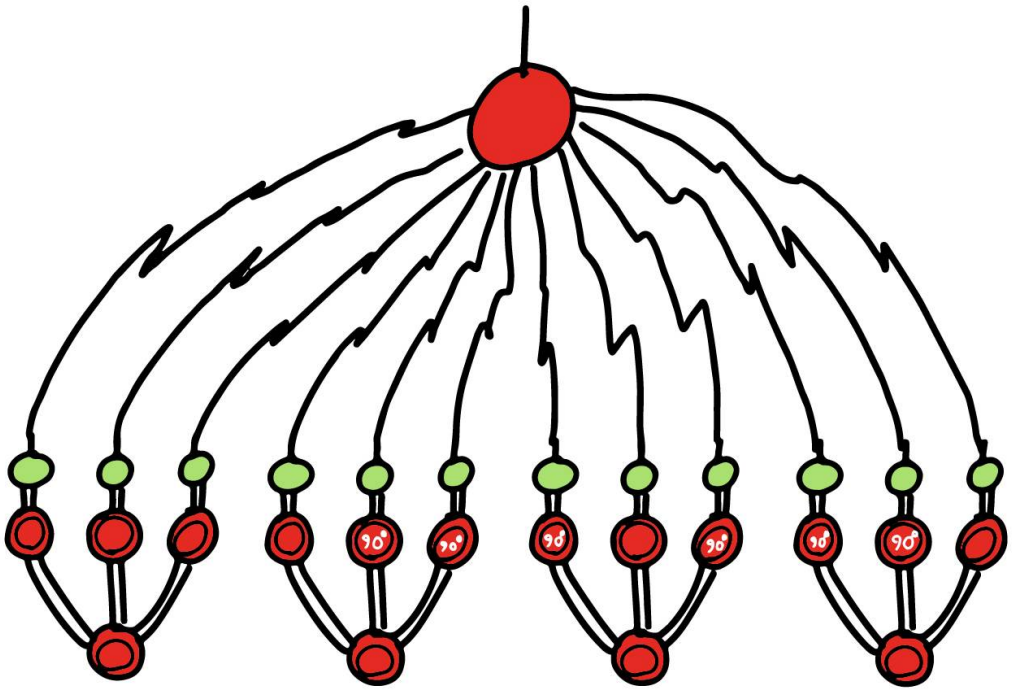


This probably looks like a big pile of spaghetti, but the idea behind it is really simple: just follow the wires and you'll know whether each friend sees sprinkles or not.

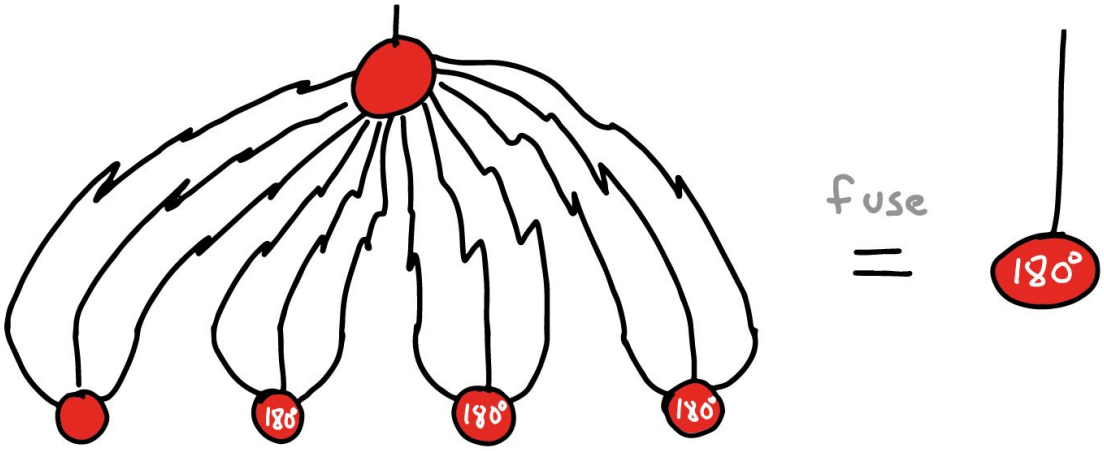
Big spiders and lightning strikes

OK, so we've simplified our picture of the four scenarios twice: once using quantum spiders, to see what quantum theory tells us, and once with sprinkle states, to see what we would expect our friends to observe if quantum particles were actually like cakes.

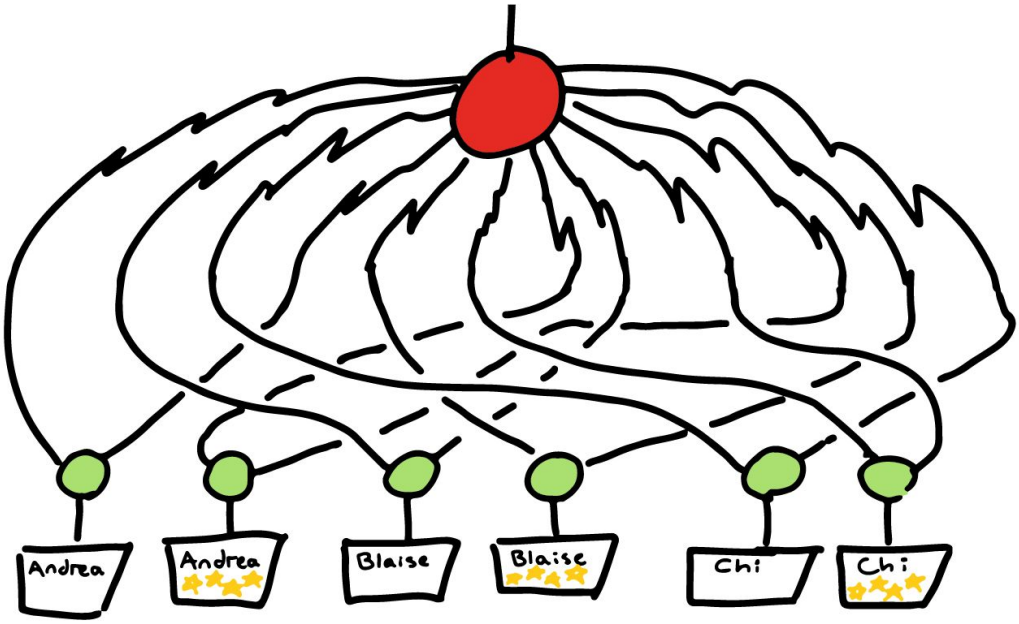
To see whether the two pictures are the same, we apply a big red spider to the top of our four scenarios:



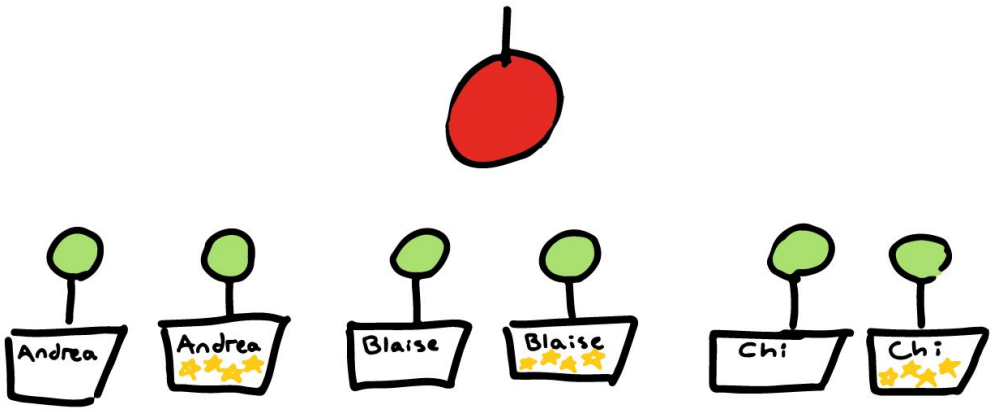
In our quantum world picture we get:



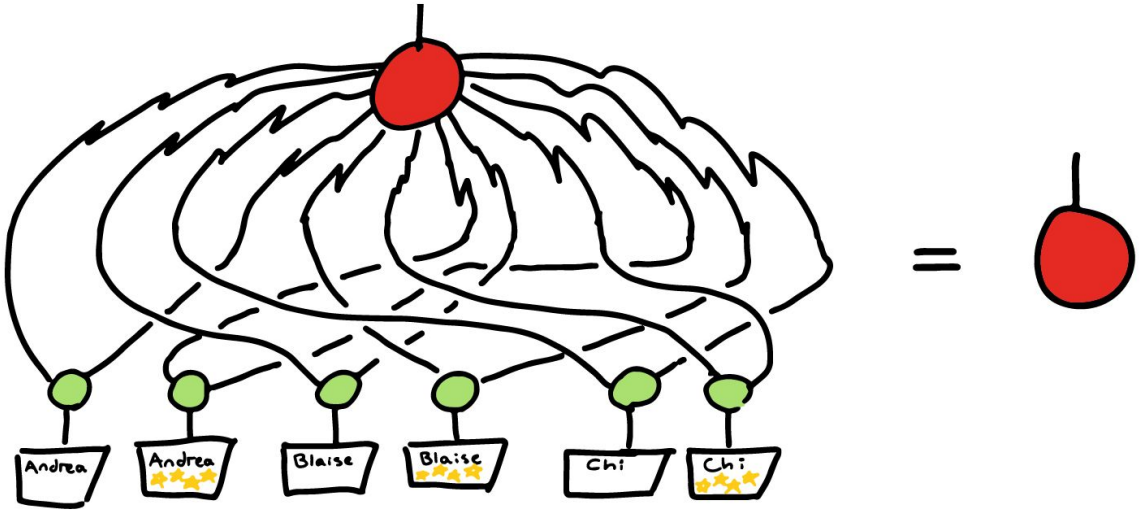
In our cakes picture we instead get:



We see two wires between each green spider and the big red spider, so they all fall off by leg-chopping:



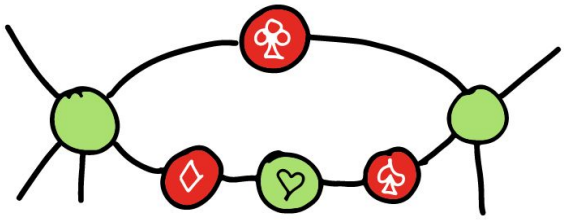
The sprinkle states are then discarded using the green spider tests, so the cake picture leads us to:



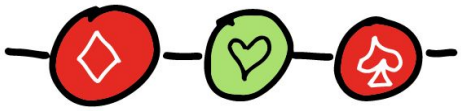
BOOOOM! Big lightning strikes! This is not the same thing we got from our quantum world picture: quantum particles give us a 1 bit, while cakes give us a 0 bit. Sadly, quantum particles are nothing like cakes!

5.3
The last of the spider rules

To really get everything out of pictures and make the most of them, we need to understand one more thing about spiders. Suppose you have three phase gates of three colours stuck between two spiders:



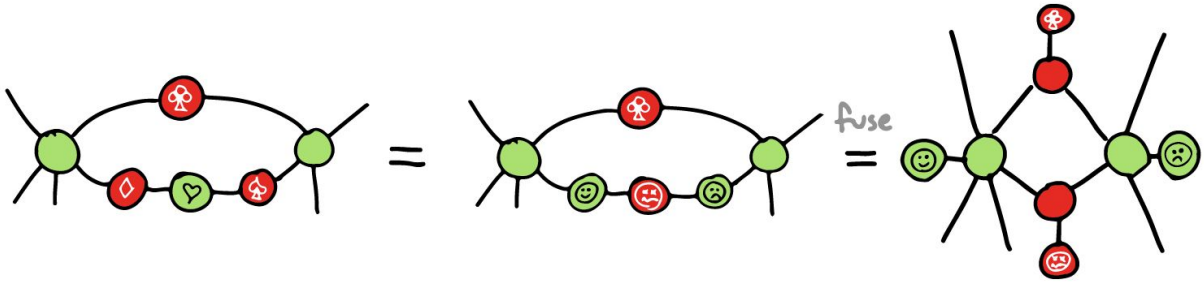
Neither leg-chopping nor square-popping can help us here... But it isn't over yet... The last of the spider rules is here to save us! It states that if we have three phase gates of alternating colours:



we can always swap the colours by changing the three original phases to three other phases (usually different from the original ones):



Using this rule, we get ourselves a square to pop!



This is just one example of the power of this new rule. Now that we know it, all the equations of quantum theory are in our power. That means that all the ugly, incomprehensible things that physicists write:

$$\begin{pmatrix} 1+i & e^{-i\frac{\pi}{4}} & 2 \\ e^{i\frac{\pi}{4}} & \gamma & \sin\beta \\ x & \cos\beta & 1-i \end{pmatrix}$$

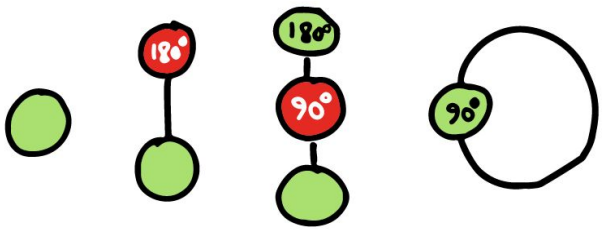
can be obtained by only using spiders. It also means that spiders can be used to make all kinds of cheap quantum circuits, saving time and money for everyone who builds and uses quantum computers.

6.
PROBABILITIES FROM PICTURES

If you open any ordinary book on quantum theory and try to understand something from it, you'd better be studying mathematics or physics at university. Assuming you do, you'll notice that it immediately talks about probabilities. Probabilities are already hard to understand in everyday life: building a new picture of the world based on them is even harder. This is why we haven't mentioned probabilities at all in this book. That is, until now...

But not to worry! The probabilities we need were already present in the pictures we have been playing with. We don't need to add anything new; we just need to know how to spot them. That's going to be easy.

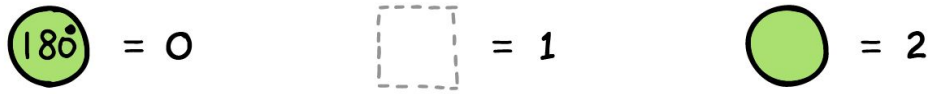
Probabilities are a kind of number. In the world of diagrams, numbers are diagrams with no inputs or outputs, like these:



Now, you probably think of numbers as things like these:

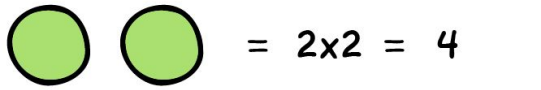
1 2 3 4 5 . . .

So, we need to figure out which diagrams correspond to those numbers. Here are a few examples:



(There are some very good reasons for this correspondence, but we don't need to go into that.)

When we have several such number diagrams together, the corresponding numbers are multiplied together. This lets us come up with new numbers:

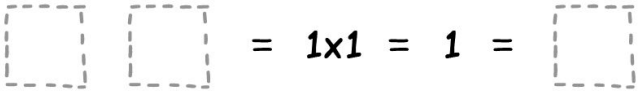




It also lets us discover alternative diagrams for numbers we already knew:



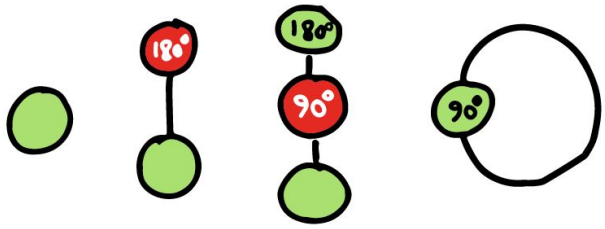
This also shows why the number 1 corresponding to the empty diagram makes so much sense:



The paper is mostly empty, so there are lots of 1's flying all around the place, but they don't change our number diagrams at all.

In the previous chapters we learned that we could use spider fusion and leg-chopping to

figure out when two diagrams—and in particular two numbers—are equal. The same tricks can be used to find new diagrams for the same number:



Another way to get numbers is to discard the output of a state:



This number is the “probability” to get that state and is always 1 for a sure-state. But wait a minute... What does “probability” actually mean? After flipping a coin, there are two possible maybe-states:



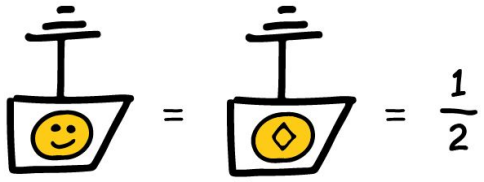
After throwing a die, there are six possible maybe-states:



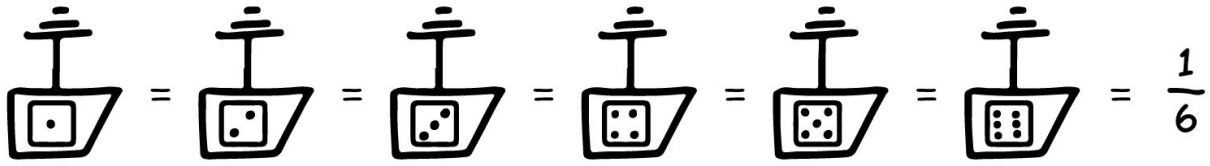
After playing on the lottery, there are again two possible maybe-states:



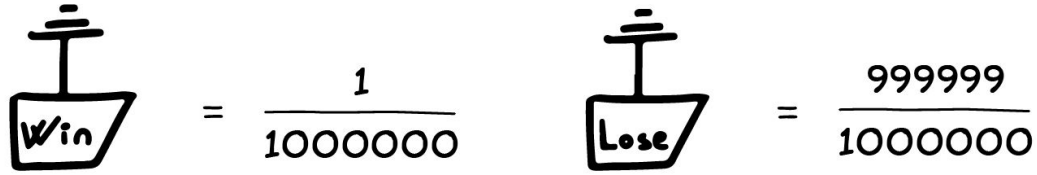
If we flip a coin many times, we will get roughly the same number of heads as tails in the long run. This means that the probability of getting heads and the probability of getting tails is the same, namely 1-in-2:



If we roll a die many times, we will get roughly the same number of each of the six faces. This means that the probability of each face is the same, namely 1-in-6:



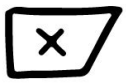
When playing the lottery, on the other hand, the two outcomes don't have the same probability at all! Winning the lottery is less like 1-in-2 and more like 1-in-a-million:



That's probabilities for you.

6.1
The mathematics of probabilities

Numbers are diagrams with no inputs or outputs:



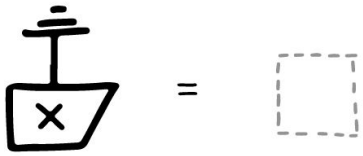
The following two examples are important:



Any state has a probability, obtained by discarding its output:

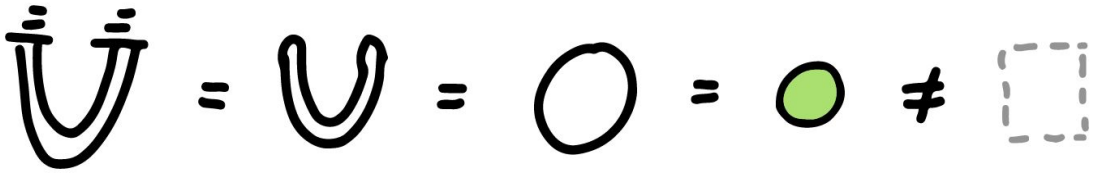


Sure-states always have probability 1 (the same as 1-in-1):



6.2
No more cheating!

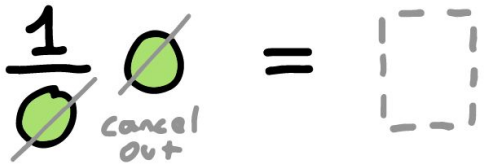
So far, if a diagram without input or output showed up in our pictures, we simply ignored it. We were cheating a bit, ignoring some probabilities for the sake of simplicity and using all states as if they were sure-states. But no more cheating! For example, quantum cups—as we have written them so far—were not really sure-states:



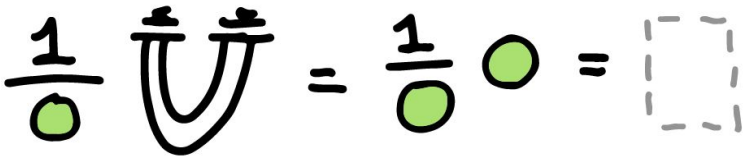
Oops! We can fix this by attaching a number to the state:



If the green dot is the number 2, 1 over the green dot is the number 1/2. All it does is cancel a green dot when it encounters one:



With that number attached to it, our cup state is now a sure-state:



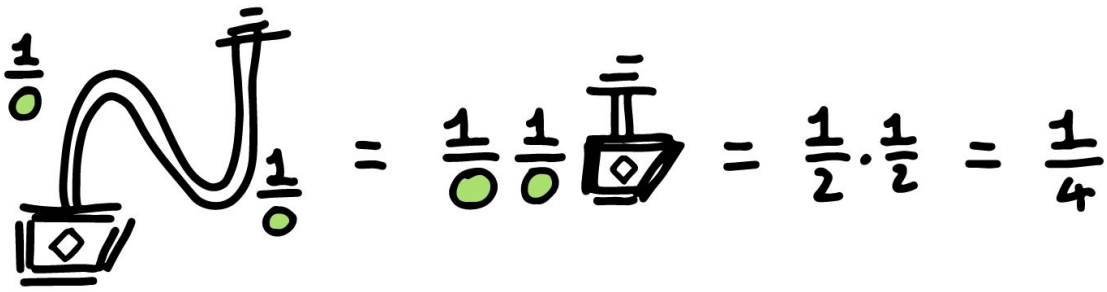
The quantum cap test changes similarly, and now has the same number attached to it:



OK. Time to compute some probabilities with diagrams!

6.3
Quantum teleporting, Part IV

We know that teleportation with the cap test happens if we are “lucky”. What’s the probability of being lucky? To figure it out, we start with a sure-state ◇ at Andrea’s end, teleport it with our quantum cap test and discard it to find the probability that the state arrived at Blaise’s end:



There it is: the probability of being lucky in teleportation is 1-in-4. Good to know!

6.4
Quantum gate computing, Part II

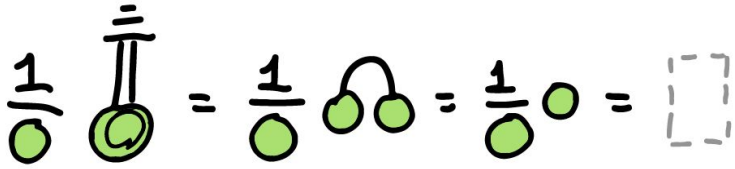
As it turns out, we did not just cheat about states and boxes... We also cheated on some equations. Sorry about that! Let's fix them quickly, shall we? Take the equations for copying 0° spider states:



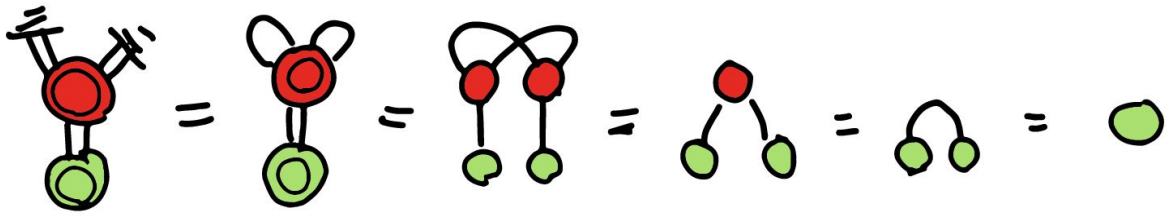
The first thing we do is turn the states involved into sure-states:



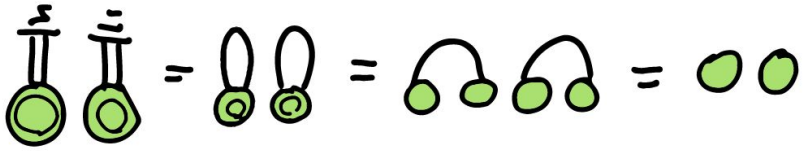
This indeed works as we'd expect:



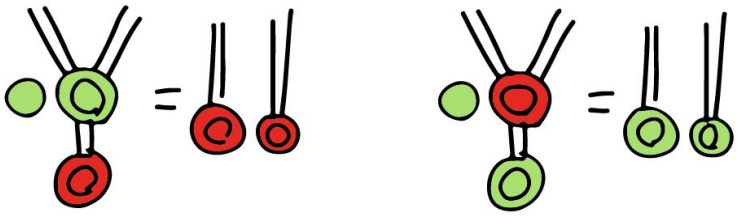
Now, to understand what's wrong with the copy equations, let's discard the outputs of both sides of one of them. If they are correct, we must get the same number in both cases. Discarding the left side, we get the number 2:



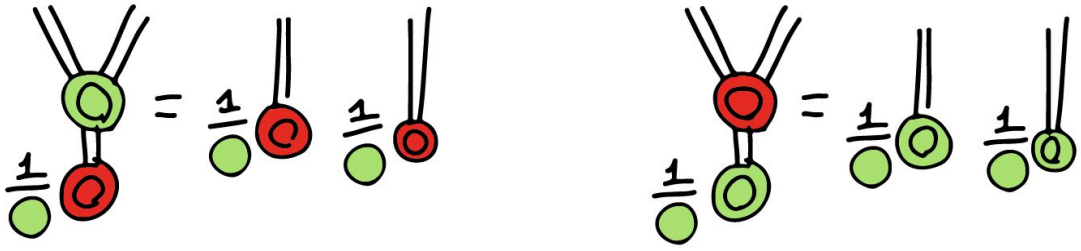
Discarding the right side, we get the number 4:



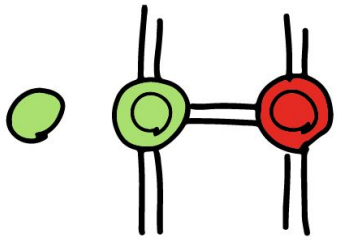
OK, something is wrong here! We need to add a green dot to our equation to balance things out:



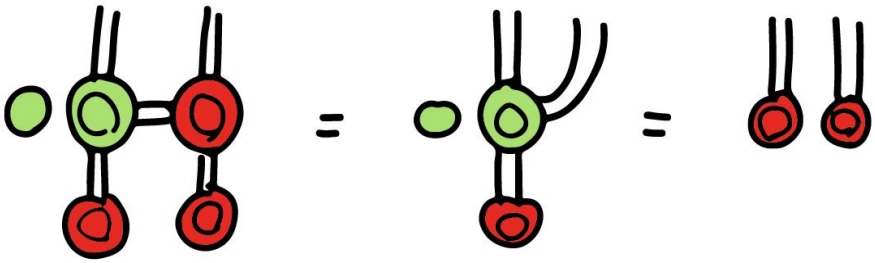
We now divide both sides of each equation by 4, that is, by two green dots. As there is one green dot in the picture, we drop that one, and divide what remains by a single green dot:



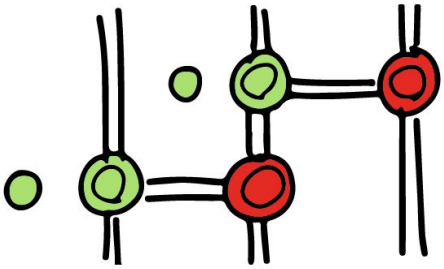
This shows that the above is exactly the same as using our new sure-states in the copy equations. The CNOT gate also needs a number to be a proper sure-box:



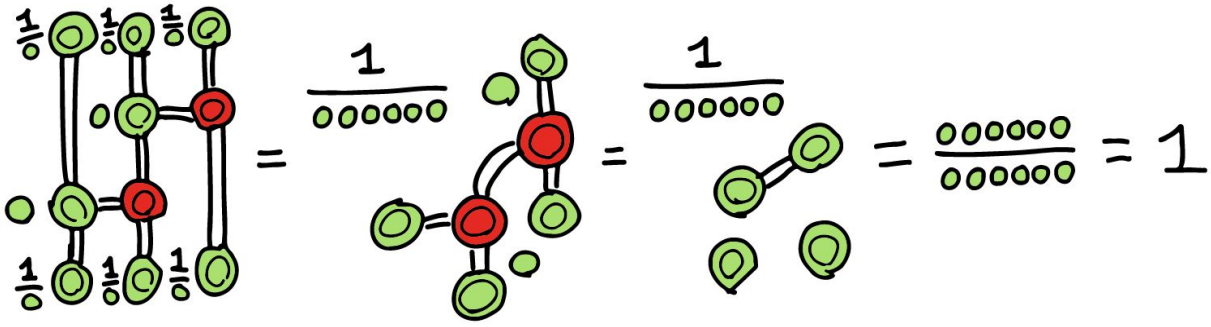
One way to see this is to stick two 0° red spider states into the input legs. We expect two 0° red spider states to come out, and indeed this happens if we have the extra number around to use in the copy equation:



Finally, we compute probabilities from a quantum circuit:



What is the probability that the green tests will happen on the output legs when we stick green spider states into the input legs?



That’s 1, which means it always happens. But to see this we need to be careful about carrying our legless spiders around.

6.5
Quantum uncertainty, Part II

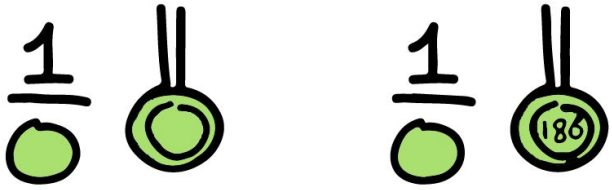
The most famous example of probabilities being fundamental in the quantum world is given by the desire to keep track of both:

1. where particles are, and
2. how fast particles move

For ordinary particles like cars, trains and planes, it is easy to keep track of both at the same time. For quantum particles, it turns out that keeping track of both is—to a degree—impossible. To see this, let's go back to our tinted glasses: green glasses show us where particles are, and red glasses show us how fast particles move. Think of the following spider states as "here" and "not here":

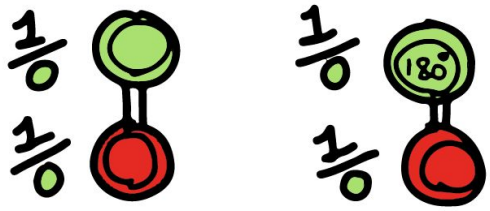


We can use green measurements to tell us whether the quantum particle is "here" or "not here". Think of the following spider states as "not moving" and "moving":

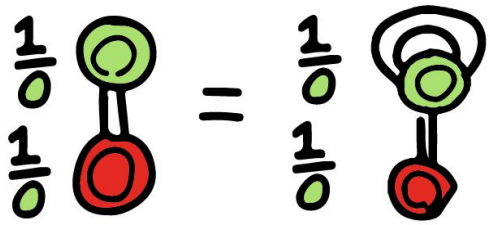


We can use red measurements to tell us whether the quantum particle is "not moving" or "moving".

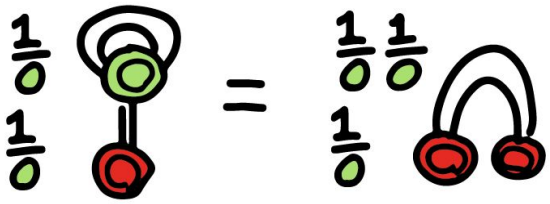
Suppose now that the quantum particle is “here”. Can we tell whether it’s moving or not? If we apply the “not moving” and “moving” tests to our particle, we get the following numbers:



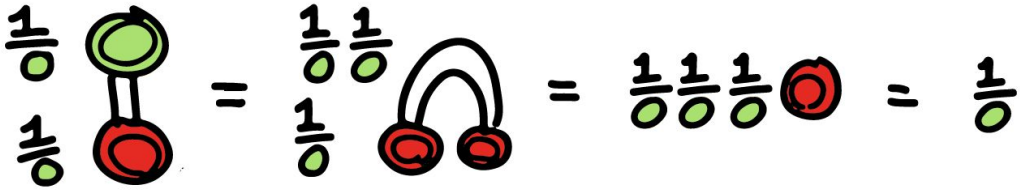
We haven’t seen them before, so we use some clever tricks to figure out what they are. By spider fusion, we have the following equation:



We can then copy the red spider state through:



After fusing the red spiders, we get to a number we already know, showing that the probability of the quantum particle “not moving” is 1-in-2:



The same probability comes out for the quantum particle “moving”, 1-in-2. What does

this mean? It means that if we know for certain where a quantum particle is, we know nothing about where it is going. Similarly, if we know for certain where a particle is going, we know nothing about where it is. This is one of the main features of quantum particles, and it puzzled physicists at the beginning of the last century. We just re-discovered it by moving some red and green spiders around. Isn't that cool?!

7.
HOW PROFESSIONALS NAME THINGS

In this final chapter we share a bit about the history of our pictures for the quantum world, and how professional physicists name the things we talked about in this book.

The first to draw wires and boxes was 2020 Nobel Prize winner Roger Penrose. He did so in the early 1970s: more than 50 years ago! Penrose pictures were not invented to describe the quantum world, but rather Einstein's theory of relativity.

In fact, our story starts much earlier, in 1935: it is at this point that John von Neumann started searching for new ways to talk about the quantum world. He had just published the first book containing the mathematics needed to describe quantum things, and after a mere three years, he didn't like that description anymore. Also in 1935, Erwin Schrödinger—yes, the one that is not always nice to cats—provided a different description of the same quantum phenomena, which encourages us to think about them in terms of wires and boxes.

Around 2003, our research community devoted itself to developing a pictorial theory of the quantum world, with an early paper by Bob, your author and guide on wonderful journey, titled "Kindergarten Quantum Mechanics", and professionals called this new field Categorical Quantum Mechanics. It doesn't quite sound like it has much to do with pictures, right? In fact, the most influential of these early papers, by Samson Abramsky and Bob, didn't feature any of the spiders we know and love.

Around 2006, spiders entered the picture, in a work by Bob, Dusko Pavlovic and Eric Paquette: but they were just one colour, and they were used to describe ordinary

particles rather than quantum particles. They were also given a complicated mathematical name, “special commutative dagger Frobenius algebras”. Had you met us at the time, you would have realised the kind of circus this was! Luckily, today even professionals call them spiders.

The green and red spiders, leg-chopping, square-popping, and all the other rules we learned about are part of what professionals call the ZX calculus: as we saw, this is much like arithmetic, but for quantum things. It took a while to kick in, but it is now being widely adopted in the quantum computing industry. The ZX calculus was invented by Bob and Ross Duncan in 2007: Ross, just like Bob, works at Quantinuum, trying to make quantum computers a reality. Over the next decade, people in Oxford, UK and Nancy, France worked hard on the ZX calculus, and showed that all of the quantum world can be described in pictures alone.

The story about single versus doubled wires and maybe-boxes vs sure-boxes also took a while to reach primetime, from 2006 to 2012. Many people were involved in this effort: Peter Selinger, Bob, Dusko, Eric, Aleks Kissinger and Quanlong Wang, just to name a few.

Once this was all done, Bob and Aleks wrote the big brother “dodo book” and Oxford University introduced the world’s first quantum computing course taught using pictures alone. The dodo book is comprehensive—enough to teach several courses!—but it is also 900 pages long, tailored for university students and professionals. That is where this book finally comes in: we selected the most important topics and presented them in such a way as to make them accessible to a much wider audience.

Much of the physics presented here was known before the pictures, therefore quite a bit more painstakingly. Those that did so include cryptographers Bennett, Brassard & Ekert, teleporters Bennett, Brassard, Crepeau, Jozsa, Peres & Wootters, broadcasters Barnum, Caves, Fuchs, Jozsa & Schumacher, swappers Zukowski, Zeilinger, Horne &

Ekert, and measurement based theorists Raussendorf, Briegel and Perdrix.

Much of the maths presented here, before it was pictures, is known as Australian category theory, not joking! It was done by Kelly, Walters, Street, Lack, and also Lambek, Bénabou, Carboni, and Joyal.

We'll now go through the remaining words used in this book and how professionals name things instead:

quantum lottery ↔ quantum uncertainty
rotating ↔ transposing
maybe-... ↔ postselected
sure-... ↔ causal
suretest ↔ demolition measurement
vertical flip ↔ adjoint
horizontal flip ↔ conjugate
flip-undoable ↔ isometry
doubly-flip-undoable ↔ unitarity
cup-fusion ↔ entanglement swapping
colour change box ↔ Hadamard gate
quantum test comp. ↔ measurement based quantum comp.
no-texting ↔ no-signaling
ordinary ↔ classical
ordinary cup state ↔ perfect correlation
quantum cup state ↔ Bell-state
phase-erasure ↔ decoherence
leaky ↔ mixed / open
non-leaky ↔ pure / closed
no quantum-copy ↔ no broadcasting
inventing a secret language ↔ quantum cryptography

Other words stay the same, like state, test (also called effect), teleportation, phase, phase-gadget and non-locality.

Hope you had fun! Bye.

AFTERWORD

I first came across Bob Coecke more than 15 years ago when his paper “Kindergarten Quantum Mechanics” was brought to my attention. I am a latecomer to quantum computing and quantum information theory. My primary interest in quantum mechanics in my younger years and well into my 40’s was the “why” rather than the “what” (i.e. theories such as Qbism that try to explain why we experience quantum mechanics in ways that are confusing and often counter-intuitive). Like so many of us who encounter quantum mechanics or quantum information theory, I have been engrossed by the compelling debates around the various theories that have pre-occupied quantum theorists and philosophers since the infamous Einstein-Bohr debates that date back almost a century.

Things changed for me a few years ago, when quantum computing became my career and I started to care just as much about the “what” as the “why” – and I started to become practically involved in how quantum computers might be used to solve some of the world’s most intractable problems. For the past 8 years working with quantum computers and quantum algorithms has been my full-time job.

A couple of years after I started working in quantum computing I came across “Picturing Quantum Processes” (the so called “Big Brother” of this work) - a book that was quickly gaining cult status among younger quantum physicists. I bought it more out of curiosity than anything else, and when it arrived, I quickly found myself unable to tear myself away, as I dissolved into an alternative universe with three or four days merging into one long, exciting journey of discovery.

It has taken over 40 years of engineering advances to get to the point where we can truly acknowledge the advent of early-stage quantum processors, and with that, a realistic expectation that quantum computing is slowly becoming a reality. It is now only a

question of “when”, and not “if”, quantum computers will solve real-world problems and have an impact on our daily lives.

This confluence of circumstances – engineering and theoretical advances in quantum mechanics, quantum information theory and quantum computers – makes “Quantum in Pictures” relevant for reasons that matter far beyond academia.

This book is also destined for cult status. As I write, “samizdat” versions are already being shared among quantum computing enthusiasts.

“Quantum in Pictures” will inspire people to consider jumping into the deep end to learn about quantum mechanics and think more carefully about how they get involved with quantum computing.

A direct consequence of the approach to quantum physics articulated and so creatively presented in this book, via a formalism that ties together von Neumann, Schrödinger, and Penrose (by way of Feynman and Dirac), will be that Bob and Stefano create a foundation that supports the interests and aspirations of a new generation of both generalists as well as students of quantum physics. This book will give many the impetus to investigate and learn about the subject and not have to confront or worry about the crippling lack of confidence engendered by traditional teaching approaches.

“Quantum in Pictures” is a book that takes us beyond the “why” questions that mesmerised the giants of quantum physics a century ago, to the “what” questions being tackled by today’s pioneers of quantum information technology. It is the perfect jumping-off point for every reader who feels ready to make the leap many have already made: from the “why” of what quantum tells us about the universe to the “what”.

Ilyas Khan

COLOPHON

Quantinuum

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Chapter 3

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Chapter 4

Aerial view of the CERN site just outside Geneva, with the Jura mountains in the background. The large circle shows the line of the LEP tunnel, 27 km in circumference, the small circle shows the SPS tunnel, 7 km in circumference. The crossed line indicates the the border between France and Switzerland. Image CERN-SI-8701973, © 1987-2022 CERN (License: CC-BY-4.0), https://cds.cern.ch/record/39027

Chapter 5

The photo of Albert Einstein sticking out his tongue when asked for a birthday pose, was shot on Einstein's 72nd birthday, 14th March 1951 by photographer Arthur Sasse for UPI/CORBIS.

A photo of an old ENIAC computer, Getty images